

(12) **United States Patent**
Burak et al.

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(54) **BULK ACOUSTIC WAVE RESONATOR WITH
PATTERNED LAYER STRUCTURES,
DEVICES AND SYSTEMS**

(56) **References Cited**

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U.S.C. 154(b) by 0 days.

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Primary Examiner — Bryan P Gordon

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(Continued)

(51) **Int. Cl.**
H03H 9/13 (2006.01)
H03H 3/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H03H 9/02259** (2013.01); **H03H 3/02**
(2013.01); **H03H 9/02015** (2013.01);
(Continued)

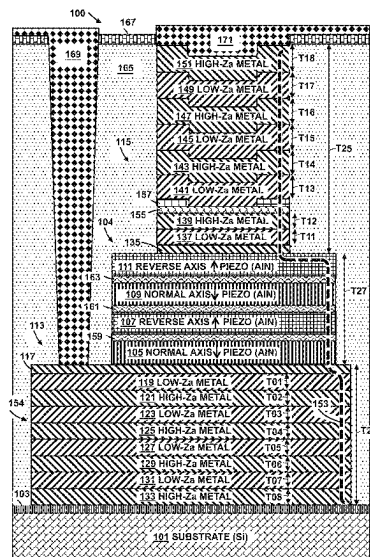
(58) **Field of Classification Search**
CPC H03H 9/02259; H03H 9/17; H03H 9/173;
H03H 9/175

See application file for complete search history.

(57) **ABSTRACT**

Techniques for improving Bulk Acoustic Wave (BAW) resonator structures are disclosed, including filters, oscillators and systems that may include such devices. First and second layers of piezoelectric material may be acoustically coupled with one another to have a piezoelectrically excitable resonance mode. The first layer of piezoelectric material may have a first piezoelectric axis orientation, and the second layer of piezoelectric material may have a second piezoelectric axis orientation that substantially opposes the first piezoelectric axis orientation of the first layer of piezoelectric material. An acoustic reflector electrode may include a pair of top metal electrode layers electrically and acoustically coupled with the first and second layer of piezoelectric material to excite the piezoelectrically excitable resonance mode at a resonant frequency of the BAW resonator. The acoustic reflector may include a patterned layer.

34 Claims, 35 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/380,011, filed on Jul. 20, 2021, now Pat. No. 11,863,153, said application No. 17/564,216 is a continuation of application No. PCT/US2020/043733, filed on Jul. 27, 2020, said application No. 17/380,011 is a continuation of application No. 16/940,172, filed on Jul. 27, 2020, now Pat. No. 11,101,783.

- (60) Provisional application No. 62/881,085, filed on Jul. 31, 2019, provisional application No. 62/881,074, filed on Jul. 31, 2019, provisional application No. 62/881,091, filed on Jul. 31, 2019, provisional application No. 62/881,087, filed on Jul. 31, 2019, provisional application No. 62/881,061, filed on Jul. 31, 2019, provisional application No. 62/881,094, filed on Jul. 31, 2019, provisional application No. 62/881,077, filed on Jul. 31, 2019.

(51) **Int. Cl.**

H03H 9/02 (2006.01)
H03H 9/17 (2006.01)
H03H 9/205 (2006.01)
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H03H 9/56 (2006.01)

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FIG. 1A

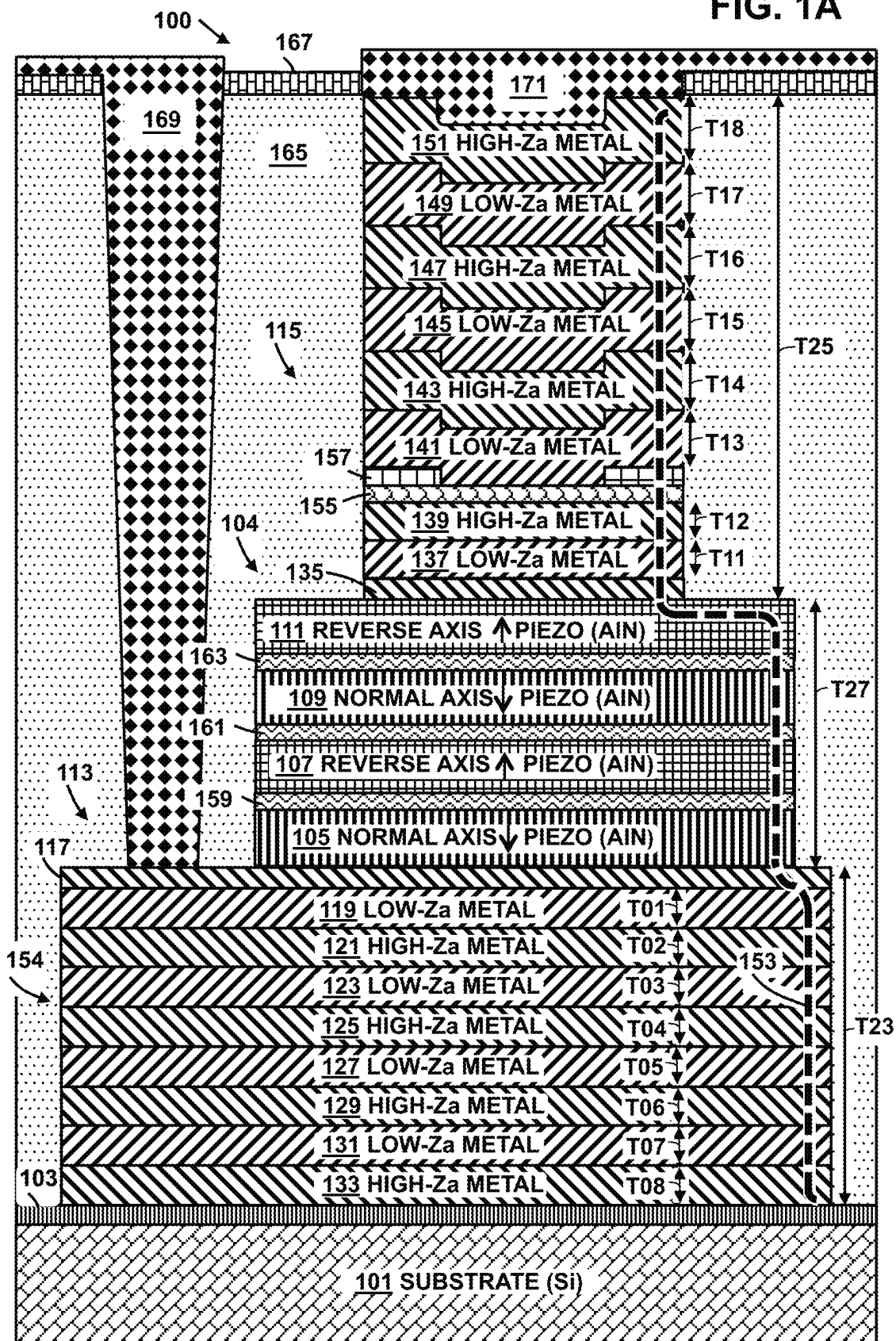
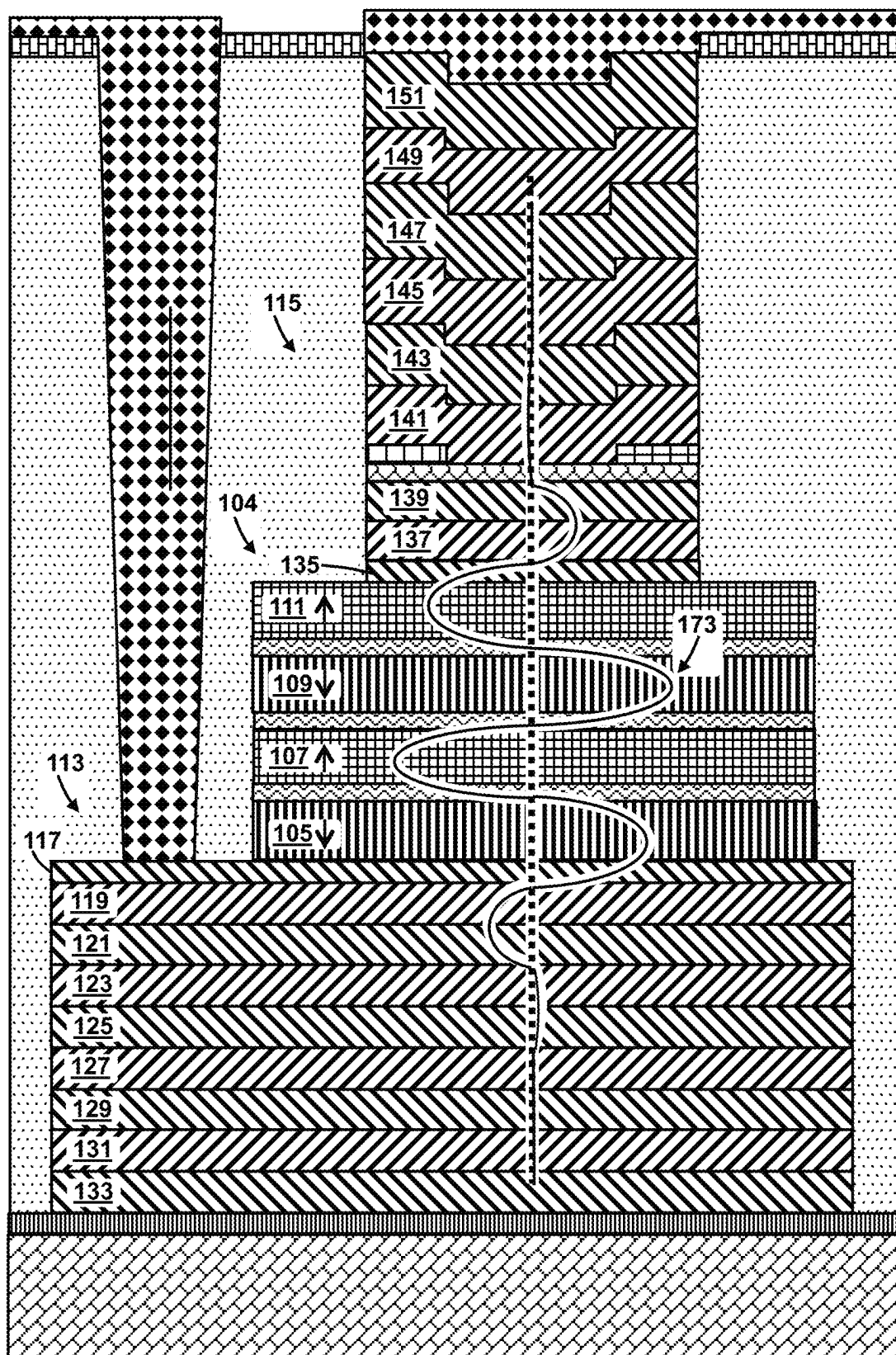
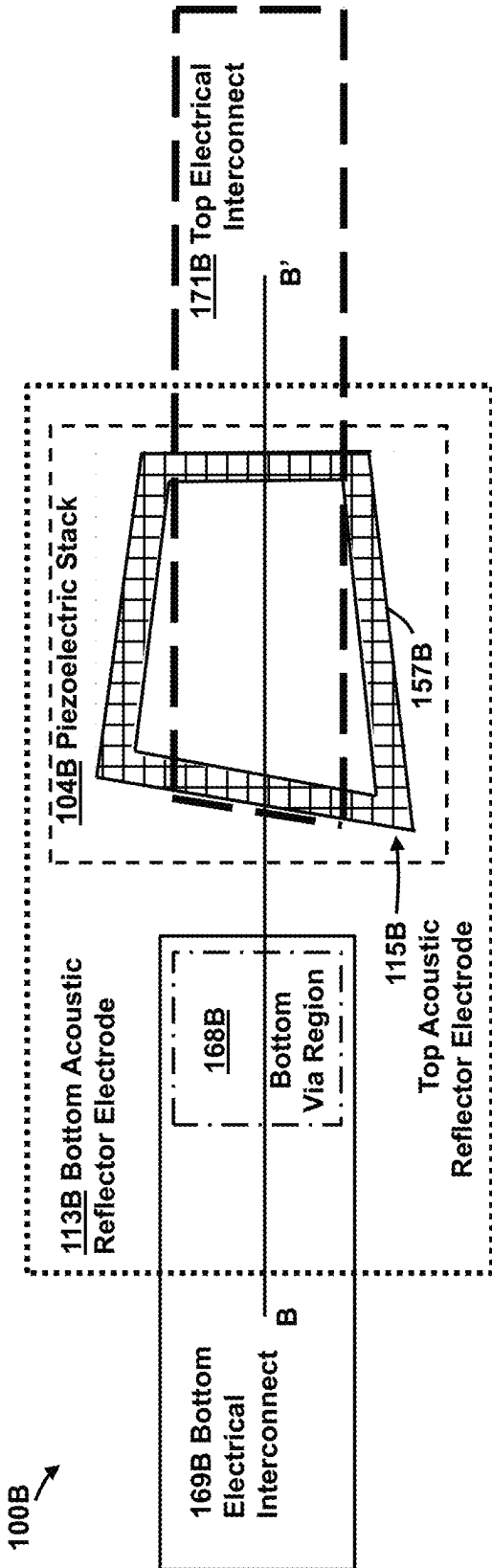
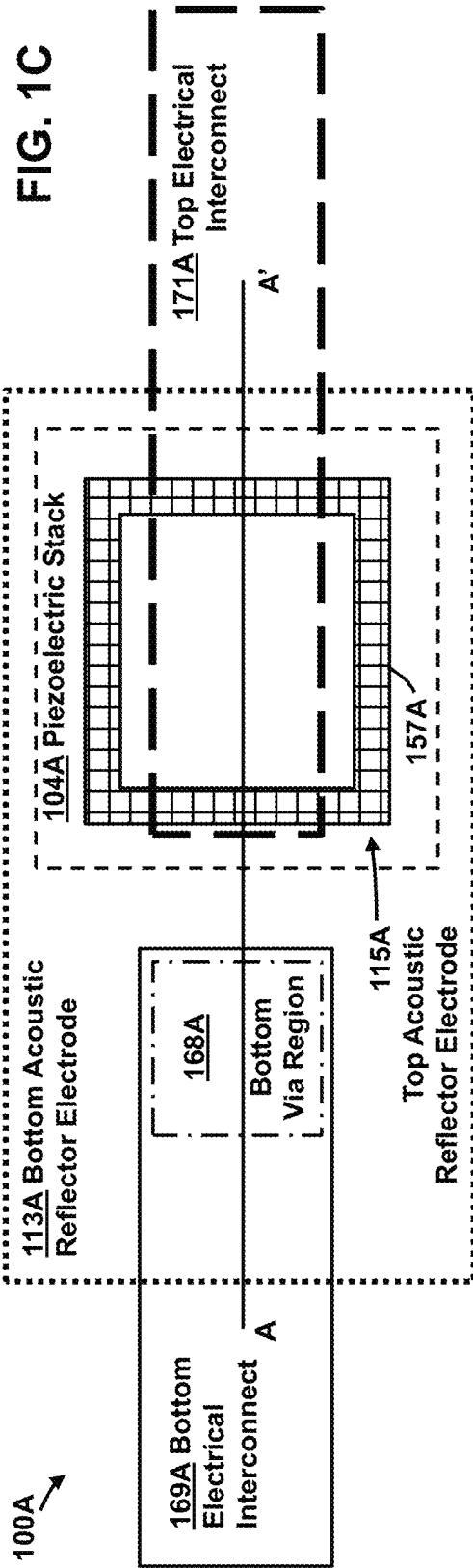


FIG. 1B





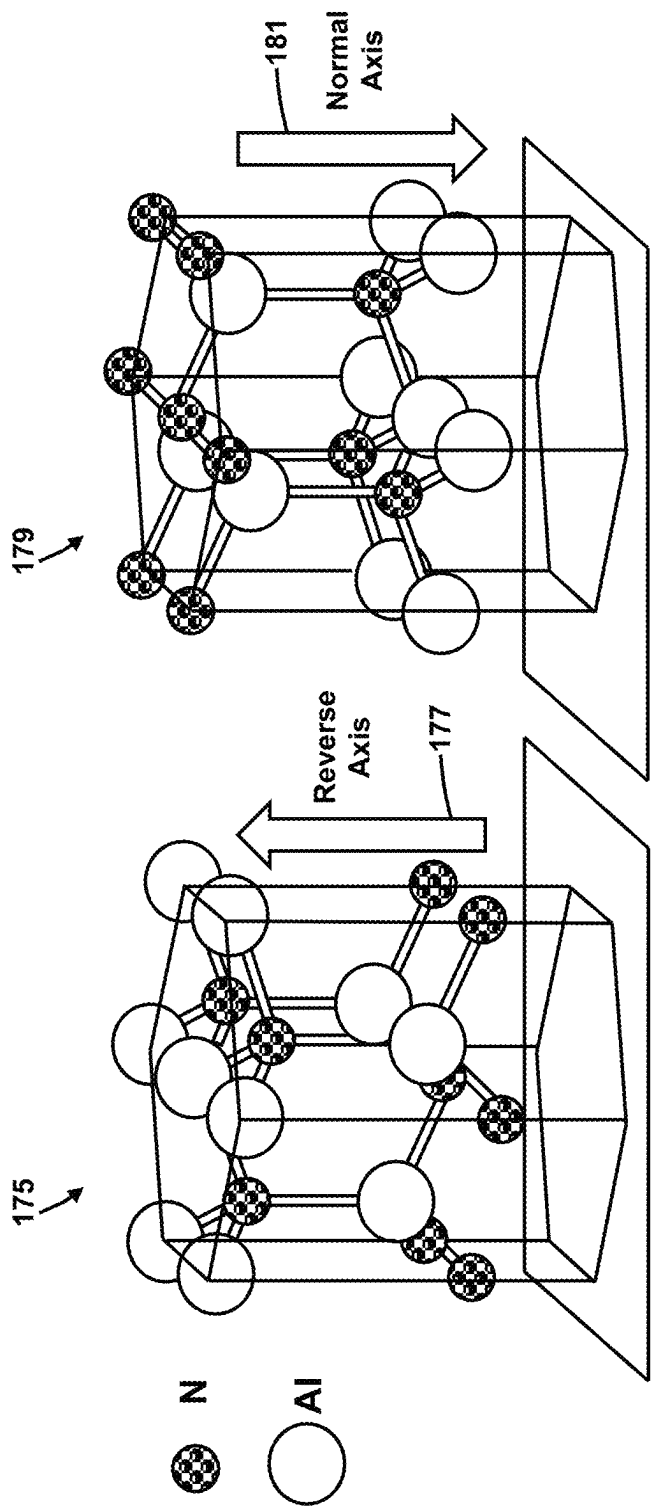
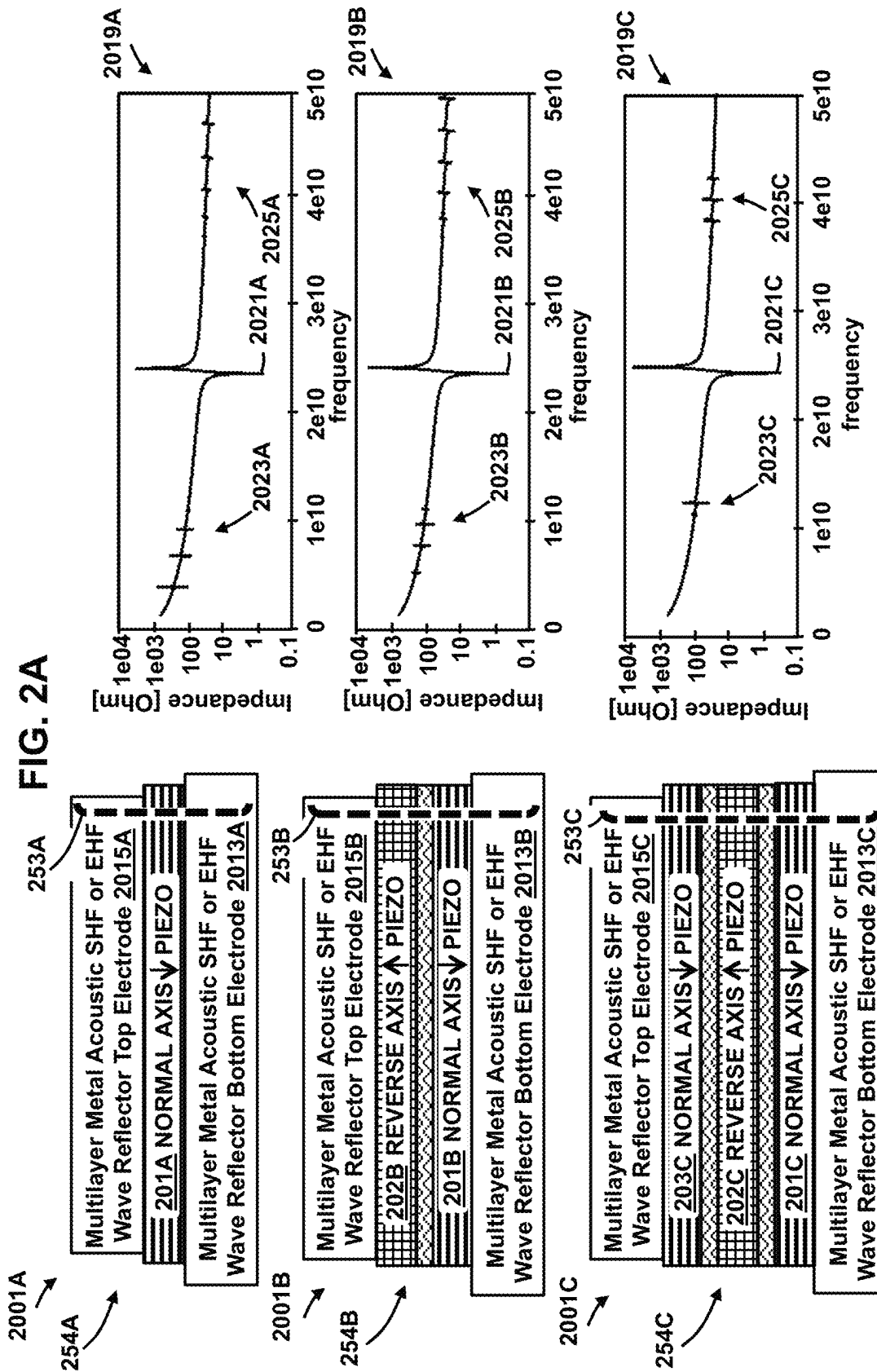


FIG. 1E

FIG. 1D



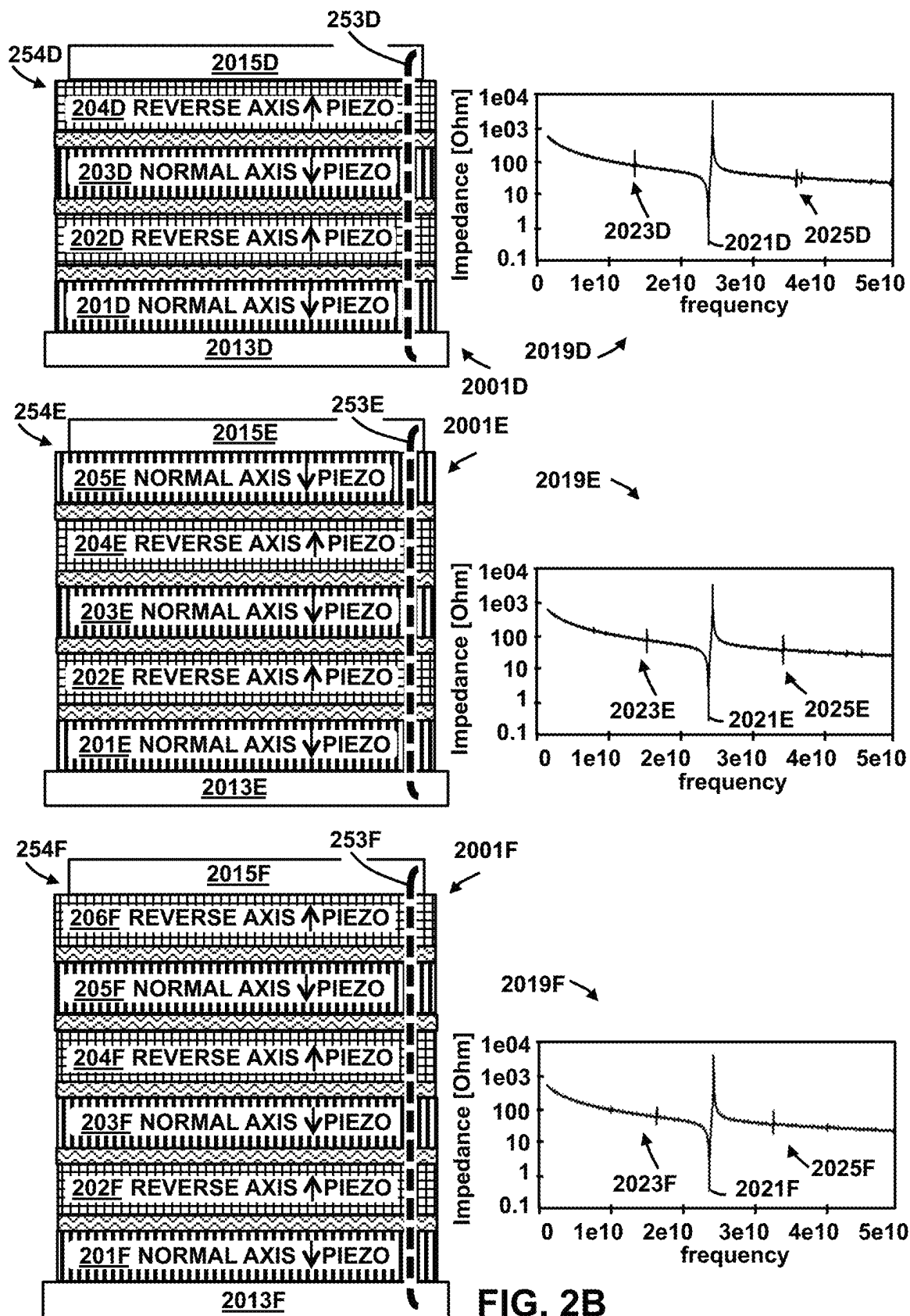
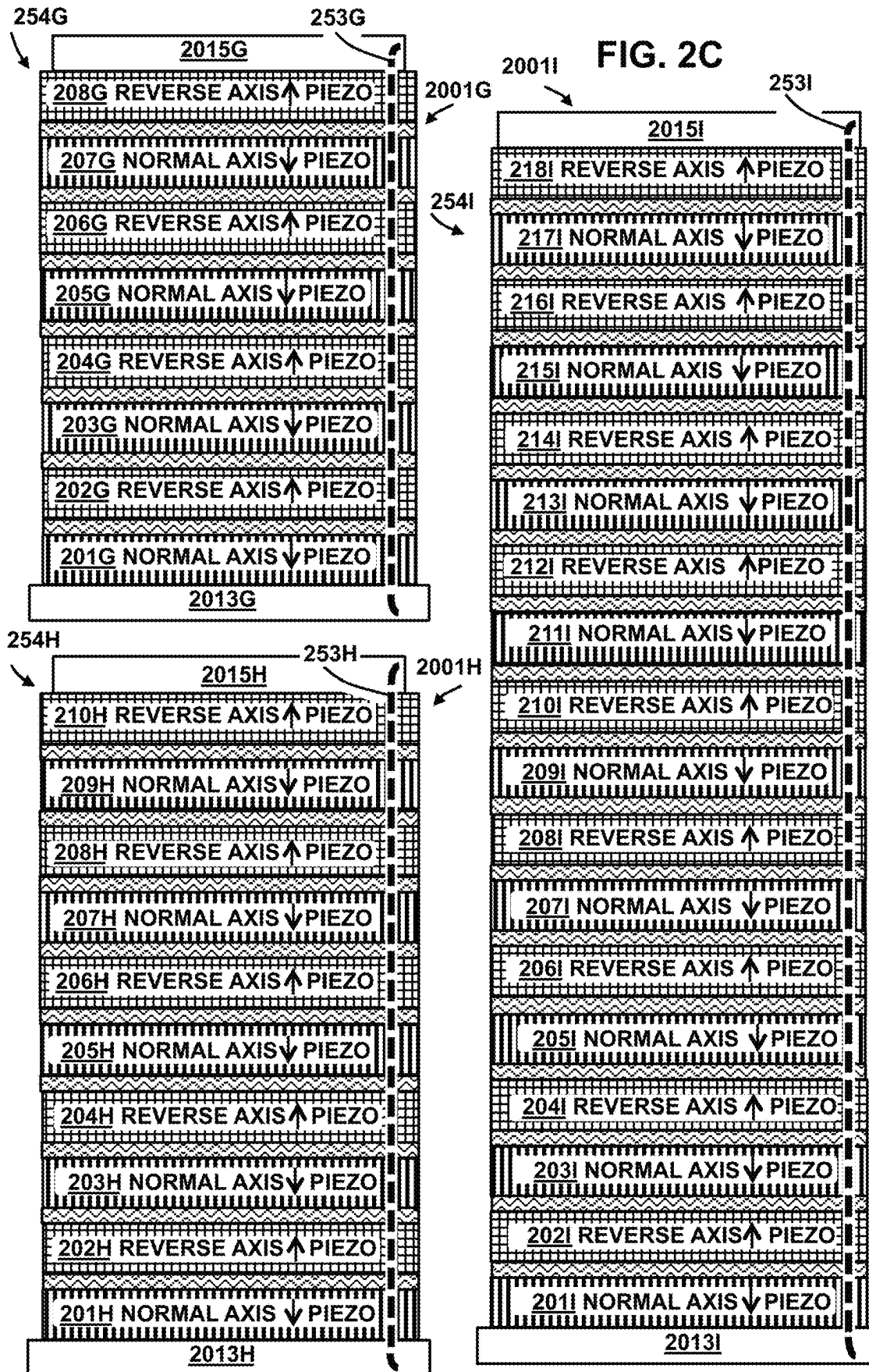


FIG. 2B



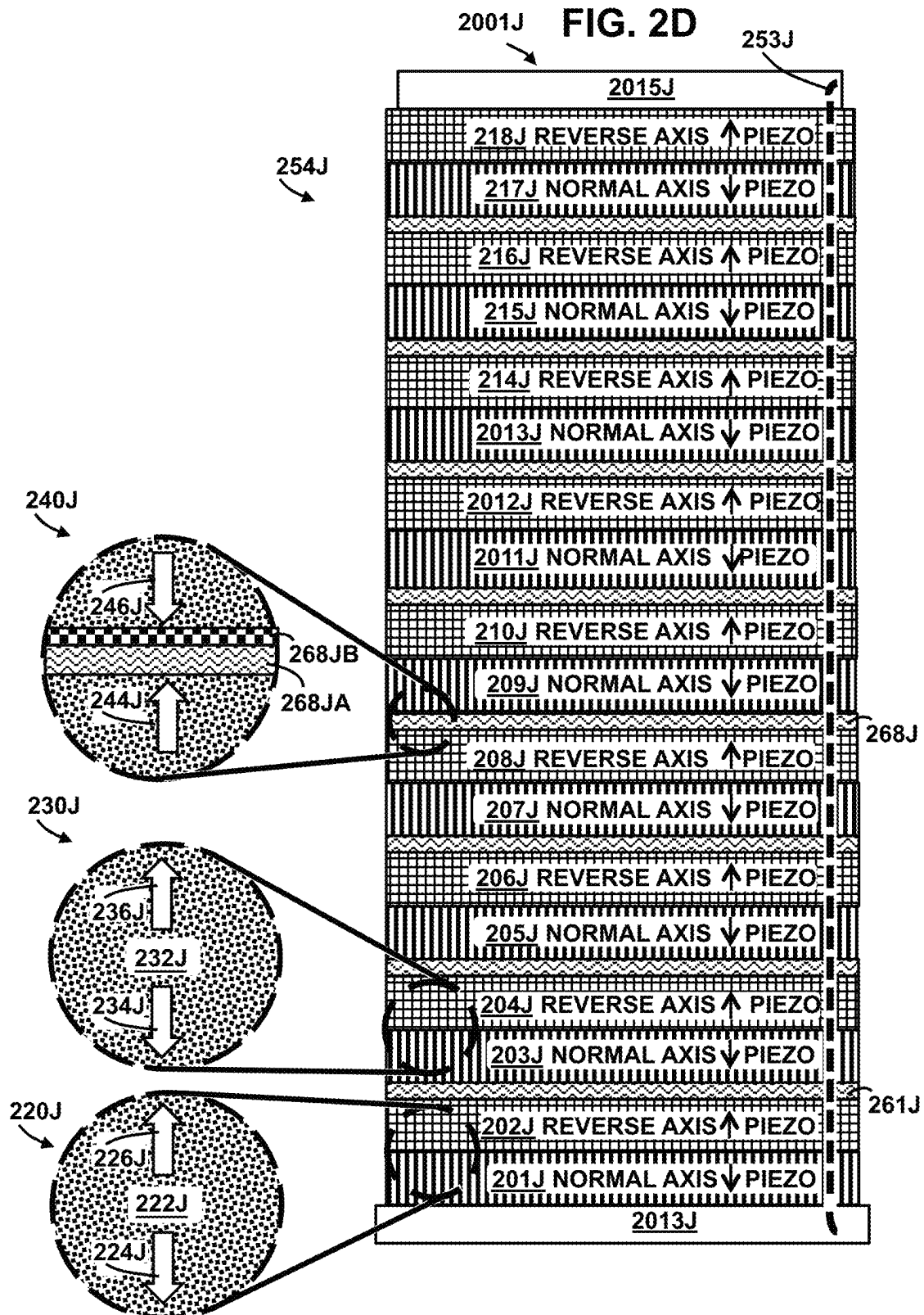
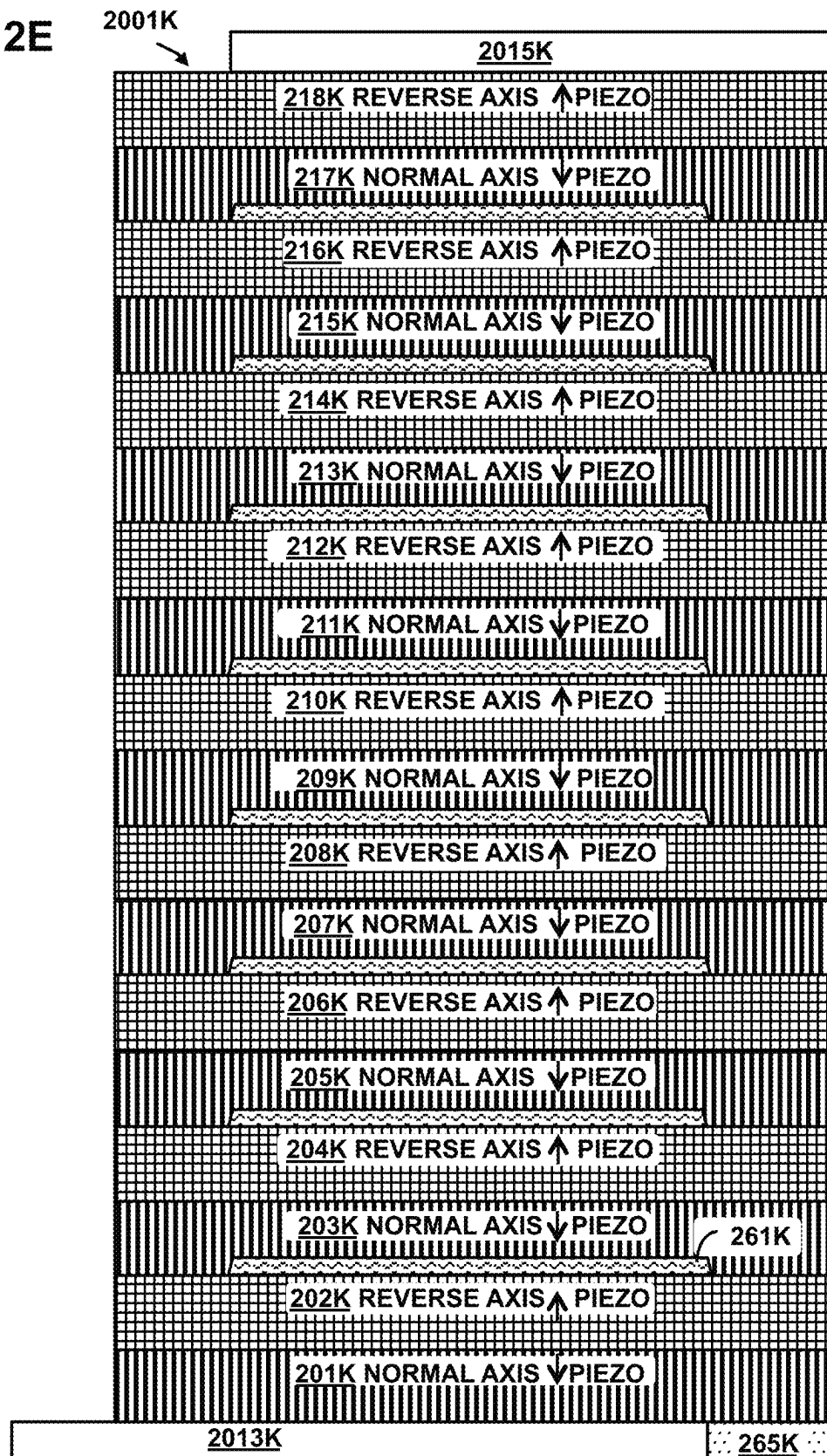
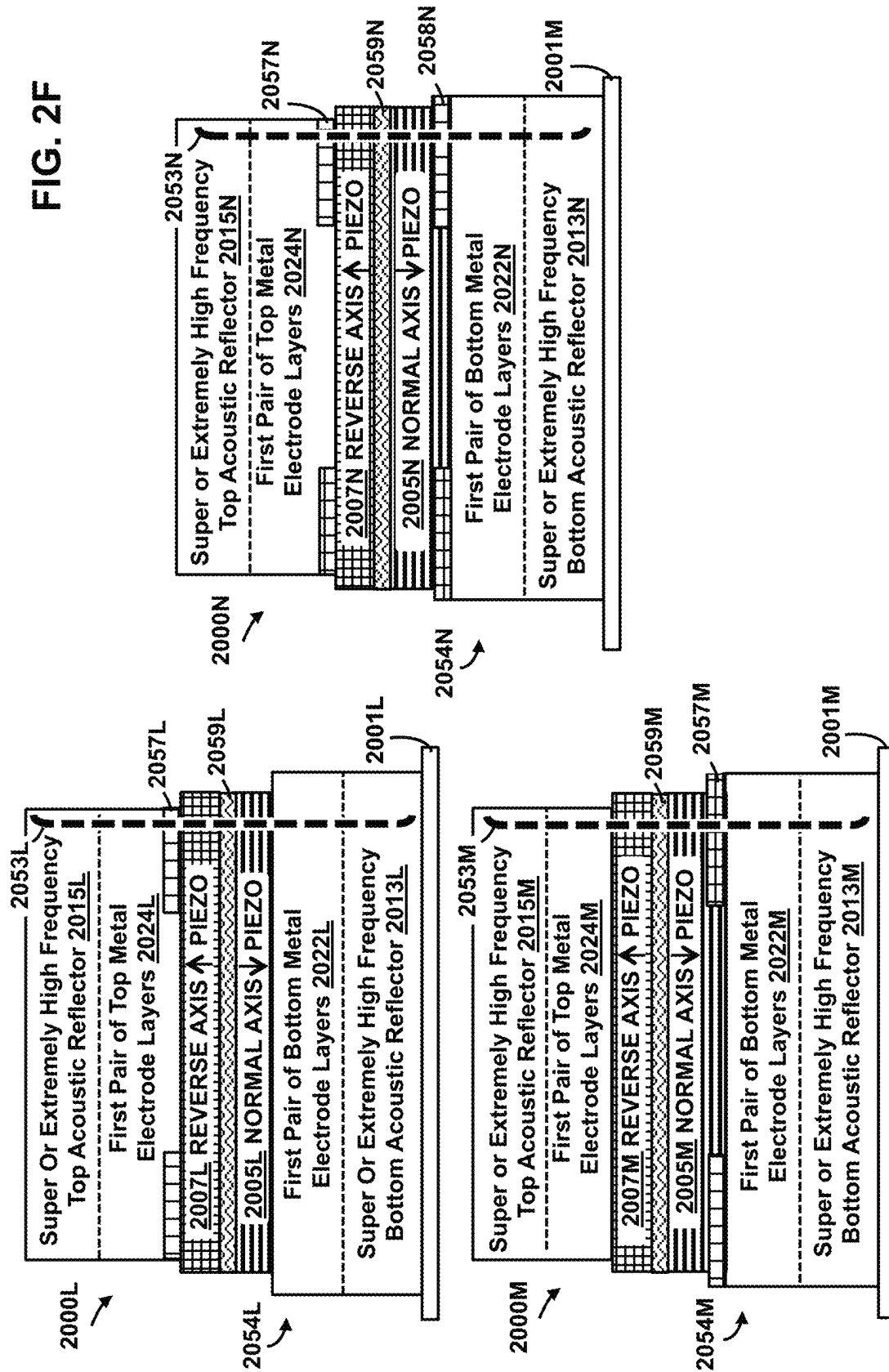
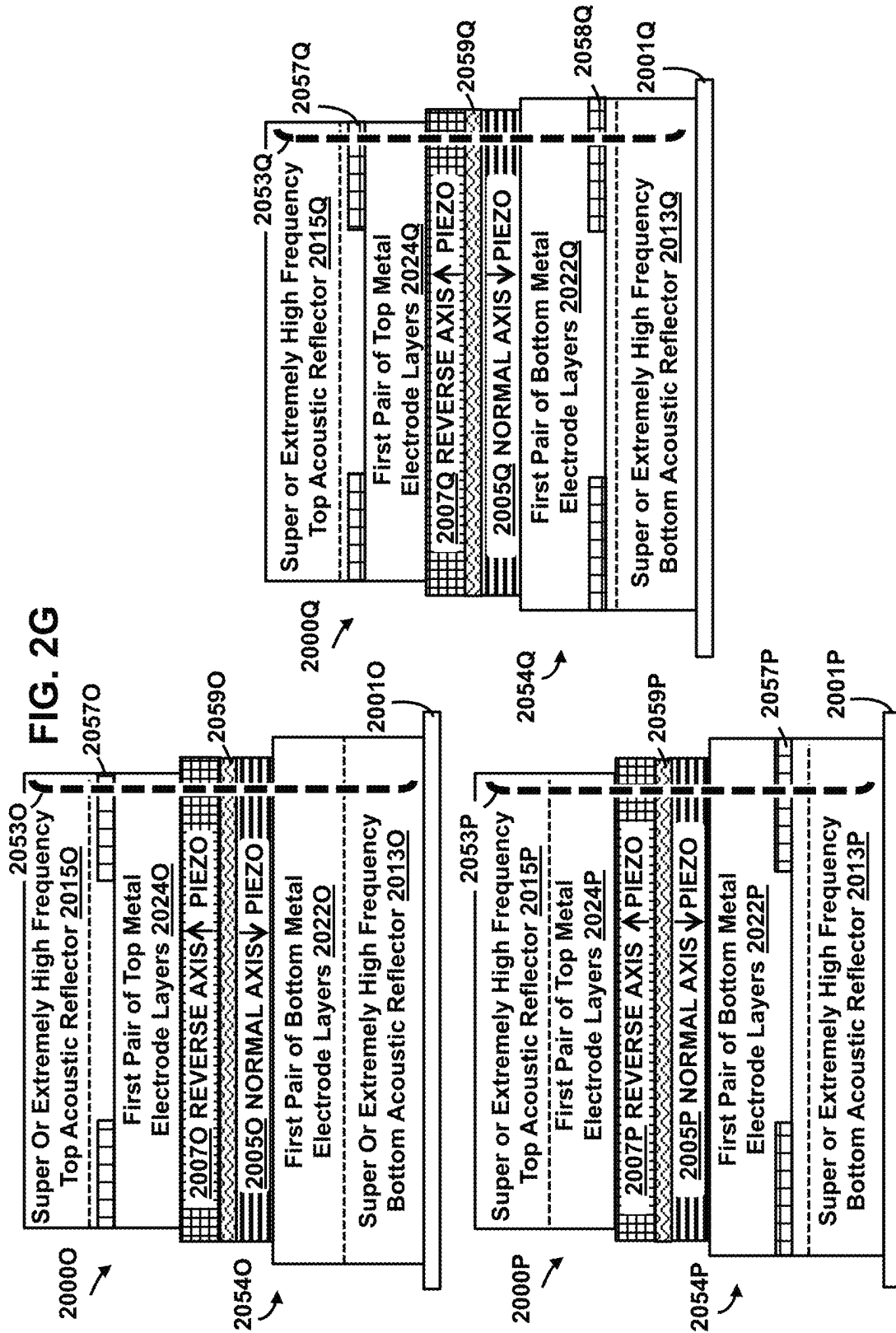
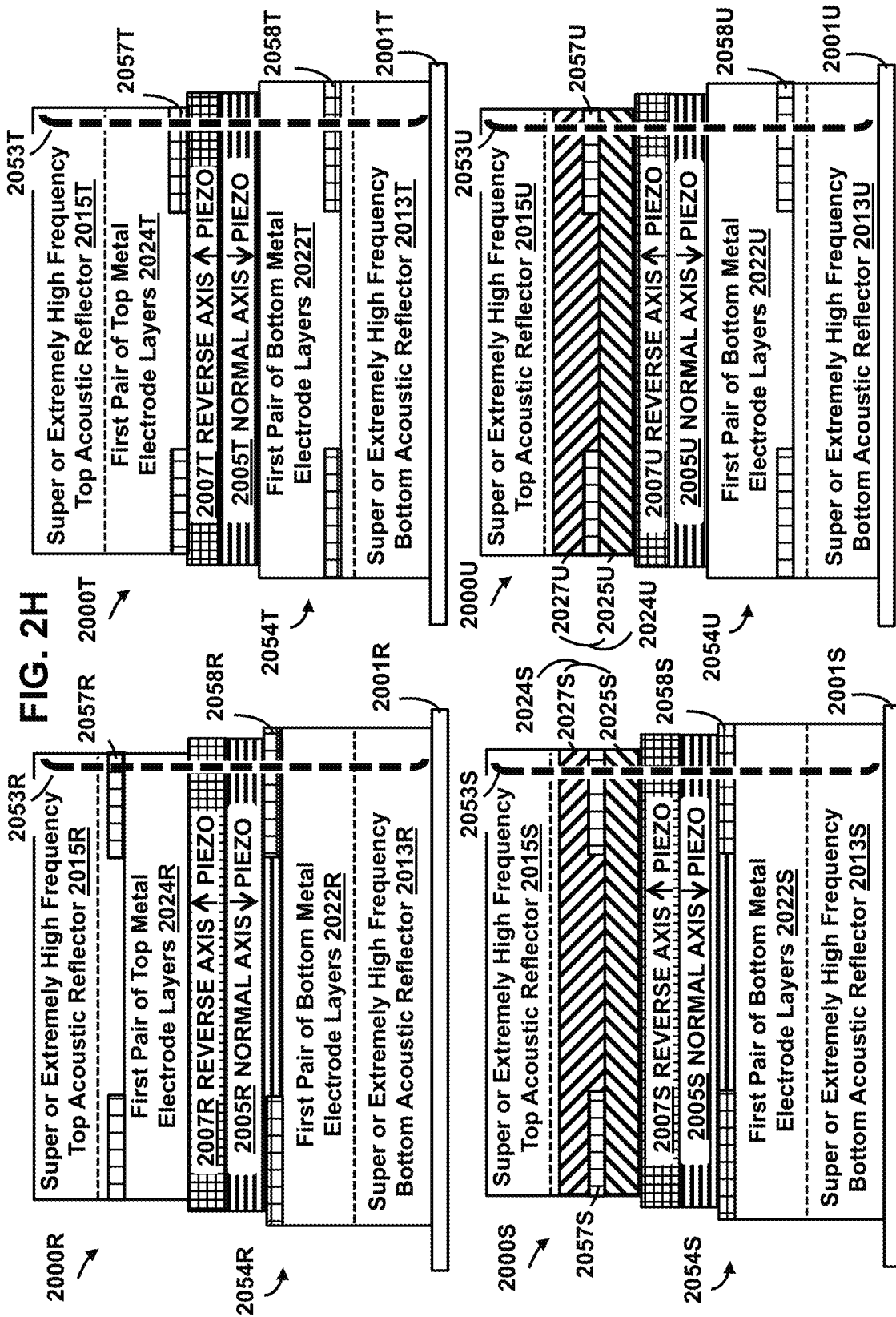


FIG. 2E









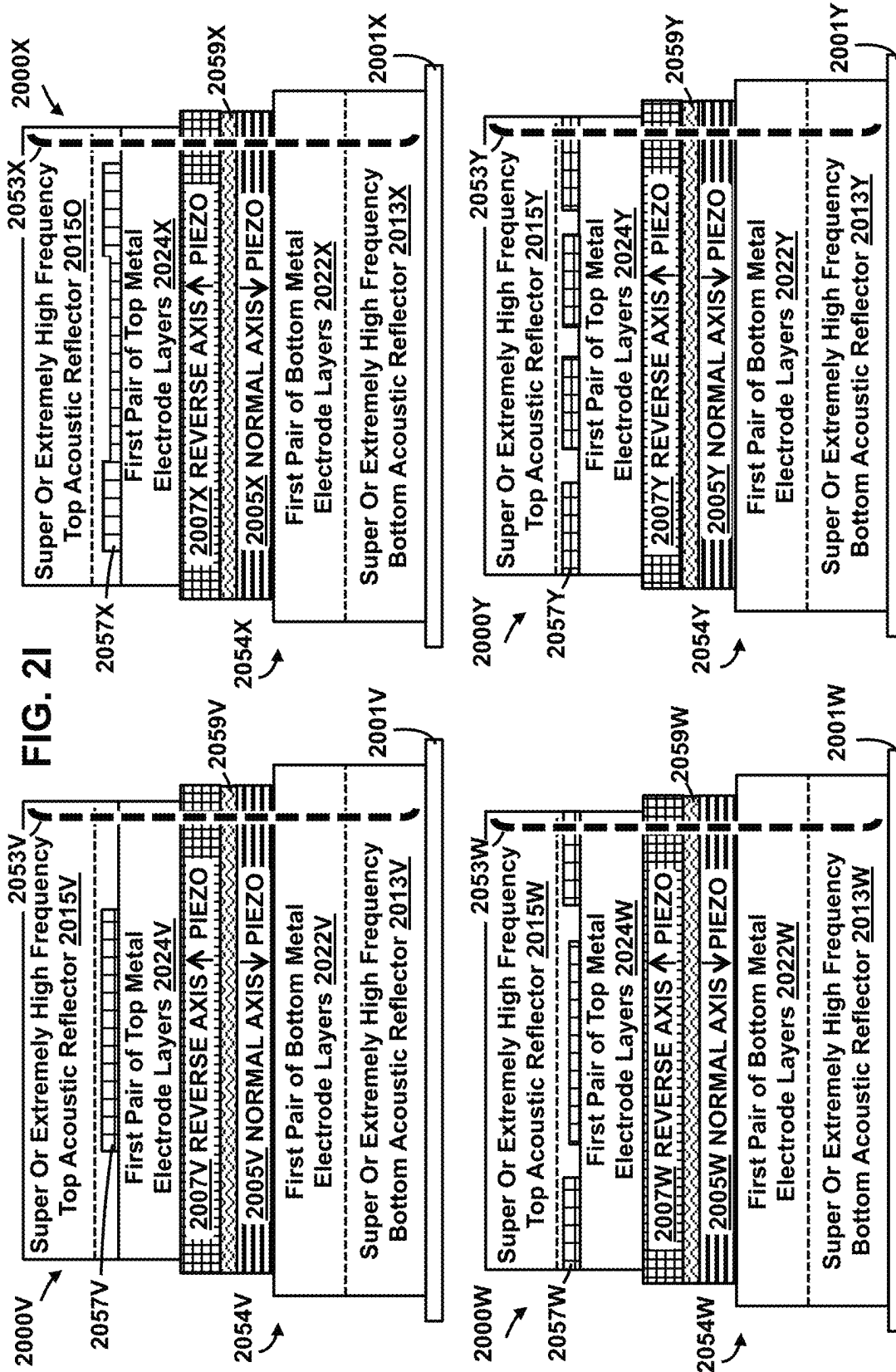


FIG. 2J

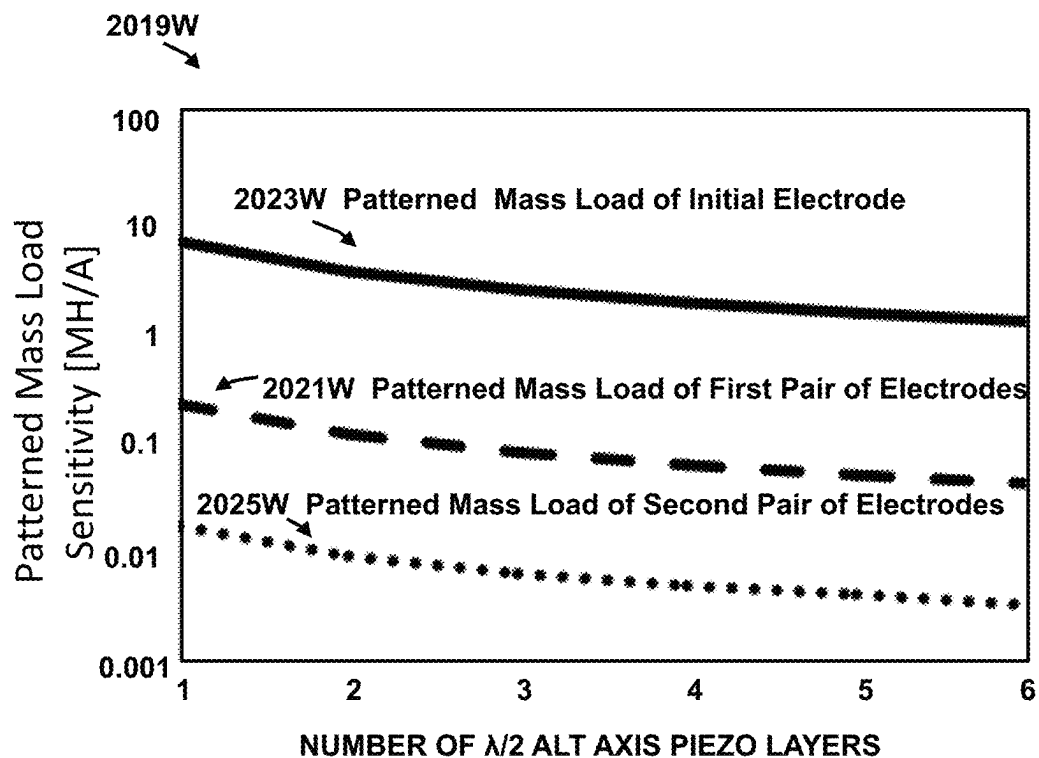
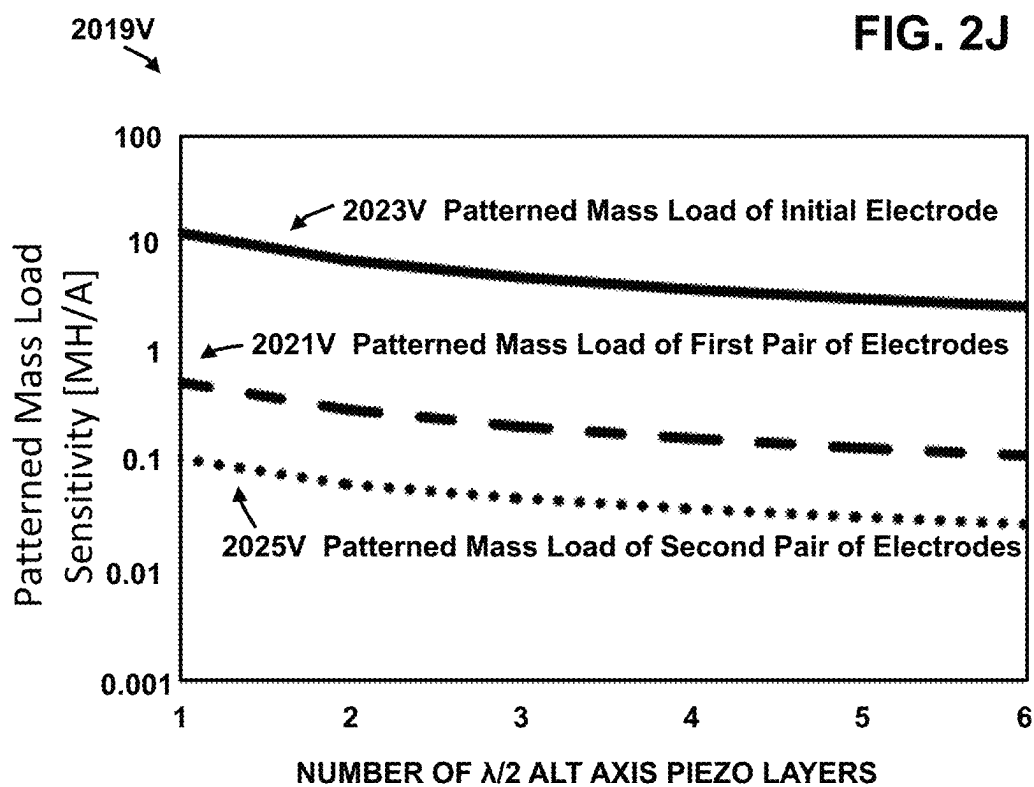


FIG. 3A

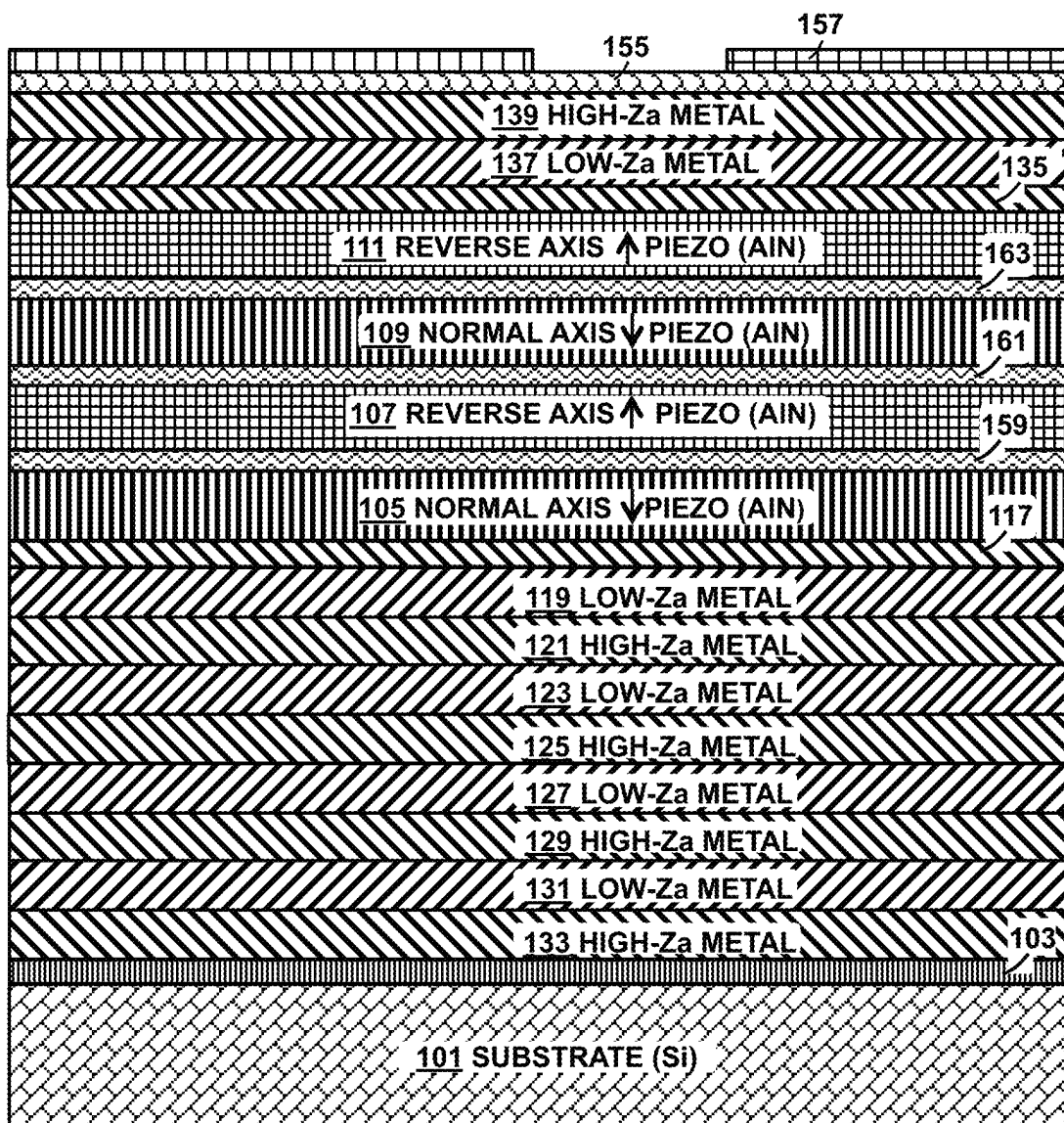


FIG. 3B

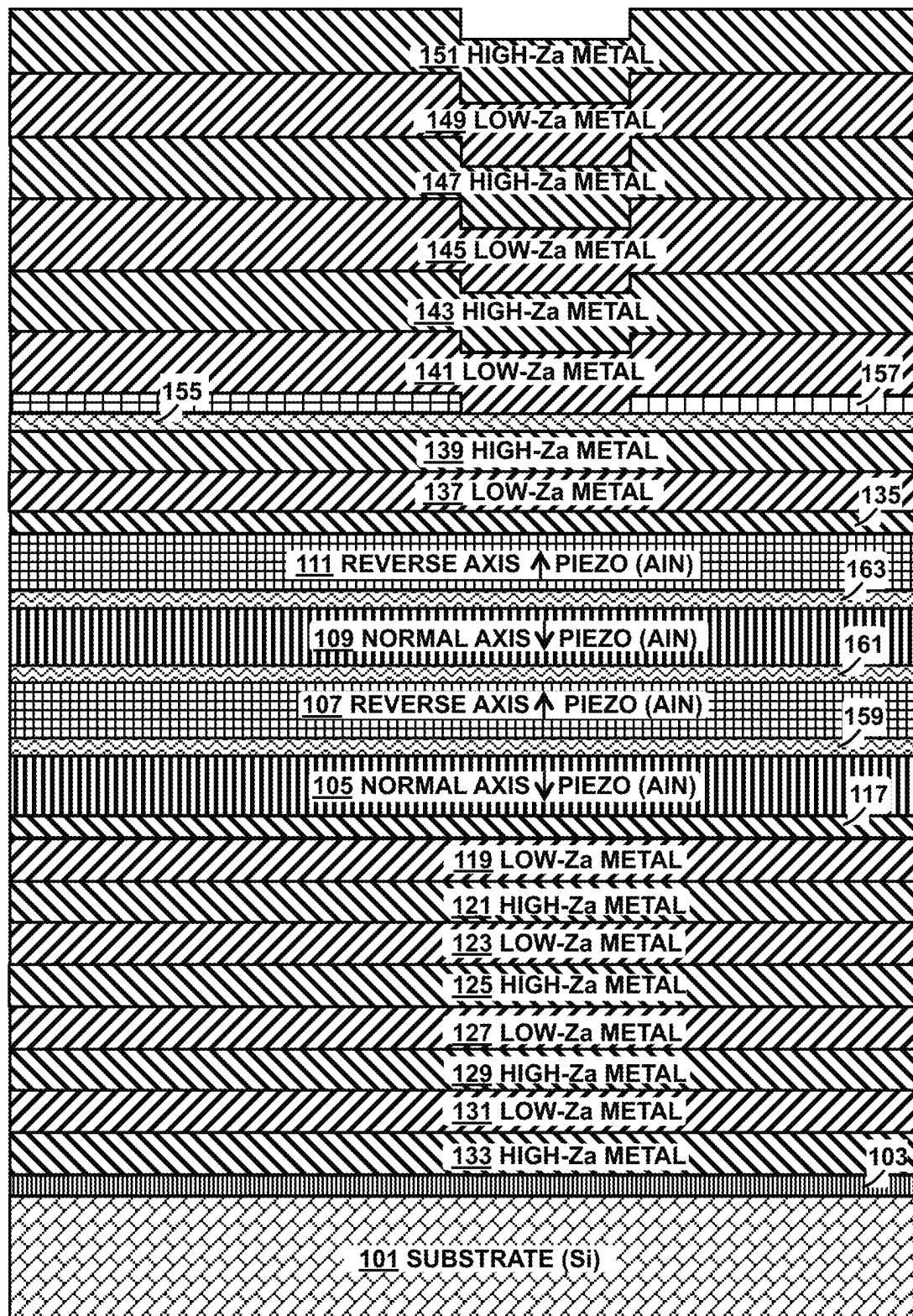


FIG. 3C

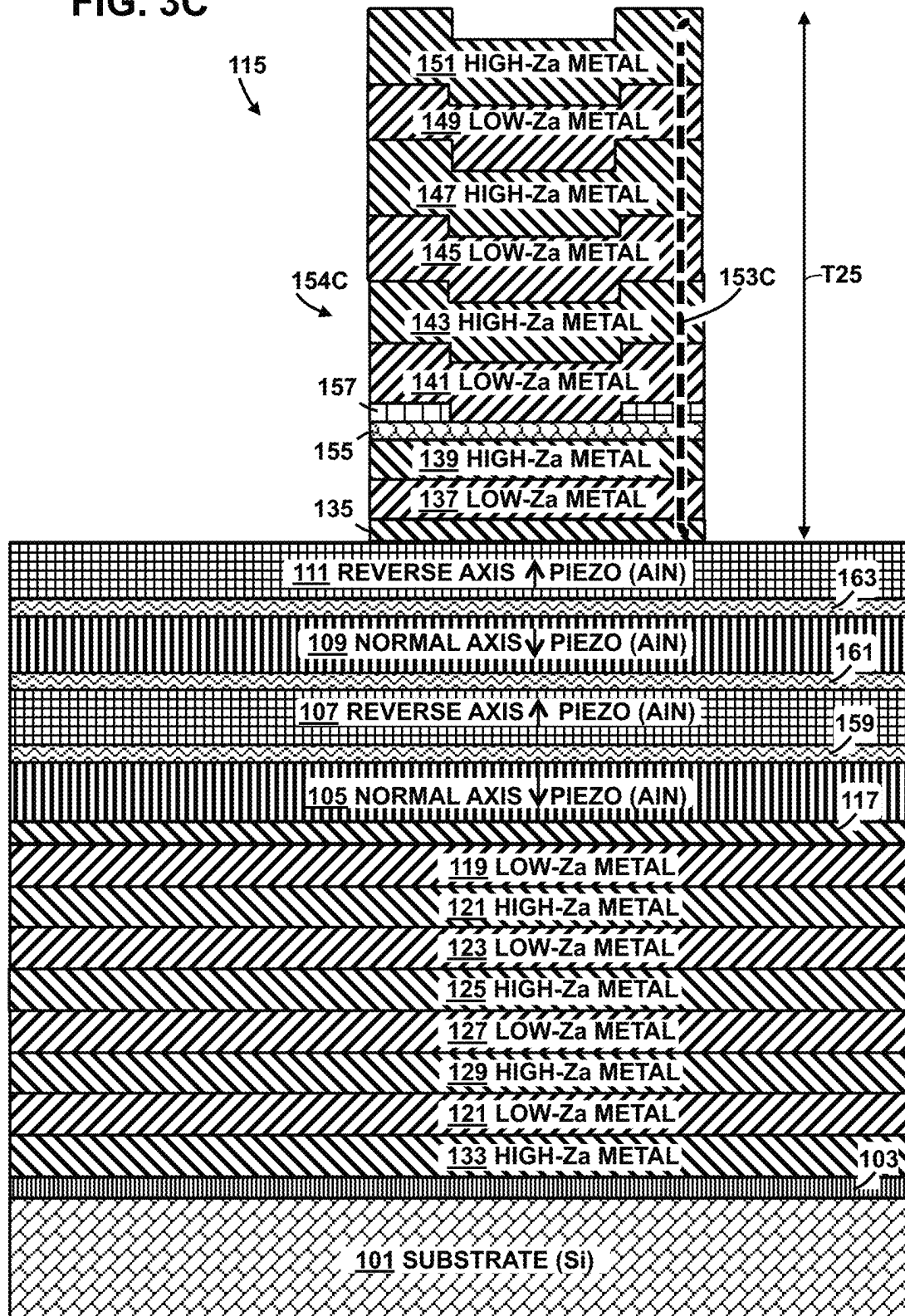
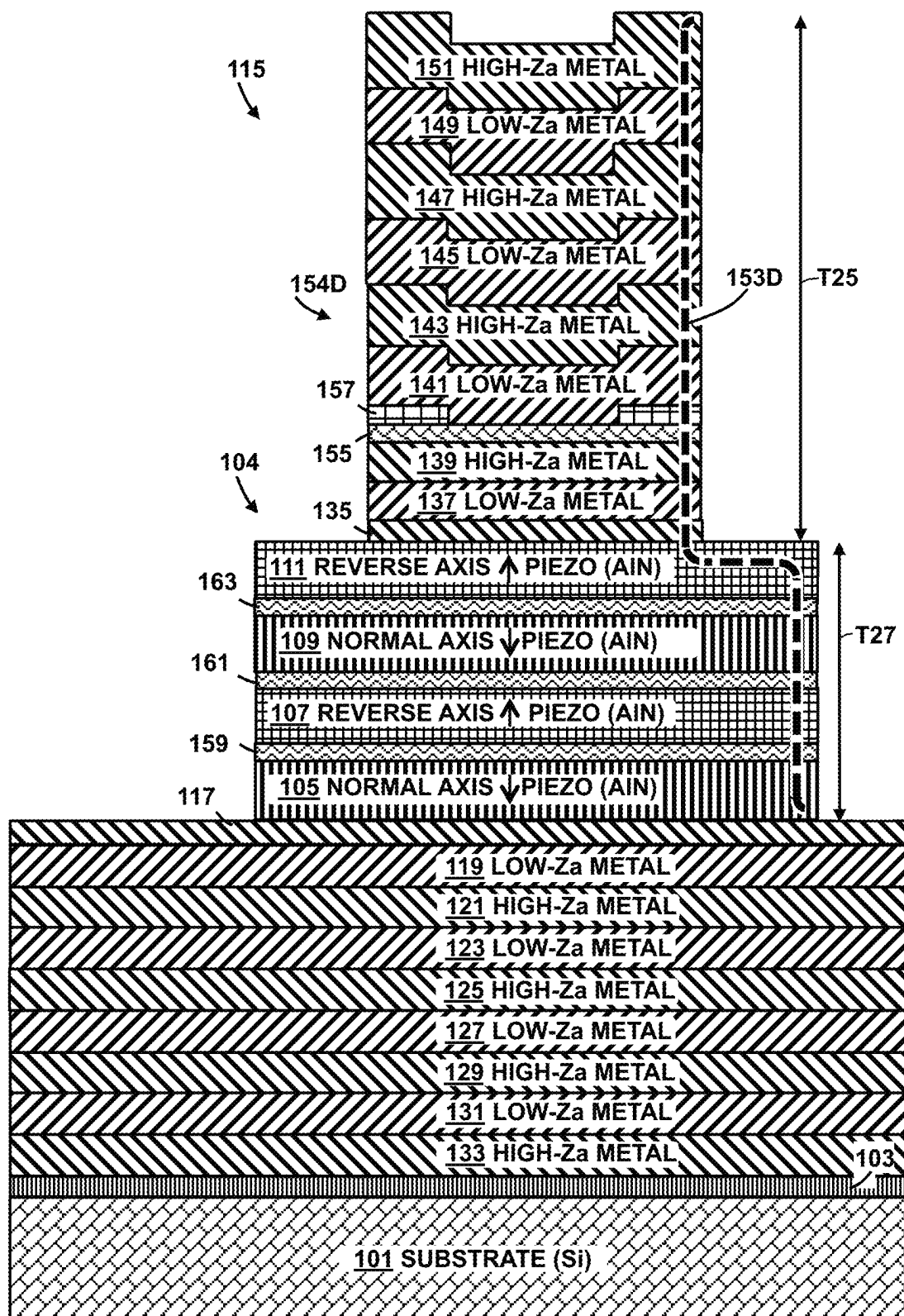
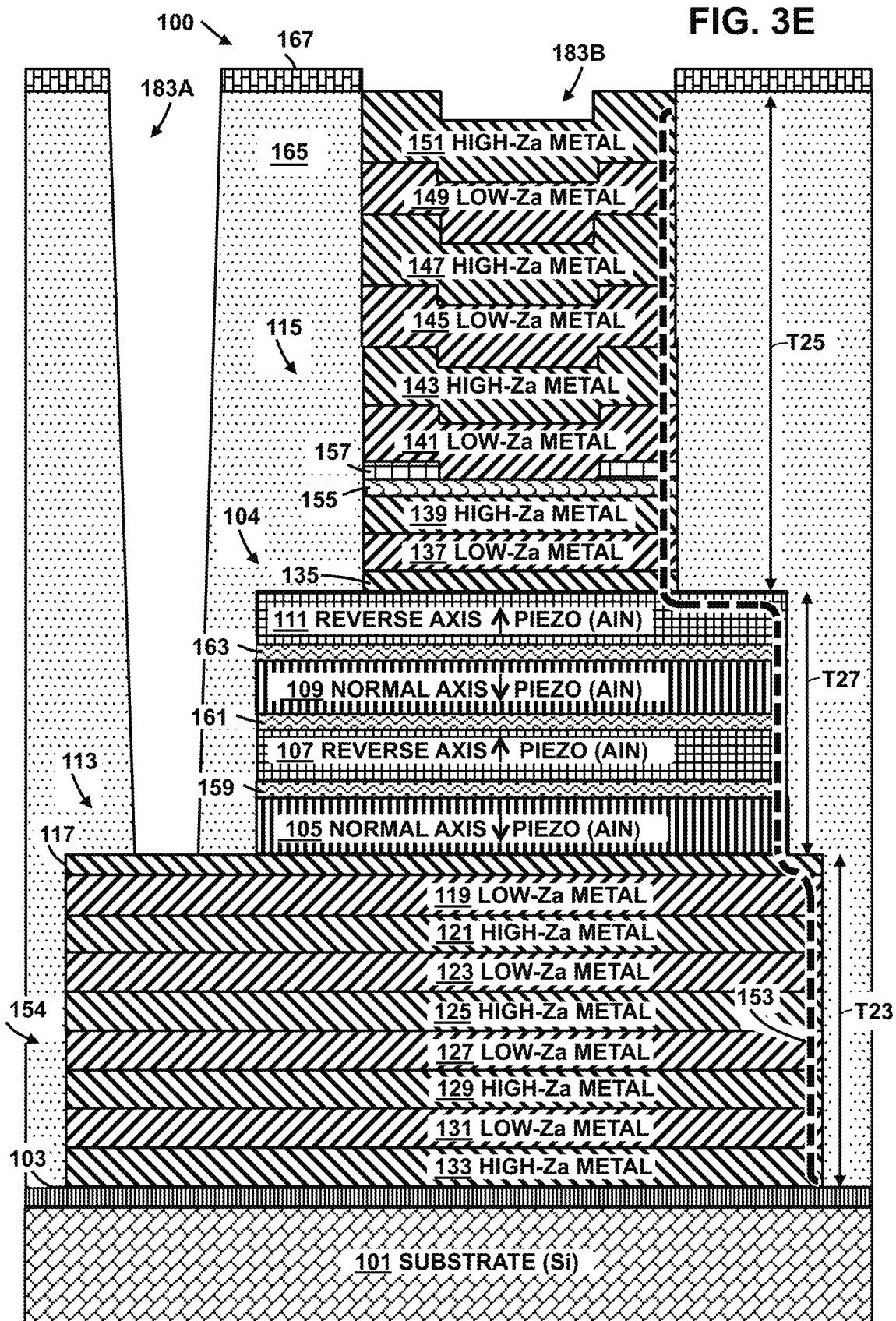


FIG. 3D





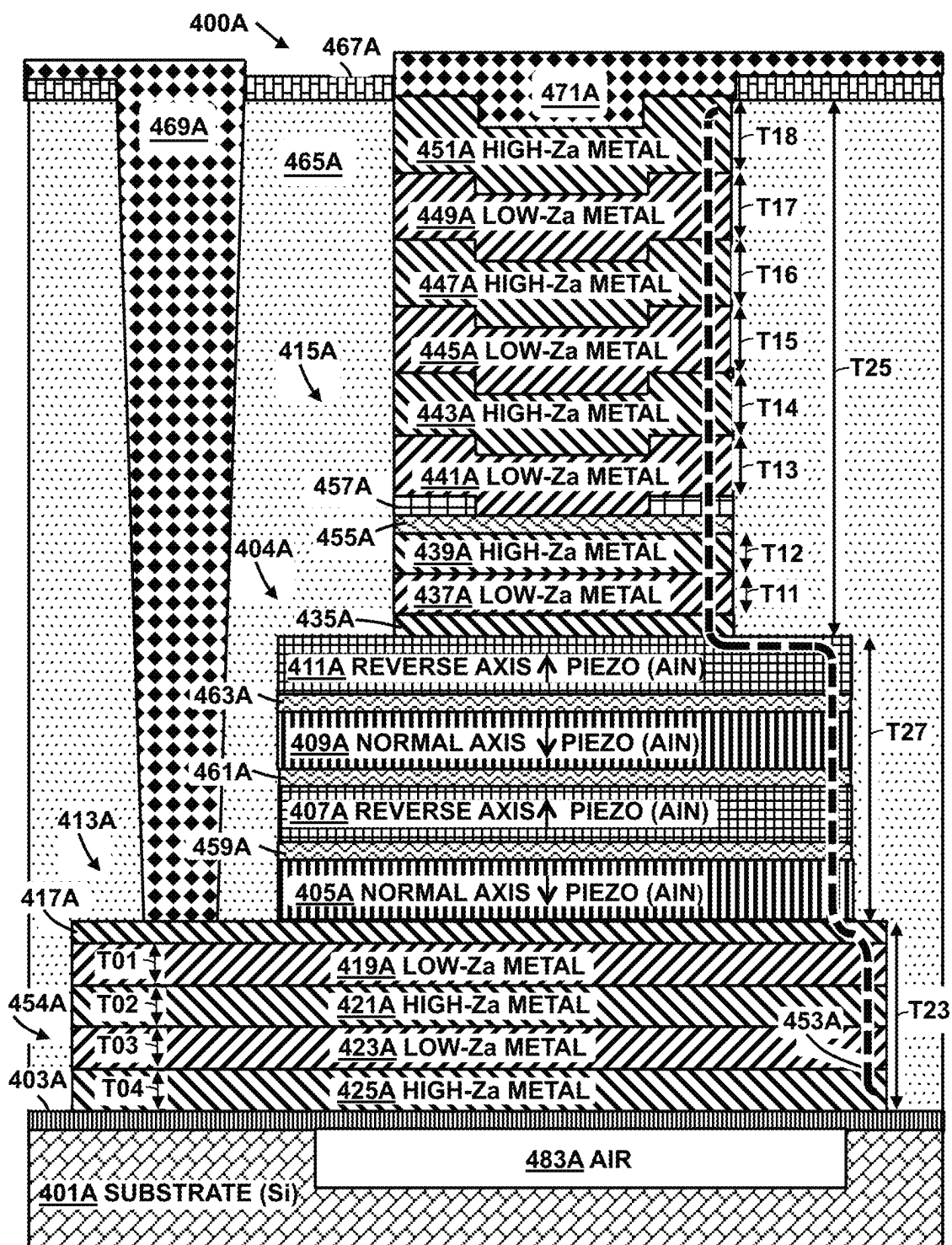


FIG. 4A

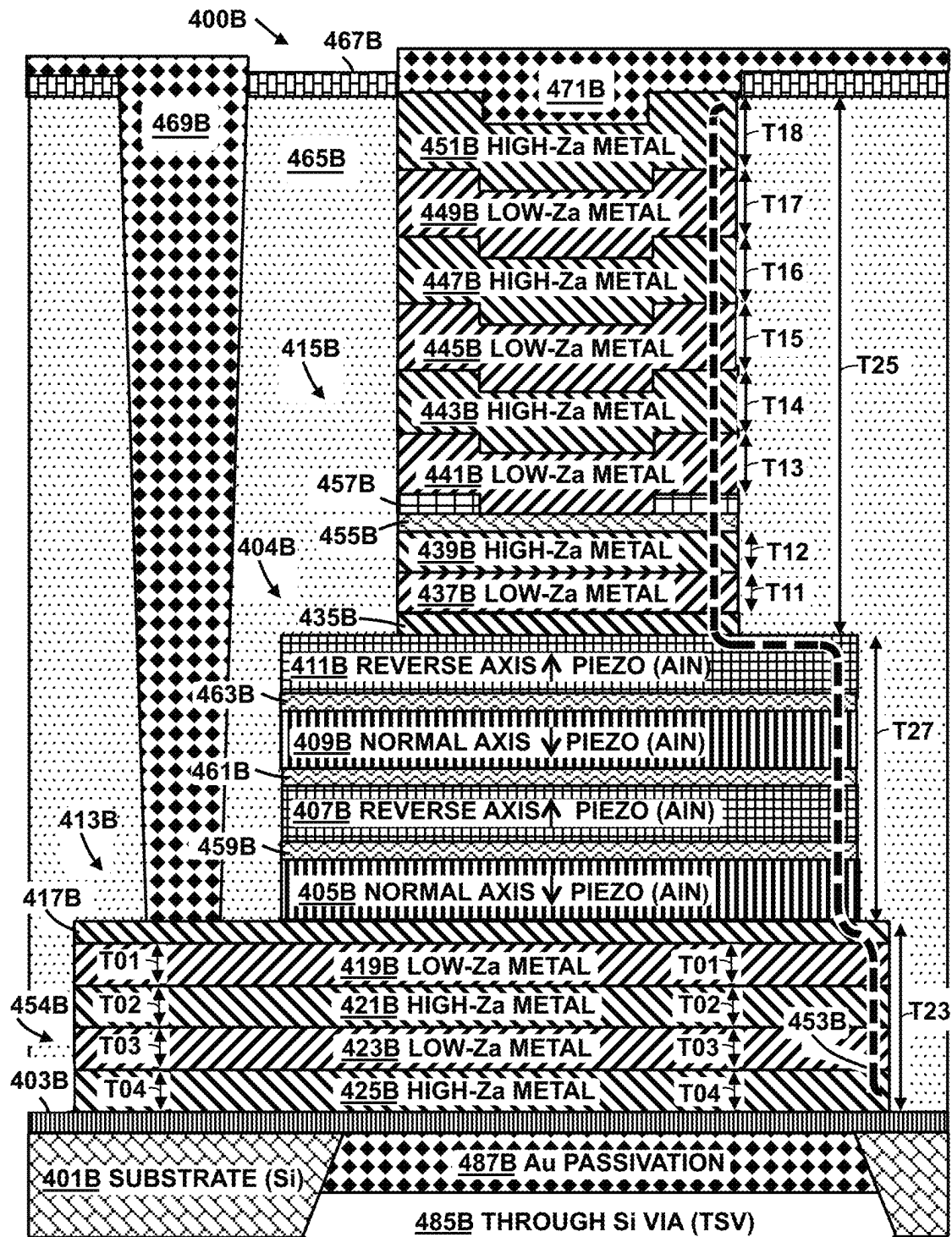


FIG. 4B

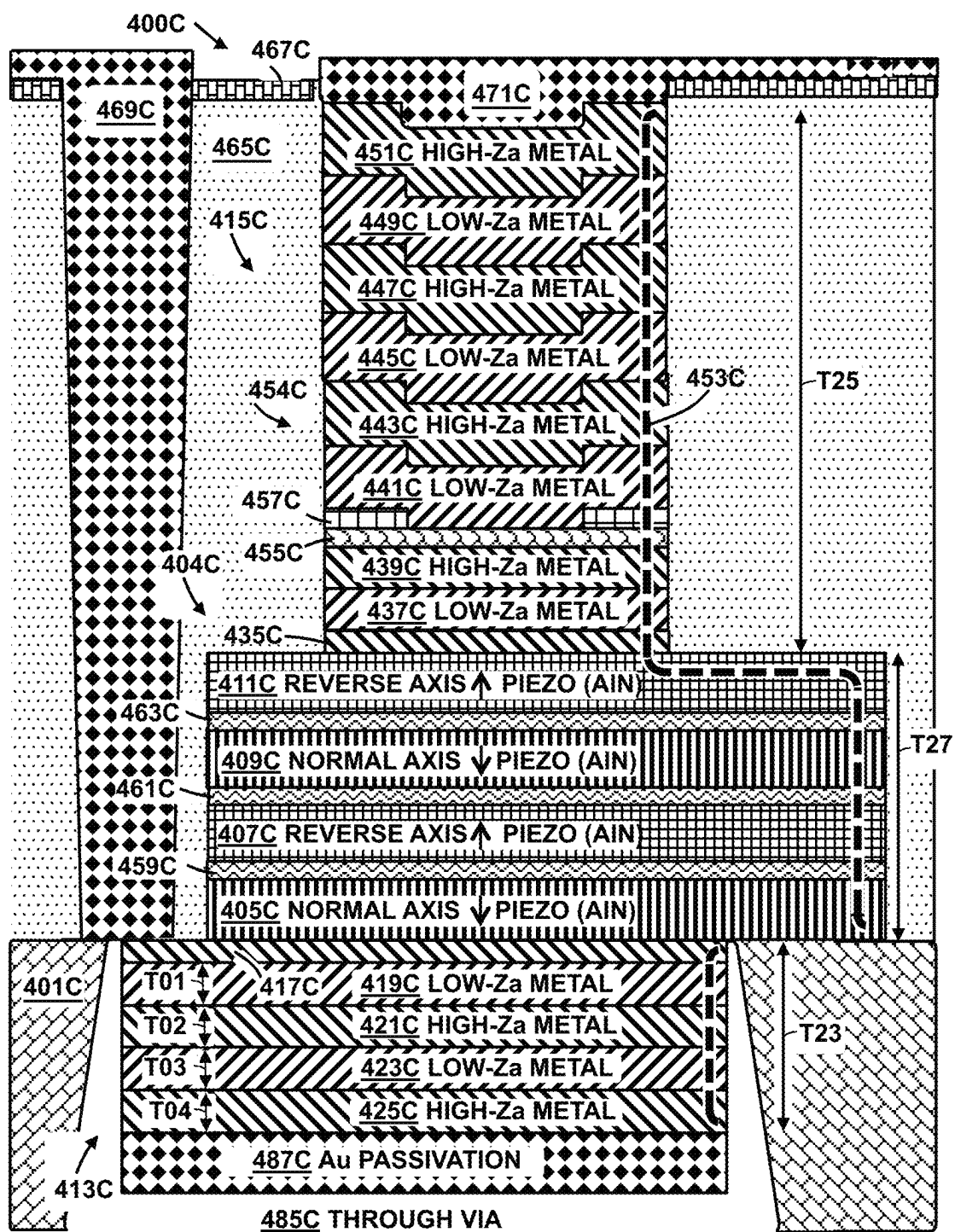


FIG. 4C

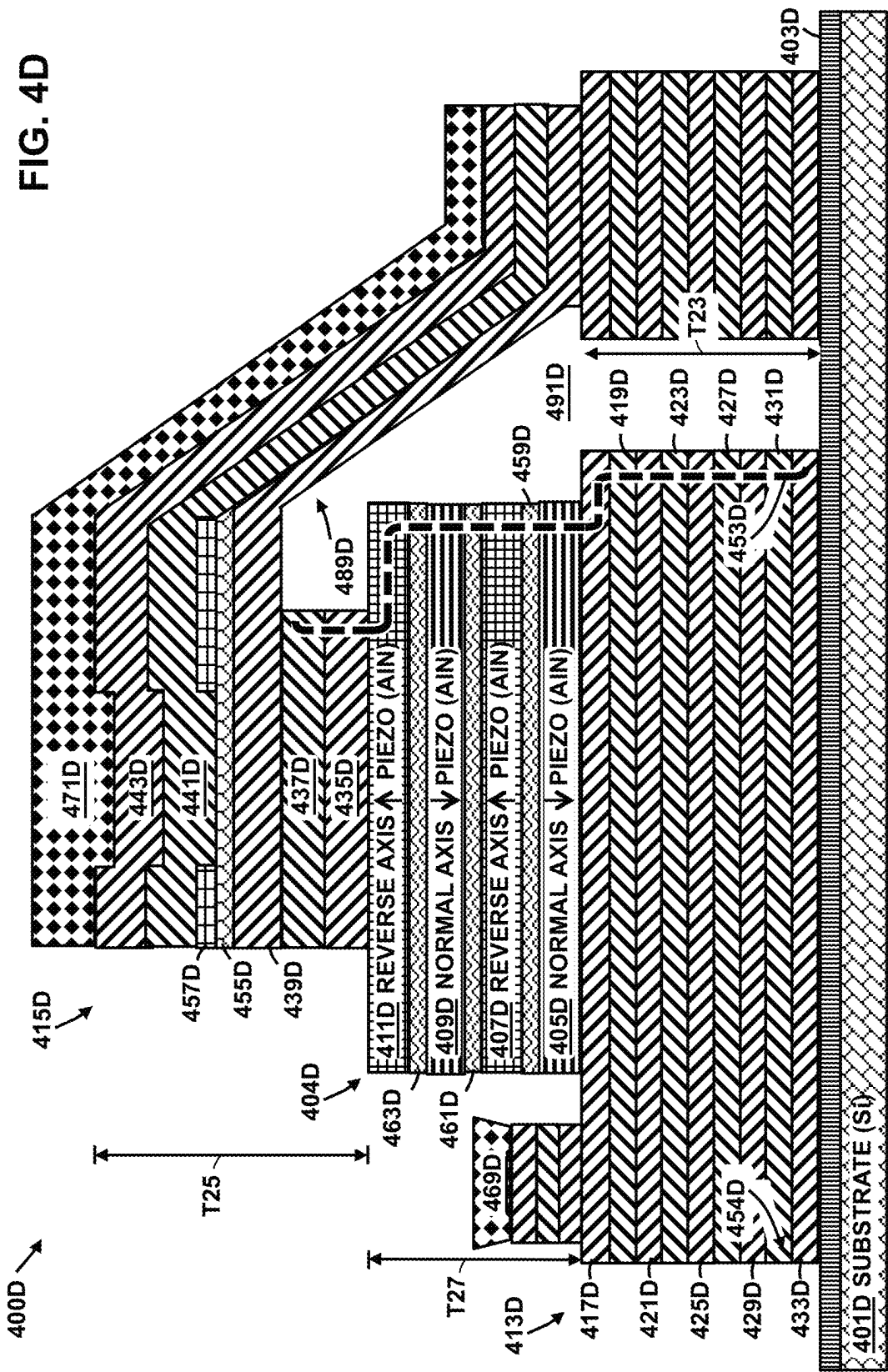
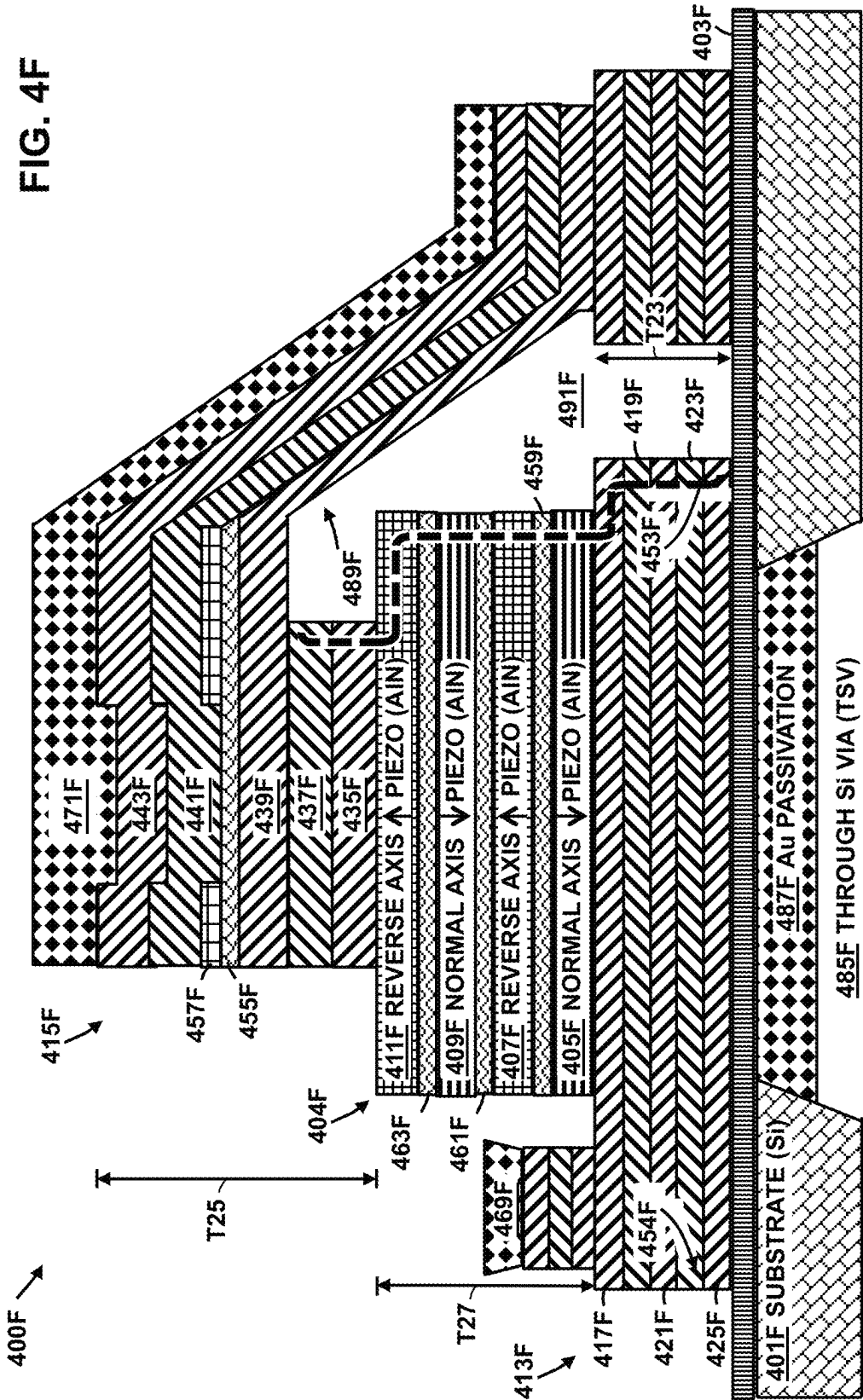


FIG. 4F



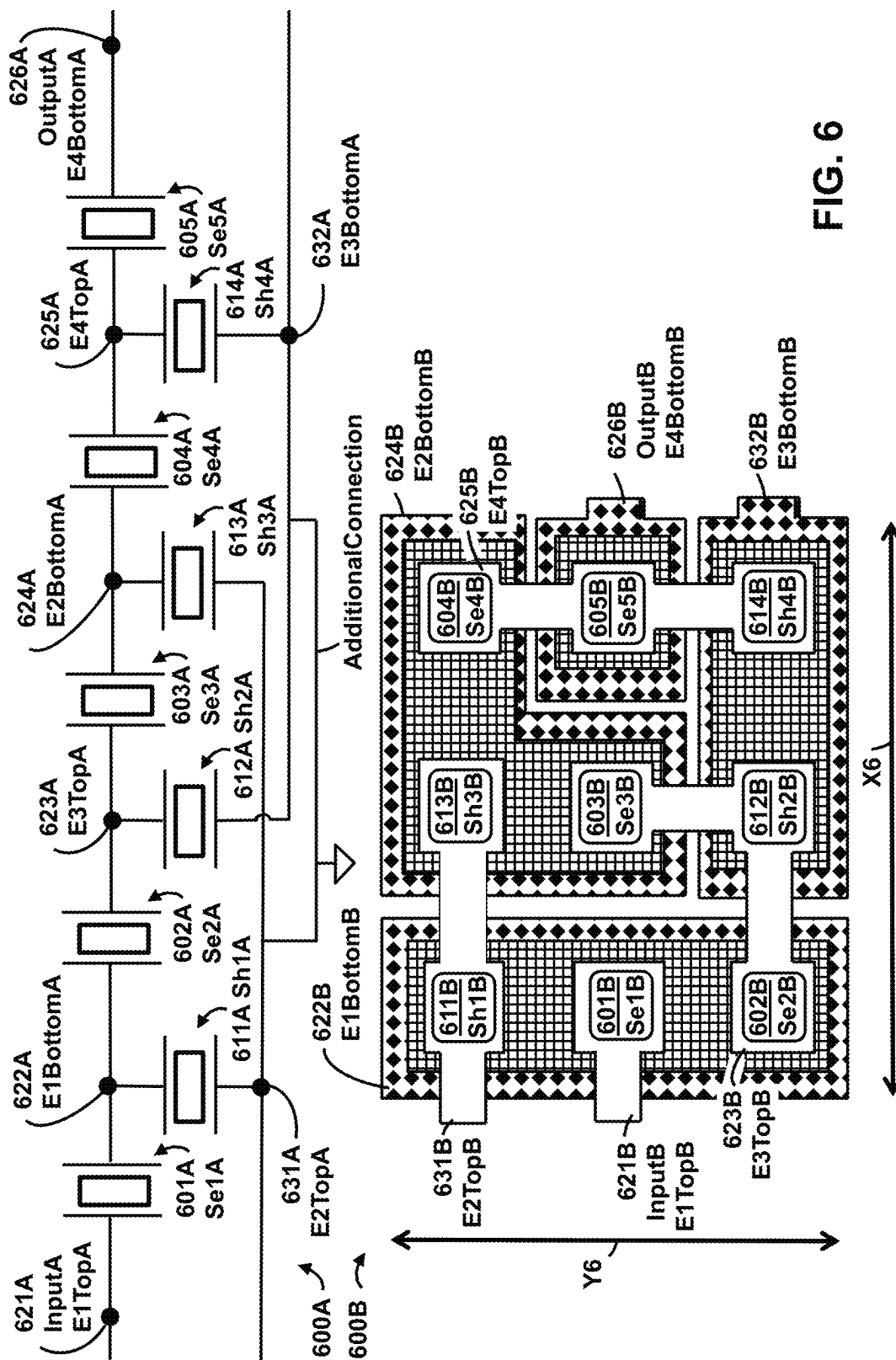


FIG. 6

FIG. 7

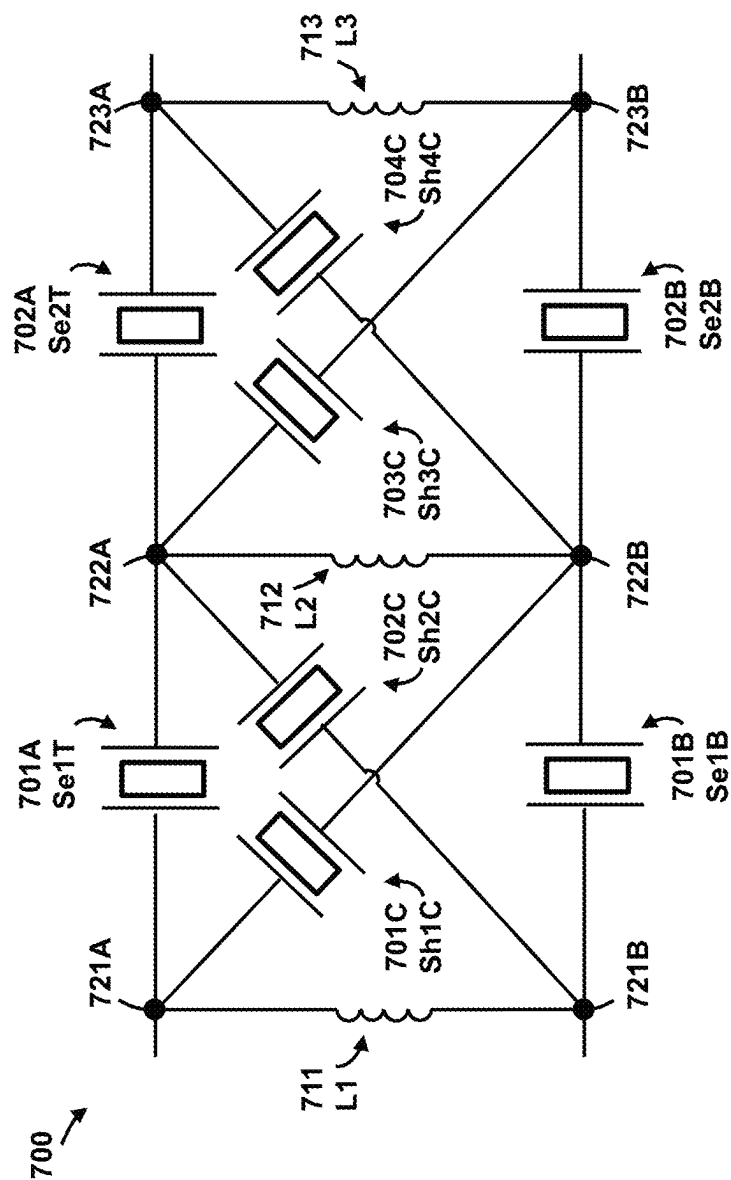


FIG. 8A

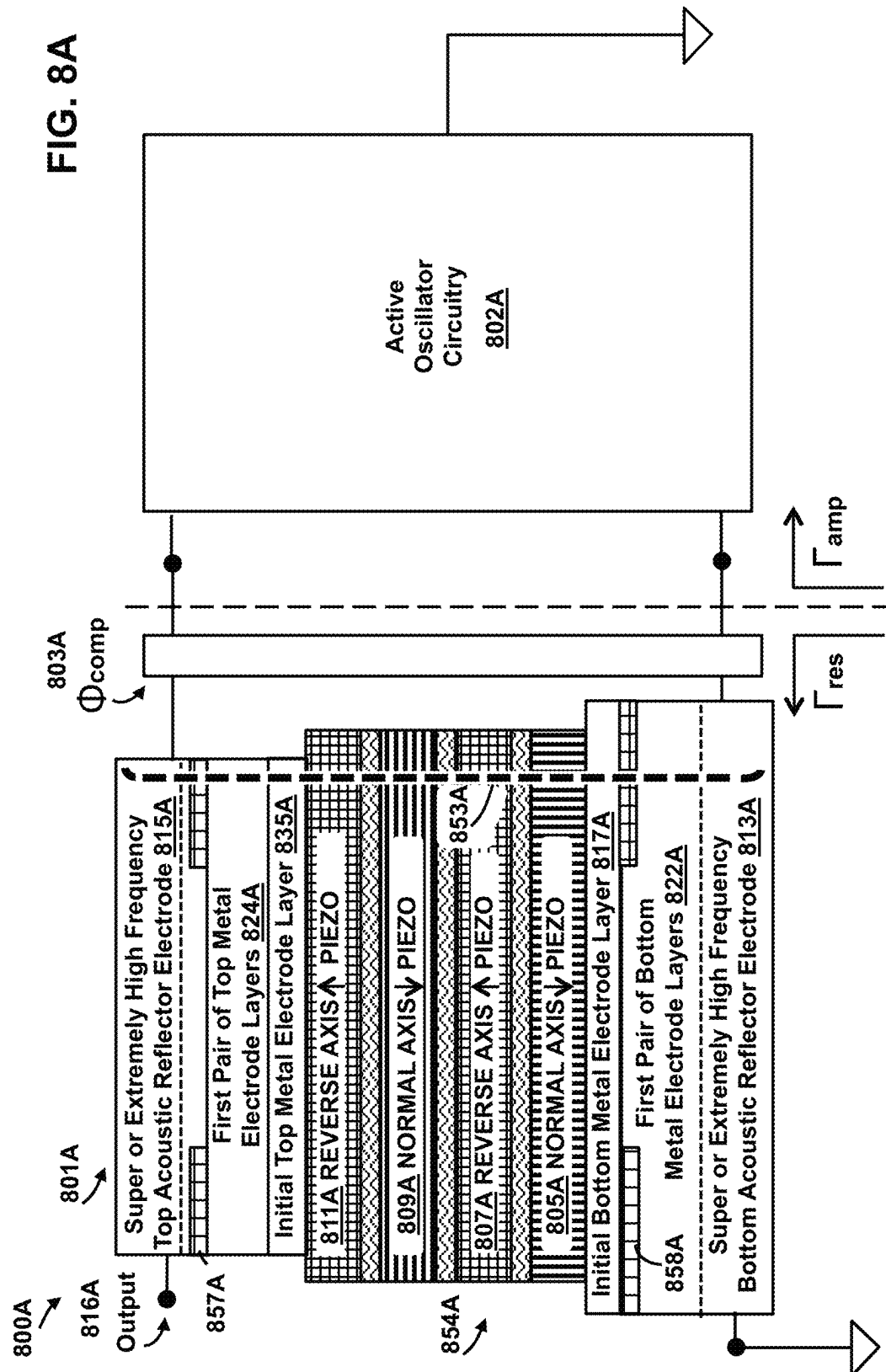
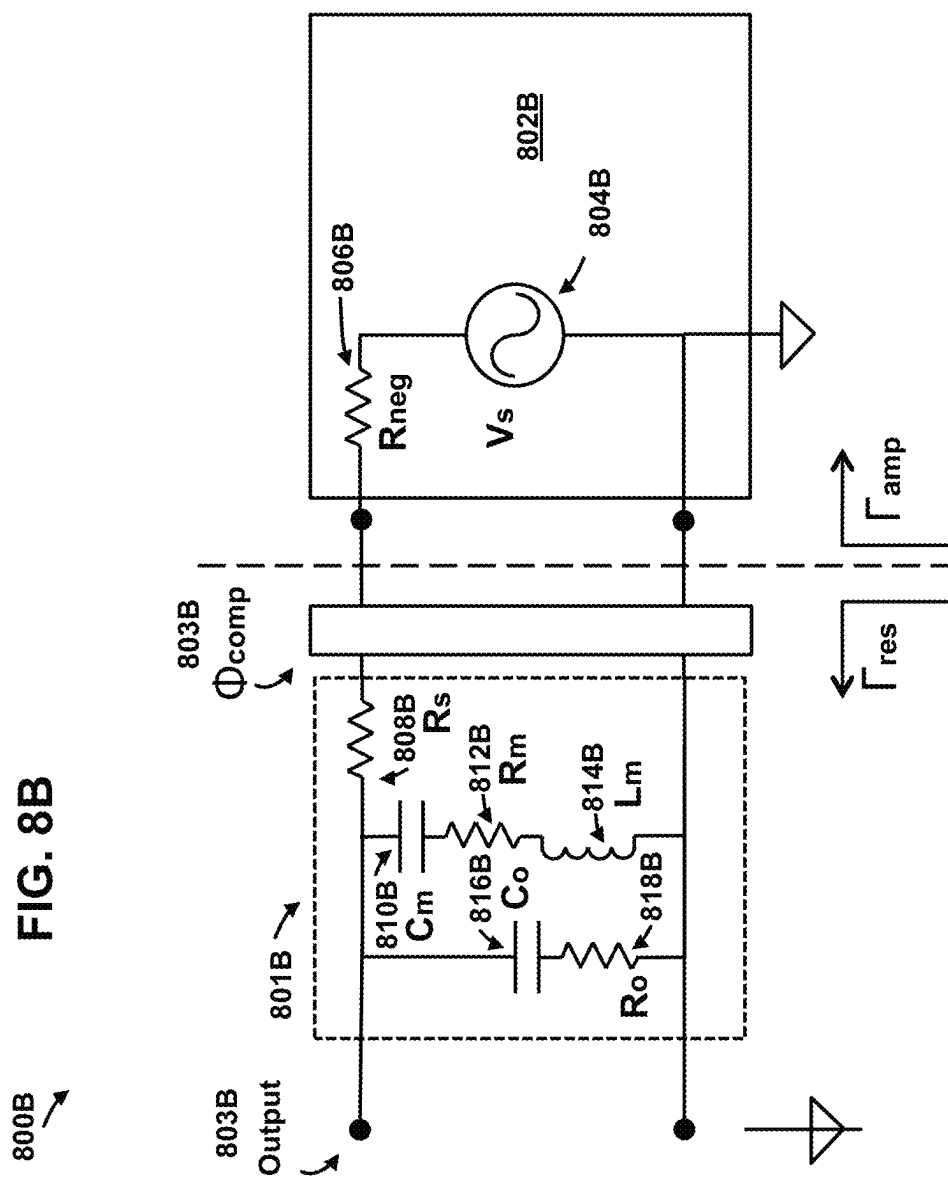
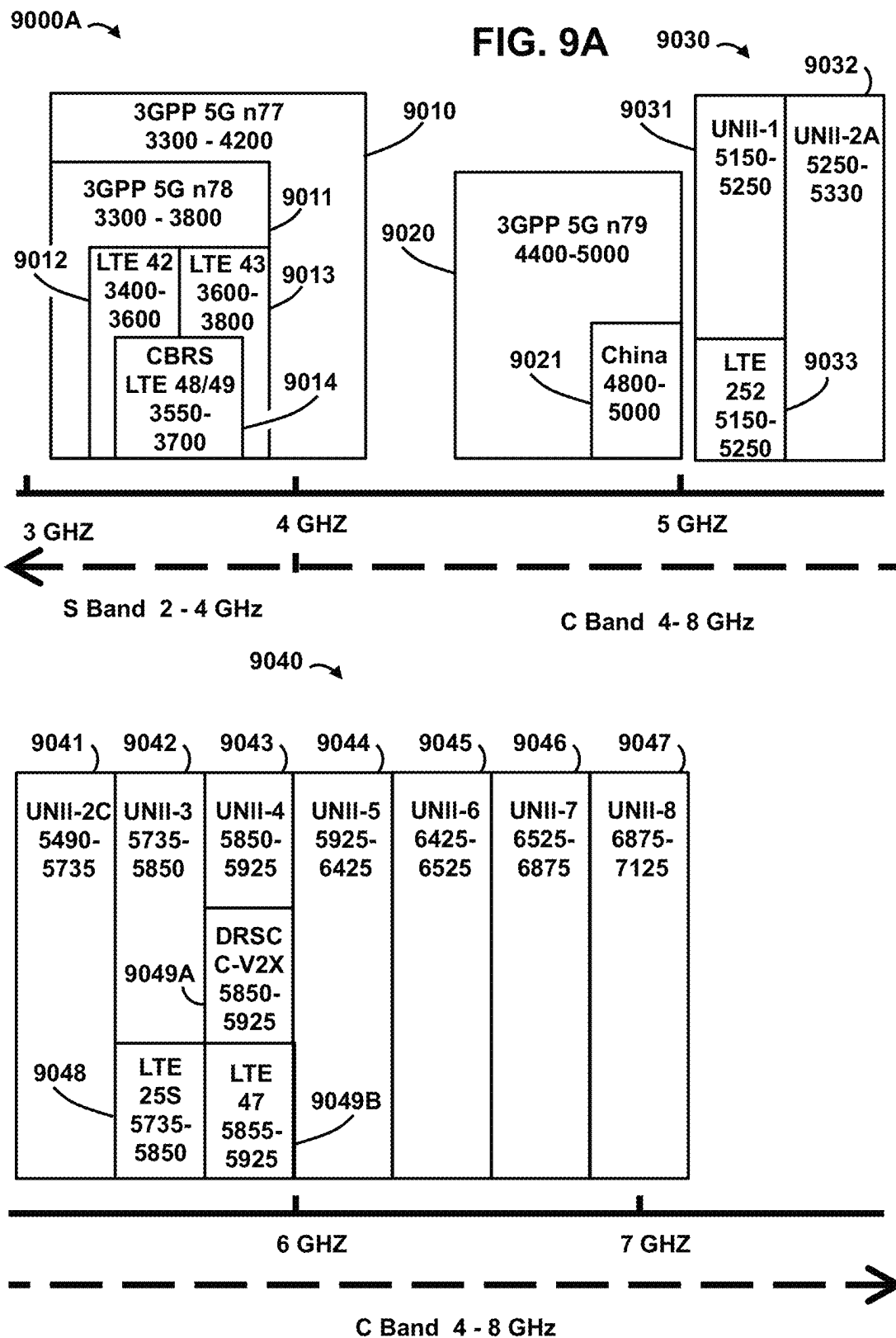
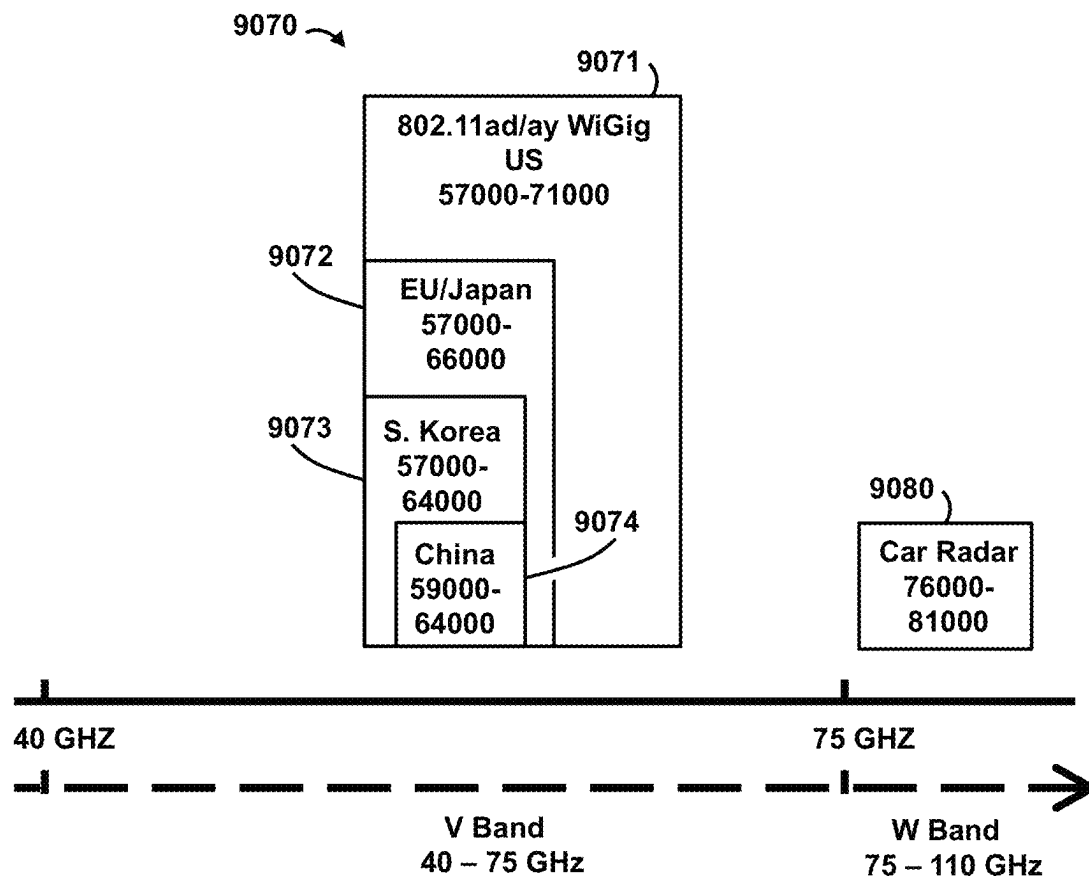
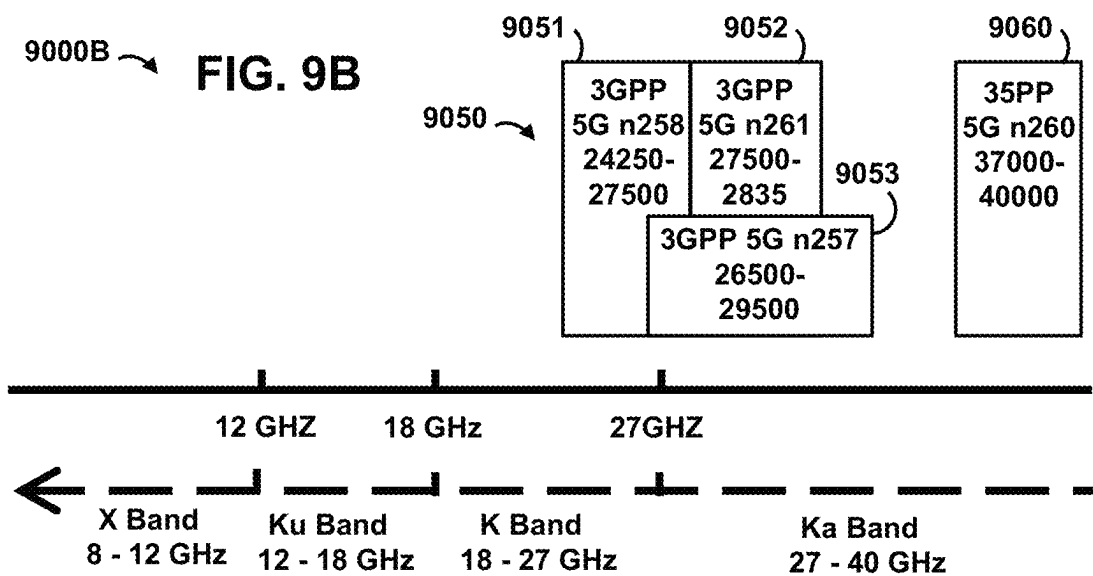


FIG. 8B







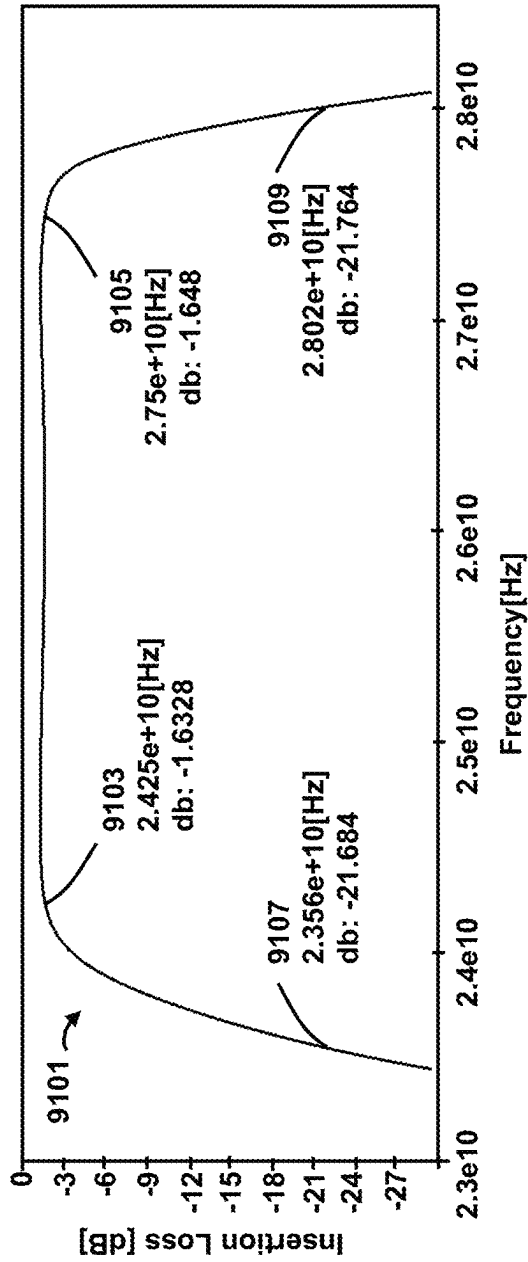


FIG. 9C

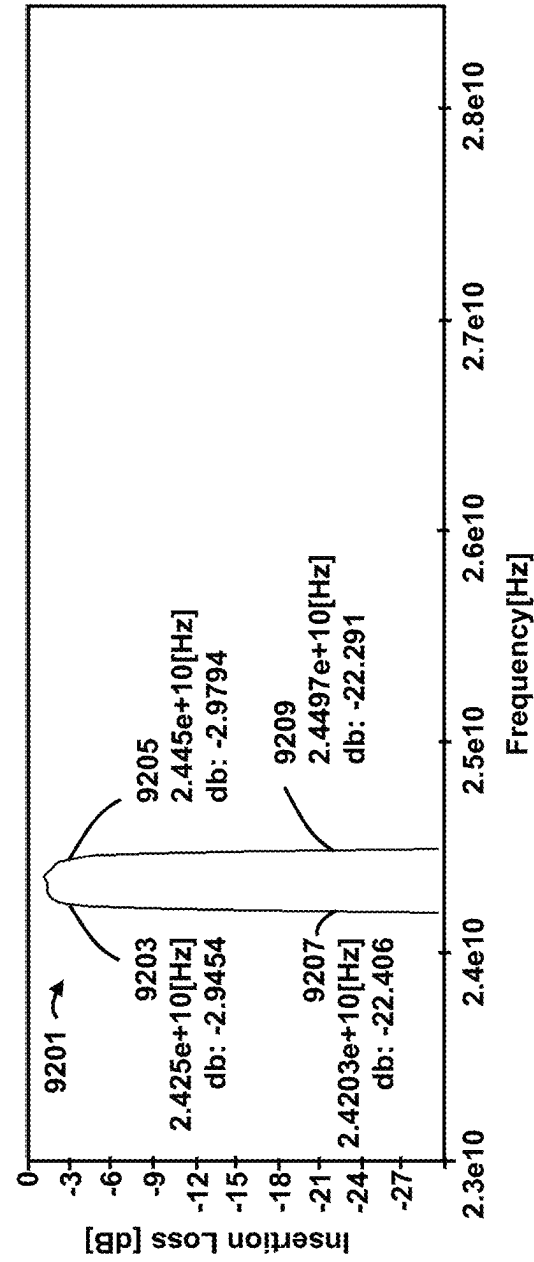


FIG. 9D

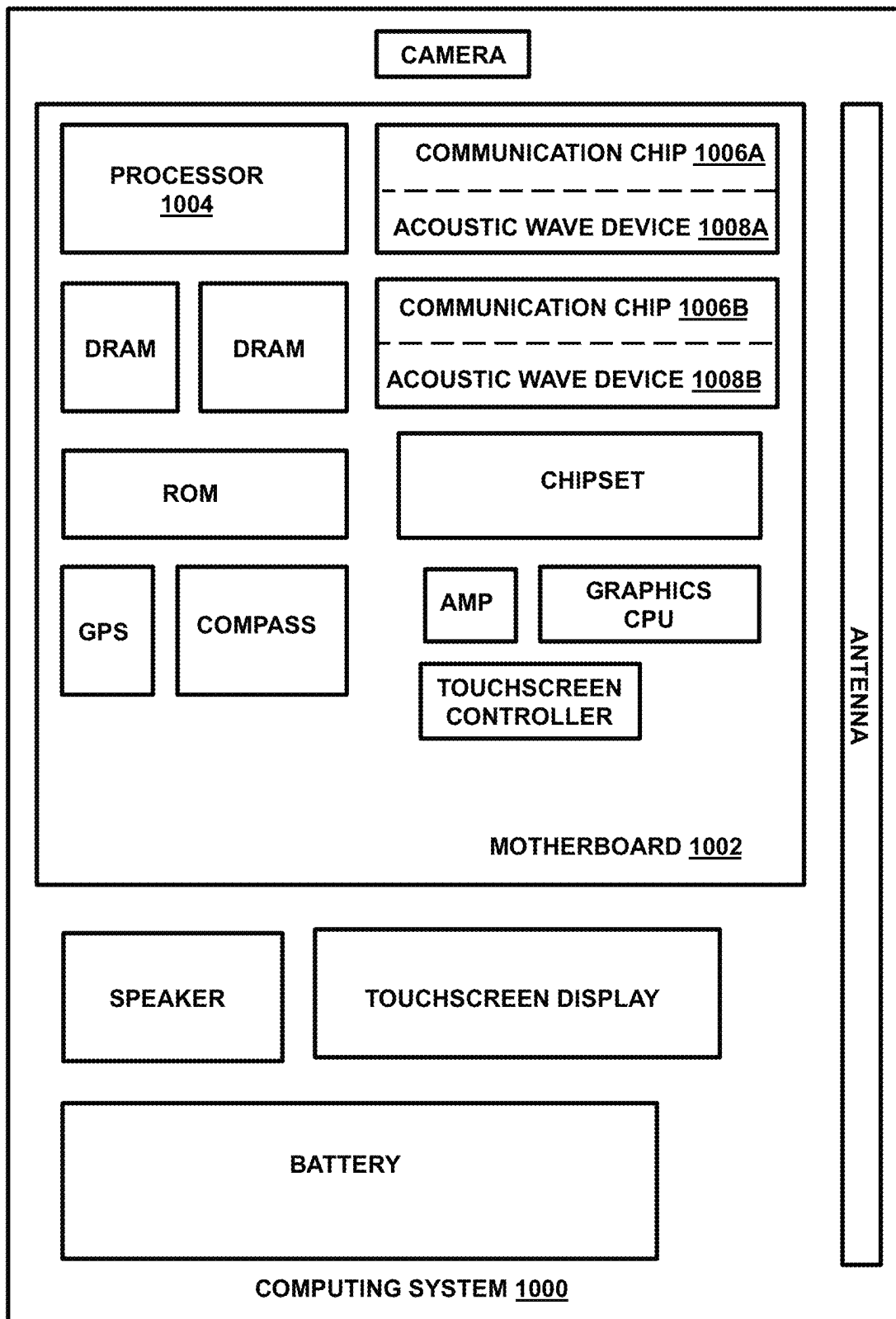


FIG. 10

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BULK ACOUSTIC WAVE RESONATOR WITH PATTERNED LAYER STRUCTURES, DEVICES AND SYSTEMS

PRIORITY CLAIM

This application arises from a continuation of U.S. patent application Ser. No. 17/564,216 filed Dec. 29, 2021, entitled “BULK ACOUSTIC WAVE (BAW) RESONATOR WITH PATTERNED LAYER STRUCTURES, DEVICES AND SYSTEMS”, which is a continuation of and claims priority to:

PCT Application No. PCTUS2020043733 filed Jul. 27, 2020, titled “BULK ACOUSTIC WAVE (BAW) RESONATOR WITH PATTERNED LAYER STRUCTURES, DEVICES AND SYSTEMS”, which claims priority to the following provisional patent applications: U.S. Provisional Patent Application Ser. No. 62/881,061, entitled “BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,074, entitled “ACOUSTIC DEVICE STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,077, entitled “DOPED BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,085, entitled “BULK ACOUSTIC WAVE (BAW) RESONATOR WITH PATTERNED LAYER STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,087, entitled “BULK ACOUSTIC WAVE (BAW) REFLECTOR AND RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,091, entitled “MASS LOADED BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019; and

U.S. Provisional Patent Application Ser. No. 62/881,094, entitled “TEMPERATURE COMPENSATING BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019.

This application is also a continuation of U.S. patent application Ser. No. 17/380,011 filed Jul. 20, 2021, entitled “STRUCTURES, ACOUSTIC WAVE RESONATORS, DEVICES AND SYSTEMS TO SENSE A TARGET VARIABLE”, which in turn is a continuation of U.S. patent application Ser. No. 16/940,172 filed Jul. 27, 2020 (issued as U.S. Pat. No. 11,101,783 on Aug. 24, 2021), entitled “STRUCTURES, ACOUSTIC WAVE RESONATORS, DEVICES AND SYSTEMS TO SENSE A TARGET VARIABLE, INCLUDING AS A NON-LIMITING EXAMPLE CORONAVIRUSES”, which in turn claims priority to the following U.S. Provisional Patent Applications:

U.S. Provisional Patent Application Ser. No. 62/881,061, entitled “BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,074, entitled “ACOUSTIC DEVICE STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

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U.S. Provisional Patent Application Ser. No. 62/881,077, entitled “DOPED BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,085, entitled “BULK ACOUSTIC WAVE (BAW) RESONATOR WITH PATTERNED LAYER STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,087, entitled “BULK ACOUSTIC WAVE (BAW) REFLECTOR AND RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019;

U.S. Provisional Patent Application Ser. No. 62/881,091, entitled “MASS LOADED BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019; and

U.S. Provisional Patent Application Ser. No. 62/881,094, entitled “TEMPERATURE COMPENSATING BULK ACOUSTIC WAVE (BAW) RESONATOR STRUCTURES, DEVICES AND SYSTEMS” and filed on Jul. 31, 2019.

Each of the patent applications identified above is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to acoustic resonators and to devices and to systems comprising acoustic resonators.

BACKGROUND

Bulk Acoustic Wave (BAW) resonators have enjoyed commercial success in filter applications. For example, 4G cellular phones that operate on fourth generation broadband cellular networks typically include a large number of BAW filters for various different frequency bands of the 4G network. In addition to BAW resonators and filters, also included in 4G phones are filters using Surface Acoustic Wave (SAW) resonators, typically for lower frequency band filters. SAW based resonators and filters are generally easier to fabricate than BAW based filters and resonators. However, performance of SAW based resonators and filters may decline if attempts are made to use them for higher 4G frequency bands. Accordingly, even though BAW based filters and resonators are relatively more difficult to fabricate than SAW based filters and resonators, they can be included in 4G cellular phones to provide better performance in higher 4G frequency bands what is provided by SAW based filters and resonators.

5G cellular phones can operate on newer, fifth generation broadband cellular networks. 5G frequencies include some frequencies that are much higher frequency than 4G frequencies. Such relatively higher 5G frequencies can transport data at relatively faster speeds than what can be provided over relatively lower 4G frequencies. However, previously known SAW and BAW based resonators and filters have encountered performance problems when attempts were made to use them at relatively higher 5G frequencies. Many learned engineering scholars have studied these problems, but have not found solutions. For example, performance problems cited for previously known SAW and BAW based resonators and filters include scaling issues and significant increases in acoustic losses at high frequencies.

From the above, it is seen that techniques for improving Bulk Acoustic Wave (BAW) resonator structures are highly

desirable, for example for operation over frequencies higher than 4G frequencies, in particular for filters, oscillators and systems that can include such devices.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a diagram that illustrates an example bulk acoustic wave resonator structure.

FIG. 1B is a simplified view of FIG. 1A that illustrates acoustic stress profile during electrical operation of the bulk acoustic wave resonator structure shown in FIG. 1A.

FIG. 1C shows a simplified top plan view of a bulk acoustic wave resonator structure corresponding to the cross sectional view of FIG. 1A, and also shows another simplified top plan view of an alternative bulk acoustic wave resonator structure.

FIG. 1D is a perspective view of an illustrative model of a crystal structure of AlN in piezoelectric material of layers in FIG. 1A having reverse axis orientation of negative polarization.

FIG. 1E is a perspective view of an illustrative model of a crystal structure of AlN in piezoelectric material of layers in FIG. 1A having normal axis orientation of positive polarization.

FIGS. 2A and 2B show a further simplified view of a bulk acoustic wave resonator similar to the bulk acoustic wave resonator structure shown in FIG. 1A along with its corresponding impedance versus frequency response during its electrical operation, as well as alternative bulk acoustic wave resonator structures with differing numbers of alternating axis piezoelectric layers, and their respective corresponding impedance versus frequency response during electrical operation, as predicted by simulation.

FIG. 2C shows additional alternative bulk acoustic wave resonator structures with additional numbers of alternating axis piezoelectric layers.

FIGS. 2D and 2E show more additional alternative bulk acoustic wave resonator structures.

FIGS. 2F, 2G, 2H and 2I show simplified diagrams of bulk acoustic wave resonator structures comprising respective pairs of acoustic reflector electrodes including one or more respective patterned layers.

FIG. 2J shows two diagrams for different patterned mass load materials and different patterned layer electrode placement shown with bulk acoustic wave resonator patterned layer sensitivity versus number of alternating axis half wavelength thickness piezoelectric layers, as predicted by simulation.

FIGS. 3A through 3E illustrate example integrated circuit structures used to form the example bulk acoustic wave resonator structure of FIG. 1A. Note that although AlN is used as an example piezoelectric layer material, the present disclosure is not intended to be so limited. For example, in some embodiments, the piezoelectric layer material may include other group III material-nitride (III-N) compounds (e.g., any combination of one or more of gallium, indium, and aluminum with nitrogen), and further, any of the foregoing may include doping, for example, of Scandium and/or Magnesium doping.

FIGS. 4A through 4G show alternative example bulk acoustic wave resonators to the example bulk acoustic wave resonator structures shown in FIG. 1A.

FIG. 5 shows a schematic of an example ladder filter using three series resonators of the bulk acoustic wave resonator structure of FIG. 1A, and two mass loaded shunt

resonators of the bulk acoustic wave resonator structure of FIG. 1A, along with a simplified view of the three series resonators.

FIG. 6 shows a schematic of an example ladder filter using five series resonators of the bulk acoustic wave resonator structure of FIG. 1A, and four mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. 1A, along with a simplified top view of the nine resonators interconnected in the example ladder filter, and lateral dimensions of the example ladder filter.

FIG. 7 shows a schematic of example inductors modifying an example lattice filter using a first pair of series resonators of the bulk acoustic wave resonator structure of FIG. 1A, a second pair of series resonators of the bulk acoustic wave resonator structure of FIG. 1A and two pairs of cross coupled mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. 1A.

FIG. 8A shows an example oscillator using the bulk acoustic wave resonator structure of FIG. 1A.

FIG. 8B shows a schematic of an example circuit implementation of the oscillator shown in FIG. 8A.

FIGS. 9A and 9B are simplified diagrams of a frequency spectrum illustrating application frequencies and application frequency bands of the example bulk acoustic wave resonators shown in FIG. 1A and FIGS. 4A through 4G, and the example filters shown in FIGS. 5 through 7, and the example oscillators shown in FIGS. 8A and 8B.

FIGS. 9C and 9D are diagrams illustrating respective simulated bandpass characteristics of insertion loss versus frequency for example filters.

FIG. 10 illustrates a computing system implemented with integrated circuit structures or devices formed using the techniques disclosed herein, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

Non-limiting embodiments will be described by way of example with reference to the accompanying figures, which are schematic and are not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment shown where illustration is not necessary to allow understanding by those of ordinary skill in the art. In the specification, as well as in the claims, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively. Further, relative terms, such as “above,” “below,” “top,” “bottom,” “upper” and “lower” are used to describe the various elements’ relationships to one another, as illustrated in the accompanying drawings. It is understood that these relative terms are intended to encompass different orientations of the device and/or elements in addition to the orientation depicted in the drawings. For example, if the device were inverted with respect to the view in the drawings, an element described as “above” another element, for example, would now be below that element. The term “compensating” is to be understood as including “substantially compensating”. The terms “oppose”, “opposes” and “opposing” are to be understood as including “substantially oppose”, “substantially opposes” and “substantially opposing” respectively. Further, as used

in the specification and appended claims, and in addition to their ordinary meanings, the terms “substantial” or “substantially” mean to within acceptable limits or degree. For example, “substantially cancelled” means that one skilled in the art would consider the cancellation to be acceptable. As used in the specification and the appended claims and in addition to its ordinary meaning, the term “approximately” or “about” means to within an acceptable limit or amount to one of ordinary skill in the art. For example, “approximately the same” means that one of ordinary skill in the art would consider the items being compared to be the same. As used in the specification and appended claims, the terms “a”, “an” and “the” include both singular and plural referents, unless the context clearly dictates otherwise. Thus, for example, “a device” includes one device and plural devices. As used herein, the International Telecommunication Union (ITU) defines Super High Frequency (SHF) as extending between three Gigahertz (3 GHz) and thirty Gigahertz (30 GHz). The ITU defines Extremely High Frequency (EHF) as extending between thirty Gigahertz (30 GHz) and three hundred Gigahertz (300 GHz).

FIG. 1A is a diagram that illustrates an example bulk acoustic wave resonator structure 100. FIGS. 4A through 4G show alternative example bulk acoustic wave resonators, 400A through 400G, to the example bulk acoustic wave resonator structure 100 shown in FIG. 1A. The foregoing are shown in simplified cross sectional views. The resonator structures are formed over a substrate 101, 401A through 401G (e.g., silicon substrate 101, 401A, 401B, 401D through 401F, e.g., silicon carbide substrate 401C). In some examples, the substrate may further comprise a seed layer 103, 403A, 403B, 403D through 403F, formed of, for example, aluminum nitride (AlN), or another suitable material (e.g., silicon dioxide (SiO₂), aluminum oxide (Al₂O₃), silicon nitride (Si₃N₄), amorphous silicon (a-Si), silicon carbide (SiC)), having an example thickness in a range from approximately 100 Å to approximately 1 μm on the silicon substrate.

The example resonators 100, 400A through 400G, include a respective stack 104, 404A through 404G, of an example four layers of piezoelectric material, for example, four layers of Aluminum Nitride (AlN) having a wurtzite structure. For example, FIG. 1A and FIGS. 4A through 4G show a bottom piezoelectric layer 105, 405A through 405G, a first middle piezoelectric layer 107, 407A through 407G, a second middle piezoelectric layer 109, 409A through 409G, and a top piezoelectric layer 111, 411A through 411G. A mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may comprise the respective stack 104, 404A through 404G, of the example four layers of piezoelectric material. The mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may comprise bottom piezoelectric layer 105, 405A through 405G. The mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may comprise first middle piezoelectric layer 107, 407A through 407G. The mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may comprise second middle piezoelectric layer 109, 409A through 409G. The mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may comprise top piezoelectric layer 111, 411A through 411G.

The four layers of piezoelectric material in the respective stack 104, 404A through 404G of FIG. 1A and FIGS. 4A through 4G may have an alternating axis arrangement in the respective stack 104, 404A through 404G. For example the bottom piezoelectric layer 105, 405A through 405G may

have a normal axis orientation, which is depicted in the figures using a downward directed arrow. Next in the alternating axis arrangement of the respective stack 104, 404A through 404G, the first middle piezoelectric layer 107, 407A through 407G may have a reverse axis orientation, which is depicted in the figures using an upward directed arrow. Next in the alternating axis arrangement of the respective stack 104, 404A through 404G, the second middle piezoelectric layer 109, 409A through 409G may have the normal axis orientation, which is depicted in the figures using the downward directed arrow. Next in the alternating axis arrangement of the respective stack 104, 404A through 404G, the top piezoelectric layer 111, 411A through 411G may have the reverse axis orientation, which is depicted in the figures using the upward directed arrow.

For example, polycrystalline thin film AlN may be grown in a crystallographic c-axis negative polarization, or normal axis orientation perpendicular relative to the substrate surface using reactive magnetron sputtering of an Aluminum target in a nitrogen atmosphere. However, as will be discussed in greater detail subsequently herein, changing sputtering conditions, for example by adding oxygen, may reverse the axis to a crystallographic c-axis positive polarization, or reverse axis, orientation perpendicular relative to the substrate surface.

In the example resonators 100, 400A through 400G, of FIG. 1A and FIGS. 4A through 4G, the bottom piezoelectric layer 105, 405A through 405G, may have a piezoelectrically excitable resonance mode (e.g., main resonance mode) at a resonant frequency (e.g., main resonant frequency) of the example resonators. Similarly, the first middle piezoelectric layer 107, 407A through 407G, may have its piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. Similarly, the second middle piezoelectric layer 109, 409A through 409G, may have its piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. Similarly, the top piezoelectric layer 111, 411A through 411G, may have its piezoelectrically excitable main resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. Accordingly, the top piezoelectric layer 111, 411A through 411G, may have its piezoelectrically excitable main resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) with the bottom piezoelectric layer 105, 405A through 405G, the first middle piezoelectric layer 107, 407A through 407G, and the second middle piezoelectric layer 109, 409A through 409G.

The bottom piezoelectric layer 105, 405A through 405G, may be acoustically coupled with the first middle piezoelectric layer 107, 407A through 407G, in the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators 100, 400A through 400G. The normal axis of bottom piezoelectric layer 105, 405A through 405G, in opposing the reverse axis of the first middle piezoelectric layer 107, 407A through 407G, may cooperate for the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. The first middle piezoelectric layer 107, 407A through 407G, may be sandwiched between the bottom piezoelectric layer 105, 405A through 405G, and the second middle piezoelectric layer 109, 409A through 409G, for example, in the alternating axis arrangement in the respective stack 104, 404A through

404G. For example, the reverse axis of the first middle piezoelectric layer 107, 407A through 407G, may oppose the normal axis of the bottom piezoelectric layer 105, 405A through 405G, and the normal axis of the second middle piezoelectric layer 109, 409A-409G. In opposing the normal axis of the bottom piezoelectric layer 105, 405A through 405G, and the normal axis of the second middle piezoelectric layer 109, 409A through 409G, the reverse axis of the first middle piezoelectric layer 107, 407A through 407G, may cooperate for the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators.

The second middle piezoelectric layer 109, 409A through 409G, may be sandwiched between the first middle piezoelectric layer 107, 407A through 407G, and the top piezoelectric layer 111, 411A through 411G, for example, in the alternating axis arrangement in the respective stack 104, 404A through 404G. For example, the normal axis of the second middle piezoelectric layer 109, 409A through 409G, may oppose the reverse axis of the first middle piezoelectric layer 107, 407A through 407G, and the reverse axis of the top piezoelectric layer 111, 411A through 411G. In opposing the reverse axis of the first middle piezoelectric layer 107, 407A through 407G, and the reverse axis of the top piezoelectric layer 111, 411A through 411G, the normal axis of the second middle piezoelectric layer 109, 409A through 409G, may cooperate for the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. Similarly, the alternating axis arrangement of the bottom piezoelectric layer 105, 405A through 405G, and the first middle piezoelectric layer 107, 407A through 407G, and the second middle piezoelectric layer 109, 409A through 409G, and the top piezoelectric layer 111, 411A through 411G, in the respective stack 104, 404A through 404G may cooperate for the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the example resonators. Despite differing in their alternating axis arrangement in the respective stack 104, 404A through 404G, the bottom piezoelectric layer 105, 405A through 405G and the first middle piezoelectric layer 107, 407A through 407G, and the second middle piezoelectric layer 109, 409A through 409G, and the top piezoelectric layer 111, 411A through 411G, may all be made of the same piezoelectric material, e.g., Aluminum Nitride (AlN).

Respective layers of piezoelectric material in the stack 104, 404A through 404G, of FIG. 1A and FIGS. 4A through 4G may have respective layer thicknesses of about one half wavelength (e.g., one half acoustic wavelength) of the main resonant frequency of the example resonators. For example, respective layers of piezoelectric material in the stack 104, 404A through 404G, of FIG. 1A and FIGS. 4A through 4G may have respective layer thicknesses so that (e.g., selected so that) the respective bulk acoustic wave resonators 100, 400A through 400G may have respective resonant frequencies that are in a Super High Frequency (SHF) band or an Extremely High Frequency (EHF) band (e.g., respective resonant frequencies that are in a Super High Frequency (SHF) band, e.g., respective resonant frequencies that are in an Extremely High Frequency (EHF) band. For example, respective layers of piezoelectric material in the stack 104, 404A through 404G, of FIG. 1A and FIGS. 4A through 4G may have respective layer thicknesses so that (e.g., selected so that) the respective bulk acoustic wave resonators 100, 400A through 400G may have respective resonant frequencies that are in a millimeter wave band. For example, for a

twenty-four gigahertz (e.g., 24 GHz) main resonant frequency of the example resonators, the bottom piezoelectric layer 105, 405A through 405G, may have a layer thickness corresponding to about one half of a wavelength (e.g., one half of an acoustic wavelength) of the main resonant frequency, and may be about two thousand Angstroms (2000 Å). Similarly, the first middle piezoelectric layer 107, 407A through 407G, may have a layer thickness corresponding to the one half of the wavelength (e.g., one half of the acoustic wavelength) of the main resonant frequency; the second middle piezoelectric layer 109, 409A through 409G, may have a layer thickness corresponding to the one half of the wavelength (e.g., one half of the acoustic wavelength) of the main resonant frequency; and the top piezoelectric layer 111, 411A through 411G, may have a layer thickness corresponding to the one half of the wavelength (e.g., one half of the acoustic wavelength) of the main resonant frequency. Piezoelectric layer thickness may be scaled up or down to determine main resonant frequency.

The example resonators 100, 400A through 400G, of FIG. 1A and FIGS. 4A through 4G may comprise: a bottom acoustic reflector 113, 413A through 413G, including an acoustically reflective bottom electrode stack of a plurality of bottom metal electrode layers; and a top acoustic reflector 115, 415A through 415G, including an acoustically reflective bottom electrode stack of a plurality of top metal electrode layers. Accordingly, the bottom acoustic reflector 113, 413A through 413G, may be a bottom multilayer acoustic reflector, and the top acoustic reflector 115, 415A through 415G, may be a top multilayer acoustic reflector. The piezoelectric layer stack 104, 404A through 404G, may be sandwiched between the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G, and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G. For example, top acoustic reflector electrode 115, 415A through 415G and bottom acoustic reflector electrode 113, 413A through 413G may abut opposite sides of a resonant volume 104, 404A through 404G (e.g., piezoelectric layer stack 104, 404A through 404G) free of any interposing electrode. The piezoelectric layer stack 104, 404A through 404G, may be electrically and acoustically coupled with the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency). For example, such excitation may be done by using the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G to apply an oscillating electric field having a frequency corresponding to the resonant frequency (e.g., main resonant frequency) of the piezoelectric layer stack 104, 404A through 404G, and of the example resonators 100, 400A through 400G.

For example, the bottom piezoelectric layer 105, 405A through 405G, may be electrically and acoustically coupled with the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer 105, 405A through 405G. Further, the bottom piezoelectric layer 105, 405A

through 405G and the first middle piezoelectric layer 107, 407A through 407G, may be electrically and acoustically coupled with the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G, and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer 105, 405A through 405G, acoustically coupled with the first middle piezoelectric layer 107, 407A through 407G. Additionally, the first middle piezoelectric layer 107, 407A-407G, may be sandwiched between the bottom piezoelectric layer 105, 405A through 405G and the second middle piezoelectric layer 109, 409A through 409G, and may be electrically and acoustically coupled with the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G, and the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the first middle piezoelectric layer 107, 407A through 407G, sandwiched between the bottom piezoelectric layer 105, 405A through 405G, and the second middle piezoelectric layer 109, 409A through 409G.

The acoustically reflective bottom electrode stack of the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G, may have an alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer. For example, an initial bottom metal electrode layer 117, 417A through 417G, may comprise a relatively high acoustic impedance metal, for example, Tungsten having an acoustic impedance of about 100 MegaRayls, or for example, Molybdenum having an acoustic impedance of about 65 MegaRayls. The acoustically reflective bottom electrode stack of the plurality of bottom metal electrode layers of the bottom acoustic reflector 113, 413A through 413G may approximate a metal distributed Bragg acoustic reflector. The plurality of metal bottom electrode layers of the bottom acoustic reflector may be electrically coupled (e.g., electrically interconnected) with one another. The acoustically reflective bottom electrode stack of the plurality of bottom metal electrode layers may operate together as a multilayer (e.g., bilayer, e.g., multiple layer) bottom electrode for the bottom acoustic reflector 113, 413A through 413G.

Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective bottom electrode stack, may be a first pair of bottom metal electrode layers 119, 419A through 419G and 121, 421A through 421G. A first member 119, 419A through 419G, of the first pair of bottom metal electrode layers may comprise a relatively low acoustic impedance metal, for example, Titanium having an acoustic impedance of about 27 MegaRayls, or for example, Aluminum having an acoustic impedance of about 18 MegaRayls. A second member 121, 421A through 421G, of the first pair of bottom metal electrode layers may comprise the relatively high acoustic impedance metal, for example, Tungsten or Molybdenum. Accordingly, the first pair of bottom metal electrode layers 119, 419A through 419G, and 121, 421A through 421G, of the bottom acoustic reflector 113, 413A through 413G, may be different metals, and may have respective acoustic impedances that are different from one another so as to provide a reflective acoustic impedance mismatch at the resonant frequency (e.g., main resonant frequency). Similarly, the initial bottom metal electrode

layer 117, 417A through 417G, and the first member of the first pair of bottom metal electrode layers 119, 419A through 419G, of the bottom acoustic reflector 113, 413A through 413G, may be different metals, and may have respective acoustic impedances that are different from one another so as to provide a reflective acoustic impedance mismatch at the resonant frequency (e.g., main resonant frequency).

Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective bottom electrode stack, a second pair of bottom metal electrode layers 123, 423A through 423G, and 125, 425A through 425G, may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal. Accordingly, the initial bottom metal electrode layer 117, 417A through 417G, and members of the first and second pairs of bottom metal electrode layers 119, 419A through 419G, 121, 421A through 421G, 123, 423A through 423G, 125, 425A through 425G, may have respective acoustic impedances in the alternating arrangement to provide a corresponding plurality of reflective acoustic impedance mismatches.

Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective bottom electrode stack, a third pair of bottom metal electrode layers 127, 427D, 129, 429D may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal. Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective bottom electrode stack, a fourth pair of bottom metal electrode layers 131, 431D and 133, 433D may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal.

Respective thicknesses of the bottom metal electrode layers may be related to wavelength (e.g., acoustic wavelength) for the main resonant frequency of the example bulk acoustic wave resonators, 100, 400A through 400G. Further, various embodiments for resonators having relatively higher resonant frequency (higher main resonant frequency) may have relatively thinner bottom metal electrode thicknesses, e.g., scaled thinner with relatively higher resonant frequency (e.g., higher main resonant frequency). Similarly, various alternative embodiments for resonators having relatively lower resonant frequency (e.g., lower main resonant frequency) may have relatively thicker bottom metal electrode layer thicknesses, e.g., scaled thicker with relatively lower resonant frequency (e.g., lower main resonant frequency). For example, a layer thickness of the initial bottom metal electrode layer 117, 417A through 417G, may be about one eighth of a wavelength (e.g., one eighth of an acoustic wavelength) at the main resonant frequency of the example resonator. For example, if molybdenum is used as the high acoustic impedance metal and the main resonant frequency of the resonator is twenty-four gigahertz (e.g., 24 GHz), then using the one eighth of the wavelength (e.g., one eighth of the acoustic wavelength) provides the layer thickness of the initial bottom metal electrode layer 117, 417A through 417G, as about three hundred and thirty Angstroms (330 Å). In the foregoing example, the one eighth of the wavelength (e.g., the one eighth of the acoustic wavelength) at the main resonant frequency was used for determining the layer thickness of the initial bottom metal electrode layer 117, 417A-417G, but it should be understood that this layer thickness may be varied to be thicker or thinner in various other alternative example embodiments.

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Respective layer thicknesses, T01 through T08, shown in FIG. 1A for members of the pairs of bottom metal electrode layers may be about an odd multiple (e.g., 1×, 3×, etc.) of a quarter of a wavelength (e.g., one quarter of the acoustic wavelength) at the main resonant frequency of the example resonator. However, the foregoing may be varied. For example, members of the pairs of bottom metal electrode layers of the bottom acoustic reflector may have respective layer thickness that correspond to from about one eighth to about one half wavelength at the resonant frequency, or an odd multiple (e.g., 1×, 3×, etc.) thereof.

In an example, if Tungsten is used as the high acoustic impedance metal, and the main resonant frequency of the resonator is twenty-four gigahertz (e.g., 24 GHz), then using the one quarter of the wavelength (e.g., one quarter of the acoustic wavelength) provides the layer thickness of the high impedance metal electrode layer members of the pairs as about five hundred and forty Angstroms (540 Å). For example, if Titanium is used as the low acoustic impedance metal, and the main resonant frequency of the resonator is twenty-four gigahertz (e.g., 24 GHz), then using the one quarter of the wavelength (e.g., one quarter of the acoustic wavelength) provides the layer thickness of the low impedance metal electrode layer members of the pairs as about six hundred and thirty Angstroms (630 Å). Similarly, respective layer thicknesses for members of the pairs of bottom metal electrode layers shown in FIGS. 4A through 4G may likewise be about one quarter of the wavelength (e.g., one quarter of the acoustic wavelength) of the main resonant frequency of the example resonator, and these respective layer thicknesses may likewise be determined for members of the pairs of bottom metal electrode layers for the high and low acoustic impedance metals employed.

For example, the bottom piezoelectric layer 105, 405A through 405G, may be electrically and acoustically coupled with the initial bottom metal electrode layer 117, 417A through 417G, and pair(s) of bottom metal electrode layers (e.g., first pair of bottom metal electrode layers 119, 419A through 419G, 121, 421A through 421G, e.g., second pair of bottom metal electrode layers 123, 423A through 423G, 125, 425A through 425G, e.g., third pair of bottom metal electrode layers 127, 427D, 129, 429D, fourth pair of bottom metal electrode layers 131, 431D, 133, 433D), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer 105, 405A through 405G. Further, the bottom piezoelectric layer 105, 405A through 405G and the first middle piezoelectric layer 107, 407A through 407G may be electrically and acoustically coupled with the initial bottom metal electrode layer 117, 417A through 417G and pair(s) of bottom metal electrode layers (e.g., first pair of bottom metal electrode layers 119, 419A through 419G, 121, 421A through 421G, e.g., second pair of bottom metal electrode layers 123, 423A through 423G, 125, 425A through 425G, e.g., third pair of bottom metal electrode layers 127, 427D, 129, 429D), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer 105, 405A through 405G acoustically coupled with the first middle piezoelectric layer 107, 407A through 407G. Additionally, the first middle piezoelectric layer 107, 407A through 407G, may be sandwiched between the bottom piezoelectric layer 105, 405A through 405G, and the second middle piezoelectric layer 109, 409A through 409G, and may be electrically and acoustically coupled with initial bottom metal electrode layer 117, 417A through 417G, and

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pair(s) of bottom metal electrode layers (e.g., first pair of bottom metal electrode layers 119, 419A through 419G, 121, 421A through 421G, e.g., second pair of bottom metal electrode layers 123, 423A through 423G, 125, 425A through 425G, e.g., third pair of bottom metal electrode layers 127, 427D, 129, 429D), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the first middle piezoelectric layer 107, 407A through 407G, sandwiched between the bottom piezoelectric layer 105, 405A through 405G, and the second middle piezoelectric layer 109, 409A through 409G.

Another mesa structure 113, 413A through 413G, (e.g., second mesa structure 113, 413A through 413G), may comprise the bottom acoustic reflector 113, 413A through 413G. The another mesa structure 113, 413A through 413G, (e.g., second mesa structure 113, 413A through 413G), may comprise initial bottom metal electrode layer 117, 417A through 417G. The another mesa structure 113, 413A through 413G, (e.g., second mesa structure 113, 413A through 413G), may comprise one or more pair(s) of bottom metal electrode layers (e.g., first pair of bottom metal electrode layers 119, 419A through 419G, 121, 421A through 421G, e.g., second pair of bottom metal electrode layers 123, 423A through 423G, 125, 425A through 425G, e.g., third pair of bottom metal electrode layers 127, 427A, 427D, 129, 429D, e.g., fourth pair of bottom metal electrode layers 131, 431D, 133, 433D).

Similar to what has been discussed for the bottom electrode stack, likewise the top electrode stack of the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, may have the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer. For example, an initial top metal electrode layer 135, 435A through 435G, may comprise the relatively high acoustic impedance metal, for example, Tungsten or Molybdenum. The top electrode stack of the plurality of top metal electrode layers of the top acoustic reflector 115, 415A through 415G, may approximate a metal distributed Bragg acoustic reflector. The plurality of top metal electrode layers of the top acoustic reflector may be electrically coupled (e.g., electrically interconnected) with one another. The acoustically reflective top electrode stack of the plurality of top metal electrode layers may operate together as a multilayer (e.g., bilayer, e.g., multiple layer) top electrode for the top acoustic reflector 115, 415A through 415G. Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective top electrode stack, may be a first pair of top metal electrode layers 137, 437A through 437G, and 139, 439A through 439G. A first member 137, 437A through 437G, of the first pair of top metal electrode layers may comprise the relatively low acoustic impedance metal, for example, Titanium or Aluminum. A second member 139, 439A through 439G, of the first pair of top metal electrode layers may comprise the relatively high acoustic impedance metal, for example, Tungsten or Molybdenum. Accordingly, the first pair of top metal electrode layers 137, 437A through 437G, 139, 439A through 439G, of the top acoustic reflector 115, 415A through 415G, may be different metals, and may have respective acoustic impedances that are different from one another so as to provide a reflective acoustic impedance mismatch at the resonant frequency (e.g., main resonant frequency). Similarly, the initial top metal electrode layer 135, 435A through 435G, and the first member of the first pair of top metal electrode layers 137, 437A through 437G,

of the top acoustic reflector **115**, **415A** through **415G**, may be different metals, and may have respective acoustic impedances that are different from one another so as to provide a reflective acoustic impedance mismatch at the resonant frequency (e.g., main resonant frequency).

Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective top electrode stack, a second pair of top metal electrode layers **141**, **441A** through **441G**, and **143**, **443A** through **443G**, may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal. Accordingly, the initial top metal electrode layer **135**, **435A** through **435G**, and members of the first and second pairs of top metal electrode layers **137**, **437A** through **437G**, **139**, **439A** through **439G**, **141**, **441A** through **441G**, **143**, **443A** through **443G**, may have respective acoustic impedances in the alternating arrangement to provide a corresponding plurality of reflective acoustic impedance mismatches.

Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective top electrode stack, a third pair of top metal electrode layers **145**, **445A** through **445C**, and **147**, **447A** through **447C**, may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal. Next in the alternating arrangement of low acoustic impedance metal layer and high acoustic impedance metal layer of the acoustically reflective top electrode stack, a fourth pair of top metal electrode layers **149**, **449A** through **449C**, **151**, **451A** through **451C**, may respectively comprise the relatively low acoustic impedance metal and the relatively high acoustic impedance metal.

For example, the bottom piezoelectric layer **105**, **405A** through **405G**, may be electrically and acoustically coupled with the initial top metal electrode layer **135**, **435A** through **435G**, and the pair(s) of top metal electrode layers (e.g., first pair of top metal electrode layers **137**, **437A** through **437G**, **139**, **439A** through **439G**, e.g., second pair of top metal electrode layers **141**, **441A** through **441G**, **143**, **443A** through **443G**, e.g., third pair of top metal electrode layers **145**, **445A** through **445C**, **147**, **447A** through **447C**), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer **105**, **405A** through **405G**. Further, the bottom piezoelectric layer **105**, **405A** through **405G** and the first middle piezoelectric layer **107**, **407A** through **407G** may be electrically and acoustically coupled with the initial top metal electrode layer **135**, **435A** through **435G** and pair(s) of top metal electrode layers (e.g., first pair of top metal electrode layers **137**, **437A** through **437G**, **139**, **439A** through **439G**, e.g., second pair of top metal electrode layers **141**, **441A** through **441G**, **143**, **443A** through **443G**, e.g., third pair of top metal electrode layers **145**, **445A** through **445C**, **147**, **447A** through **447C**), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the bottom piezoelectric layer **105**, **405A** through **405G** acoustically coupled with the first middle piezoelectric layer **107**, **407A** through **407G**. Additionally, the first middle piezoelectric layer **107**, **407A** through **407G**, may be sandwiched between the bottom piezoelectric layer **105**, **405A** through **405G**, and the second middle piezoelectric layer **109**, **409A** through **409G**, and may be electrically and acoustically coupled with the initial top metal electrode layer **135**, **435A** through **435G**, and the pair(s) of top metal electrode layers (e.g., first

pair of top metal electrode layers **137**, **437A** through **437G**, **139**, **439A** through **439G**, e.g., second pair of top metal electrode layers **141**, **441A** through **441G**, **143**, **443A** through **443G**, e.g., third pair of top metal electrode layers **145**, **445A** through **445C**, **147**, **447A** through **447C**), to excite the piezoelectrically excitable resonance mode (e.g., main resonance mode) at the resonant frequency (e.g., main resonant frequency) of the first middle piezoelectric layer **107**, **407A** through **407G**, sandwiched between the bottom piezoelectric layer **105**, **405A** through **405G**, and the second middle piezoelectric layer **109**, **409A** through **409G**.

Yet another mesa structure **115**, **415A** through **415G**, (e.g., third mesa structure **115**, **415A** through **415G**), may comprise the top acoustic reflector **115**, **415A** through **415G**, or a portion of the top acoustic reflector **115**, **415A** through **415G**. The yet another mesa structure **115**, **415A** through **415G**, (e.g., third mesa structure **115**, **415A** through **415G**), may comprise initial top metal electrode layer **135**, **435A** through **435G**. The yet another mesa structure **115**, **415A** through **415C**, (e.g., third mesa structure **115**, **415A** through **415C**), may comprise one or more pair(s) of top metal electrode layers (e.g., first pair of top metal electrode layers **137**, **437A** through **437C**, **139**, **439A** through **439C**, e.g., second pair of top metal electrode layers **141**, **441A** through **441C**, **143**, **443A** through **443C**, e.g., third pair of top metal electrode layers **145**, **445A** through **445C**, **147**, **447A** through **447C**, e.g., fourth pair of top metal electrode layers **149**, **449A** through **449C**, **151**, **451A** through **451C**).

Like the respective layer thicknesses of the bottom metal electrode layers, respective thicknesses of the top metal electrode layers may likewise be related to wavelength (e.g., acoustic wavelength) for the main resonant frequency of the example bulk acoustic wave resonators, **100**, **400A** through **400G**. Further, various embodiments for resonators having relatively higher main resonant frequency may have relatively thinner top metal electrode thicknesses, e.g., scaled thinner with relatively higher main resonant frequency. Similarly, various alternative embodiments for resonators having relatively lower main resonant frequency may have relatively thicker top metal electrode layer thicknesses, e.g., scaled thicker with relatively lower main resonant frequency. Like the layer thickness of the initial bottom metal, a layer thickness of the initial top metal electrode layer **135**, **435A** through **435G**, may likewise be about one eighth of the wavelength (e.g., one eighth of the acoustic wavelength) of the main resonant frequency of the example resonator. For example, if molybdenum is used as the high acoustic impedance metal and the main resonant frequency of the resonator is twenty-four gigahertz (e.g., 24 GHz), then using the one eighth of the wavelength (e.g., one eighth of the acoustic wavelength) provides the layer thickness of the initial top metal electrode layer **135**, **435A** through **435G**, as about three hundred and thirty Angstroms (330 Å). In the foregoing example, the one eighth of the wavelength (e.g., one eighth of the acoustic wavelength) at the main resonant frequency was used for determining the layer thickness of the initial top metal electrode layer **135**, **435A**-**435G**, but it should be understood that this layer thickness may be varied to be thicker or thinner in various other alternative example embodiments. Respective layer thicknesses, **T11** through **T18**, shown in FIG. 1A for members of the pairs of top metal electrode layers may be about an odd multiple (e.g., 1×, 3×, etc.) of a quarter of a wavelength (e.g., one quarter of an acoustic wavelength) of the main resonant frequency of the example resonator. Similarly, respective layer thicknesses for members of the pairs of top metal electrode layers shown in FIGS. 4A through 4G may likewise be about one quarter

of a wavelength (e.g., one quarter of an acoustic wavelength) at the main resonant frequency of the example resonator multiplied by an odd multiplier (e.g., 1x, 3x, etc.), and these respective layer thicknesses may likewise be determined for members of the pairs of top metal electrode layers for the high and low acoustic impedance metals employed. However, the foregoing may be varied. For example, members of the pairs of top metal electrode layers of the top acoustic reflector may have respective layer thickness that correspond to from an odd multiple (e.g., 1x, 3x, etc.) of about one eighth to an odd multiple (e.g., 1x, 3x, etc.) of about one half wavelength at the resonant frequency.

The bottom acoustic reflector 113, 413A through 413G, may have a thickness dimension T23 extending along the stack of bottom electrode layers. For the example of the 24 GHz resonator, the thickness dimension T23 of the bottom acoustic reflector may be about five thousand Angstroms (5,000 Å). The top acoustic reflector 115, 415A through 415G, may have a thickness dimension T25 extending along the stack of top electrode layers. For the example of the 24 GHz resonator, the thickness dimension T25 of the top acoustic reflector may be about five thousand Angstroms (5,000 Å). The piezoelectric layer stack 104, 404A through 404G, may have a thickness dimension T27 extending along the piezoelectric layer stack 104, 404A through 404G. For the example of the 24 GHz resonator, the thickness dimension T27 of the piezoelectric layer stack may be about eight thousand Angstroms (8,000 Å).

In the example resonators 100, 400A through 400G, of FIG. 1A and FIGS. 4A through 4G, a notional heavy dashed line is used in depicting an etched edge region 153, 453A through 453G, associated with the example resonators 100, 400A through 400G. Similarly, a laterally opposing etched edge region 154, 454A through 454G is arranged laterally opposing or opposite from the notional heavy dashed line depicting the etched edge region 153, 453A through 453G. The etched edge region may, but need not, assist with acoustic isolation of the resonators. The etched edge region may, but need not, help with avoiding acoustic losses for the resonators. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend along the thickness dimension T27 of the piezoelectric layer stack 104, 404A through 404G. The etched edge region 153, 453A through 453G, may extend through (e.g., entirely through or partially through) the piezoelectric layer stack 104, 404A through 404G. Similarly, the laterally opposing etched edge region 154, 454A through 454G may extend through (e.g., entirely through or partially through) the piezoelectric layer stack 104, 404A through 404G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the bottom piezoelectric layer 105, 405A through 405G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the first middle piezoelectric layer 107, 407A through 407G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the second middle piezoelectric layer 109, 409A through 409G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the top piezoelectric layer 111, 411A through 411G.

The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend along the thickness dimension T23 of the bottom acoustic reflector 113, 413A through 413G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the bottom acoustic reflector 113, 413A through 413G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the initial bottom metal electrode layer 117, 417A through 417G. The etched edge region 153, 453A through 453G, (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the first pair of bottom metal electrode layers, 119, 419A through 419G, 121, 421A through 421G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the second pair of bottom metal electrode layers, 123, 423A through 423G, 125, 425A through 425G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the third pair of bottom metal electrode layers, 127, 427D, 129, 429D. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the fourth pair of bottom metal electrode layers, 131, 431D, 133, 433D.

The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend along the thickness dimension T25 of the top acoustic reflector 115, 415A through 415G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the top acoustic reflector 115, 415A through 415G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the initial top metal electrode layer 135, 435A through 435G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the first pair of top metal electrode layers, 137, 437A through 437G, 139, 439A through 439G. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the second pair of top metal electrode layers, 141, 441A through 441C, 143, 443A through 443C. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the third pair of top metal electrode layers, 145, 445A through 445C, 147, 447A through 447C. The etched edge region 153, 453A through 453G (and the laterally opposing etched edge region 154, 454A through 454G) may extend through (e.g., entirely through or partially through) the fourth pair of top metal electrode layers, 149, 449A through 449C, 151, 451A through 451C.

As mentioned previously, mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through

404G) may comprise the respective stack 104, 404A through 404G, of the example four layers of piezoelectric material. The mesa structure 104, 404A through 404G (e.g., first mesa structure 104, 404A through 404G) may extend laterally between (e.g., may be formed between) etched edge region 153, 453A through 453G and laterally opposing etched edge region 154, 454A through 454G. As mentioned previously, another mesa structure 113, 413A through 413G, (e.g., second mesa structure 113, 413A through 413G), may comprise the bottom acoustic reflector 113, 413A through 413G. The another mesa structure 113, 413A through 413G, (e.g., second mesa structure 113, 413A through 413G) may extend laterally between (e.g., may be formed between) etched edge region 153, 453A through 453G and laterally opposing etched edge region 154, 454A through 454G. As mentioned previously, yet another mesa structure 115, 415A through 415G, (e.g., third mesa structure 115, 415A through 415G), may comprise the top acoustic reflector 115, 415A through 415G or a portion of the top acoustic reflector 115, 415A through 415G. The yet another mesa structure 115, 415A through 415G, (e.g., third mesa structure 115, 415A through 415G) may extend laterally between (e.g., may be formed between) etched edge region 153, 453A through 453G and laterally opposing etched edge region 154, 454A through 454G. In some example resonators 100, 400A, 400B, 400D through 400F, the second mesa structure corresponding to the bottom acoustic reflector 113, 413A, 413B, 413D through 413F may be laterally wider than the first mesa structure corresponding to the stack 104, 404A, 404B, 404D through 404F, of the example four layers of piezoelectric material. In some example resonators 100, 400A through 400C, the first mesa structure corresponding to the stack 104, 404A through 404C, of the example four layers of piezoelectric material may be laterally wider than the third mesa structure corresponding to the top acoustic reflector 115, 415A through 415C. In some example resonators 400D through 400G, the first mesa structure corresponding to the stack 404D through 404G, of the example four layers of piezoelectric material may be laterally wider than a portion of the third mesa structure corresponding to the top acoustic reflector 415D through 415G.

A mass load layer 155, 455A through 455G, may be added to the example resonators 100, 400A through 400G. For example, filters may include series connected resonator designs and shunt connected resonator designs that may include mass load layers. For example, for ladder filter designs, the shunt resonator may include a sufficient mass load layer so that the parallel resonant frequency (F_p) of the shunt resonator approximately matches the series resonant frequency (F_s) of the series resonator design. Thus the series resonator design (without the mass load layer) may be used for the shunt resonator design, but with the addition of the mass load layer 155, 455A through 455G, for the shunt resonator design. By including the mass load layer, the design of the shunt resonator may be approximately downshifted, or reduced, in frequency relative to the series resonator by a relative amount approximately corresponding to the electromechanical coupling coefficient (K_t^2) of the shunt resonator. For the example resonators 100, 400A through 400G, the mass load layer 155, 455A through 455G, may be arranged in the top acoustic reflector 115, 415A through 415G, above the first pair of top metal electrode layers. A metal may be used for the mass load. A dense metal such as Tungsten may be used for the mass load 155, 455A through 455G. An example thickness dimension of the mass load layer 155, 455A through 455G, may be about one hundred Angstroms (100 Å).

However, it should be understood that the thickness dimension of the mass load layer 155, 455A through 455G, may be varied depending on how much mass loading is desired for a particular design and depending on which metal is used for the mass load layer. Since there may be less acoustic energy in the top acoustic reflector 115, 415A through 415G, at locations further away from the piezoelectric stack 104, 404A through 404G, there may be less acoustic energy interaction with the mass load layer, depending on the location of the mass load layer in the arrangement of the top acoustic reflector. Accordingly, in alternative arrangements where the mass load layer is further away from the piezoelectric stack 104, 404A through 404G, such alternative designs may use more mass loading (e.g., thicker mass load layer) to achieve the same effect as what is provided in more proximate mass load placement designs. Also, in other alternative arrangements the mass load layer may be arranged relatively closer to the piezoelectric stack 104, 404A through 404G. Such alternative designs may use less mass loading (e.g., thinner mass load layer). This may achieve the same or similar mass loading effect as what is provided in previously discussed mass load placement designs, in which the mass load is arranged less proximate to the piezoelectric stack 104, 404A through 404G. Similarly, since Titanium (Ti) or Aluminum (Al) is less dense than Tungsten (W) or Molybdenum (Mo), in alternative designs where Titanium or Aluminum is used for the mass load layer, a relatively thicker mass load layer of Titanium (Ti) or Aluminum (Al) is needed to produce the same mass load effect as a mass load layer of Tungsten (W) or Molybdenum (Mo) of a given mass load layer thickness. Moreover, in alternative arrangements both shunt and series resonators may be additionally mass-loaded with considerably thinner mass loading layers (e.g., having thickness of about one tenth of the thickness of a main mass loading layer) in order to achieve specific filter design goals, as may be appreciated by one skilled in the art.

The example resonators 100, 400A through 400G, of FIG. 1A and FIGS. 4A through 4G may include a plurality of lateral features 157, 457A through 457G (e.g., patterned layer 157, 457A through 457G, e.g., step mass features 157, 457A through 457G), sandwiched between two top metal electrode layers (e.g., between the second member 139, 439A through 439G, of the first pair of top metal electrode layers and the first member 141, 441A through 441G, of the second pair of top metal electrode layers) of the top acoustic reflector 115, 415A through 415G. As shown in the figures, the plurality of lateral features 157, 457A through 457G, of patterned layer 157, 457A through 457G may comprise step features 157, 457A through 457G (e.g., step mass features 157, 457A through 457G). As shown in the figures, the plurality of lateral features 157, 457A through 457G, may be arranged proximate to lateral extremities (e.g., proximate to a lateral perimeter) of the top acoustic reflector 115, 415A through 415G. At least one of the lateral features 157, 457A through 457G, may be arranged proximate to where the etched edge region 153, 453A through 453G, extends through the top acoustic reflector 115, 415A through 415G.

After the lateral features 157, 457A through 457G, are formed, they may function as a step feature template, so that subsequent top metal electrode layers formed on top of the lateral features 157, 457A through 457G, may retain step patterns imposed by step features of the lateral features 157, 457A through 457G. For example, the second pair of top metal electrode layers 141, 441A through 441G, 143, 443A through 443G, the third pair of top metal electrode layers 145, 445A through 445C, 147, 447A through 447C, and the

fourth pair of top metal electrodes **149**, **449A** through **449C**, **151**, **451A** through **451C**, may retain step patterns imposed by step features of the lateral features **157**, **457A** through **457G**. The plurality of lateral features **157**, **457A** through **457G**, may add a layer of mass loading. The plurality of lateral features **157**, **457A** through **457G**, may be made of a patterned metal layer (e.g., a patterned layer of Tungsten (W), Molybdenum (Mo), Titanium (Ti) or Aluminum (Al)). In alternative examples, the plurality of lateral features **157**, **457A** through **457G**, may be made of a patterned dielectric layer (e.g., a patterned layer of Silicon Nitride (SiN), Silicon Dioxide (SiO₂) or Silicon Carbide (SiC)). The plurality of lateral features **157**, **457A** through **457G**, may, but need not, limit parasitic lateral acoustic modes (e.g., facilitate suppression of spurious modes) of the example resonators **100**, **400A** through **400G**. Thickness of the patterned layer of the lateral features **157**, **457A** through **457G**, (e.g., thickness of the patterned layers **157**, **457A** through **457G**) may be adjusted, e.g., may be determined as desired. For example, for the 24 GHz resonator, thickness may be adjusted within a range from about fifty Angstroms (50 Å) to about five hundred Angstroms (500 Å). Lateral step width of the lateral features **157**, **457A** through **457G** (e.g., width of the step mass features **157**, **457A** through **457G**) may be adjusted down, for example, from about two microns (2 μm). The foregoing may be adjusted to balance a design goal of limiting parasitic lateral acoustic modes (e.g., facilitating suppression of spurious modes) of the example resonators **100**, **400A** through **400G** as well as increasing average quality factor above the series resonance frequency against other design considerations e.g., maintaining desired average quality factor below the series resonance frequency.

In the example bulk acoustic wave resonator **100** shown in FIG. 1A, the patterned layer **157** may comprise Tungsten (W) (e.g., the step mass feature **157** of the patterned layer may comprise Tungsten (W)). A suitable thickness of the patterned layer **157** (e.g., thickness of the step mass feature **157**) and lateral width of features of the patterned layer **157** may vary based on various design parameters e.g., material selected for the patterned layer **157**, e.g., the desired resonant frequency of the given resonant design, e.g., effectiveness in facilitating spurious mode suppression. For the example 24 GHz bulk acoustic wave resonator **100** shown in FIG. 1A in which the patterned layer comprises Tungsten (W), a suitable thickness of the patterned layer **157** (e.g., thickness of the step mass feature **157**) may be 200 Angstroms and lateral width of features of the patterned layer **157** (e.g., lateral width of the step mass feature **157**) may be 0.8 microns, may facilitate suppression of the average strength of the spurious modes in the passband by approximately fifty percent (50%), as estimated by simulation relative to similar designs without the benefit of patterned layer **157**.

The example resonators **100**, **400A** through **400G**, of FIG. 1A and FIGS. 4A through 4G may include one or more (e.g., one or a plurality of) interposer layers sandwiched between piezoelectric layers of the stack **104**, **404A** through **404G**. For example, a first interposer layer **159**, **459A** through **459G** may be sandwiched between the bottom piezoelectric layer **105**, **405A** through **405G**, and the first middle piezoelectric layer **107**, **407A** through **407G**. For example, a second interposer layer **161**, **461A** through **461G**, may be sandwiched between the first middle piezoelectric layer **107**, **407A** through **407G**, and the second middle piezoelectric layer **109**, **409A** through **409G**. For example, a third interposer layer **163**, **463A** through **463G**, may be sandwiched

between the second middle piezoelectric layer **109**, **409A** through **409G**, and the top piezoelectric layer **111**, **411A** through **411G**.

One or more (e.g., one or a plurality of) interposer layers may be metal interposer layers. The metal interposer layers may be relatively high acoustic impedance metal interposer layers (e.g., using relatively high acoustic impedance metals such as Tungsten (W) or Molybdenum (Mo)). Such metal interposer layers may (but need not) flatten stress distribution across adjacent piezoelectric layers, and may (but need not) raise effective electromechanical coupling coefficient (Kt₂) of adjacent piezoelectric layers.

Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may be dielectric interposer layers. The dielectric of the dielectric interposer layers may be a dielectric that has a positive acoustic velocity temperature coefficient, so acoustic velocity increases with increasing temperature of the dielectric. The dielectric of the dielectric interposer layers may be, for example, silicon dioxide. Dielectric interposer layers may, but need not, facilitate compensating for frequency response shifts with increasing temperature. Most materials (e.g., metals, e.g., dielectrics) generally have a negative acoustic velocity temperature coefficient, so acoustic velocity decreases with increasing temperature of such materials. Accordingly, increasing device temperature generally causes response of resonators and filters to shift downward in frequency. Including dielectric (e.g., silicon dioxide) that instead has a positive acoustic velocity temperature coefficient may facilitate countering or compensating (e.g., temperature compensating) this downward shift in frequency with increasing temperature. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may comprise metal and dielectric for respective interposer layers.

In addition to the foregoing application of metal interposer layers to raise effective electromechanical coupling coefficient (Kt₂) of adjacent piezoelectric layers, and the application of dielectric interposer layers to facilitate compensating for frequency response shifts with increasing temperature, interposer layers may, but need not, increase quality factor (Q-factor) and/or suppress irregular spectral response patterns characterized by sharp reductions in Q-factor known as “rattles”. Q-factor of a resonator is a figure of merit in which increased Q-factor indicates a lower rate of energy loss per cycle relative to the stored energy of the resonator. Increased Q-factor in resonators used in filters results in lower insertion loss and sharper roll-off in filters. The irregular spectral response patterns characterized by sharp reductions in Q-factor known as “rattles” may cause ripples in filter pass bands.

Metal and/or dielectric interposer layer of suitable thicknesses and acoustic material properties (e.g., velocity, density) may be placed at appropriate places in the stack **104**, **404A** through **404G**, of piezoelectric layers, for example, proximate to the nulls of acoustic energy distribution in the stacks (e.g., between interfaces of piezoelectric layers of opposing axis orientation). Finite Element Modeling (FEM) simulations and varying parameters in fabrication prior to subsequent testing may help to optimize interposer layer designs for the stack. Thickness of interposer layers may, but need not, be adjusted to influence increased Q-factor and/or rattle suppression. It is theorized that if the interposer layer is too thin there is no substantial effect. Thus minimum thickness for the interposer layer may be about one monolayer, or about five Angstroms (5 Å). Alternatively, if the interposer layer is too thick, rattle strength may increase rather than being suppressed. Accordingly, an upper limit of

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interposer thickness may be about five-hundred Angstroms (500 Å) for a twenty-four Gigahertz (24 GHz) resonator design, with limiting thickness scaling inversely with frequency for alternative resonator designs. It is theorized that below a series resonant frequency of resonators, F_s , Q-factor may not be systematically and significantly affected by including a single interposer layer. However, it is theorized that there may, but need not, be significant increases in Q-factor, for example from about two-thousand (2000) to about three-thousand (3000), for inclusion of two or more interposer layers.

In the example resonators **100**, **400A** through **400C**, of FIG. 1A and FIGS. 4A through 4C, a planarization layer **165**, **465A** through **465C** may be included. A suitable material may be used for planarization layer **165**, **465A** through **465C**, for example Silicon Dioxide (SiO_2), Hafnium Dioxide (HfO_2), polyimide, or BenzoCyclobutene (BCB). An isolation layer **167**, **467A** through **467C**, may also be included and arranged over the planarization layer **165**, **465A-465C**. A suitable low dielectric constant (low-k), low acoustic impedance (low-Za) material may be used for the isolation layer **167**, **467A** through **467C**, for example polyimide, or BenzoCyclobutene (BCB).

In the example resonators **100**, **400A** through **400G**, of FIG. 1A and FIGS. 4A through 4G, a bottom electrical interconnect **169**, **469A** through **469G**, may be included to interconnect electrically with (e.g., electrically contact with) the bottom acoustic reflector **113**, **413A** through **413G**, stack of the plurality of bottom metal electrode layers. A top electrical interconnect **171**, **471A** through **471G**, may be included to interconnect electrically with the top acoustic reflector **115**, **415A** through **415G**, stack of the plurality of top metal electrode layers. A suitable material may be used for the bottom electrical interconnect **169**, **469A** through **469G**, and the top electrical interconnect **171**, **471A** through **471G**, for example, gold (Au). Top electrical interconnect **171**, **471A** through **471G** may be substantially acoustically isolated from the stack **104**, **404A** through **404G** of the example four layers of piezoelectric material by the top multilayer metal acoustic reflector electrode **115**, **415A** through **415G**. Top electrical interconnect **171**, **471A** through **471G** may have dimensions selected so that the top electrical interconnect **171**, **471A** through **471G** approximates a fifty ohm electrical transmission line at the main resonant frequency of the bulk acoustic wave resonator **100**, **400A** through **400G**. Top electrical interconnect **171**, **471A** through **471G** may have a thickness that is substantially thicker than a thickness of a pair of top metal electrode layers of the top multilayer metal acoustic reflector electrode **115**, **415A** through **415G** (e.g., thicker than thickness of the first pair of top metal electrode layers **137**, **437A** through **437G**, **139**, **439A** through **439G**). Top electrical interconnect **171**, **471A** through **471G** may have a thickness within a range from about one hundred Angstroms (100 Å) to about five micrometers (5 µm). For example, top electrical interconnect **171**, **471A** through **471G** may have a thickness of about two thousand Angstroms (2000 Å).

FIG. 1B is a simplified view of FIG. 1A that illustrates an example of acoustic stress distribution during electrical operation of the bulk acoustic wave resonator structure shown in FIG. 1A. A notional curved line schematically depicts vertical (T_{zz}) stress distribution **173** through stack **104** of the example four piezoelectric layers, **105**, **107**, **109**, **111**. The stress **173** is excited by the oscillating electric field applied via the top acoustic reflector **115** stack of the plurality of top metal electrode layers **135**, **137**, **139**, **141**, **143**, **145**, **147**, **149**, **151**, and the bottom acoustic reflector

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113 stack of the plurality of bottom metal electrode layers **117**, **119**, **121**, **123**, **125**, **127**, **129**, **131**, **133**. The stress **173** has maximum values inside the stack **104** of piezoelectric layers, while exponentially tapering off within the top acoustic reflector **115** and the bottom acoustic reflector **113**. Notably, acoustic energy confined in the resonator structure **100** is proportional to stress magnitude.

As discussed previously herein, the example four piezoelectric layers, **105**, **107**, **109**, **111** in the stack **104** may have an alternating axis arrangement in the stack **104**. For example the bottom piezoelectric layer **105** may have the normal axis orientation, which is depicted in FIG. 1B using the downward directed arrow. Next in the alternating axis arrangement of the stack **104**, the first middle piezoelectric layer **107** may have the reverse axis orientation, which is depicted in FIG. 1B using the upward directed arrow. Next in the alternating axis arrangement of the stack **104**, the second middle piezoelectric layer **109** may have the normal axis orientation, which is depicted in FIG. 1B using the downward directed arrow. Next in the alternating axis arrangement of the stack **104**, the top piezoelectric layer **111** may have the reverse axis orientation, which is depicted in FIG. 1B using the upward directed arrow. For the alternating axis arrangement of the stack **104**, stress **173** excited by the applied oscillating electric field causes normal axis piezoelectric layers (e.g., bottom and second middle piezoelectric layers **105**, **109**) to be in compression, while reverse axis piezoelectric layers (e.g., first middle and top piezoelectric layers **107**, **111**) to be in extension. Accordingly, FIG. 1B shows peaks of stress **173** on the right side of the heavy dashed line to depict compression in normal axis piezoelectric layers (e.g., bottom and second middle piezoelectric layers **105**, **109**), while peaks of stress **173** are shown on the left side of the heavy dashed line to depict extension in reverse axis piezoelectric layers (e.g., first middle and top piezoelectric layers **107**, **111**).

FIG. 1C shows a simplified top plan view of a bulk acoustic wave resonator structure **100A** corresponding to the cross sectional view of FIG. 1A, and also shows another simplified top plan view of an alternative bulk acoustic wave resonator structure **100B**. The bulk acoustic wave resonator structure **100A** includes the stack **104A** of four layers of piezoelectric material e.g., having the alternating piezoelectric axis arrangement of the four layers of piezoelectric material. The stack **104A** of piezoelectric layers may be sandwiched between the bottom acoustic reflector electrode **113A** and the top acoustic reflector electrode **115A**. The bottom acoustic reflector electrode may comprise the stack of the plurality of bottom metal electrode layers of the bottom acoustic reflector electrode **113A**, e.g., having the alternating arrangement of low acoustic impedance bottom metal electrode layers and high acoustic impedance bottom metal layers. Similarly, the top acoustic reflector electrode **115A** may comprise the stack of the plurality of top metal electrode layers of the top acoustic reflector electrode **115A**, e.g., having the alternating arrangement of low acoustic impedance top metal electrode layers and high acoustic impedance top metal electrode layers. The top acoustic reflector electrode **115A** may include a patterned layer **157A**. The patterned layer **157A** may approximate a frame shape (e.g., rectangular frame shape) proximate to a perimeter (e.g., rectangular perimeter) of top acoustic reflector electrode **115A** as shown in simplified top plan view in FIG. 1C. This patterned layer **157A**, e.g., approximating the rectangular frame shape in the simplified top plan view in FIG. 1C, corresponds to the patterned layer **157** shown in simplified cross sectional view in FIG. 1A. Top electrical

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interconnect **171A** extends over (e.g., electrically contacts) top acoustic reflector electrode **115A**. Bottom electrical interconnect **169A** extends over (e.g., electrically contacts) bottom acoustic reflector electrode **113A** through bottom via region **168A**.

FIG. 1C also shows another simplified top plan view of an alternative bulk acoustic wave resonator structure **100B**. Similarly, the bulk acoustic wave resonator structure **100B** includes the stack **104B** of four layers of piezoelectric material e.g., having the alternating piezoelectric axis arrangement of the four layers of piezoelectric material. The stack **104B** of piezoelectric layers may be sandwiched between the bottom acoustic reflector electrode **113B** and the top acoustic reflector electrode **115B**. The bottom acoustic reflector electrode may comprise the stack of the plurality of bottom metal electrode layers of the bottom acoustic reflector electrode **113B**, e.g., having the alternating arrangement of low acoustic impedance bottom metal electrode layers and high acoustic impedance bottom metal layers. Similarly, the top acoustic reflector electrode **115B** may comprise the stack of the plurality of top metal electrode layers of the top acoustic reflector electrode **115B**, e.g., having the alternating arrangement of low acoustic impedance top metal electrode layers and high acoustic impedance top metal electrode layers. The top acoustic reflector electrode **115B** may include a patterned layer **157B**. The patterned layer **157B** may approximate a frame shape (e.g., apodized frame shape) proximate to a perimeter (e.g., apodized perimeter) of top acoustic reflector electrode **115B** as shown in simplified top plan view in FIG. 1C. The apodized frame shape may be a frame shape in which substantially opposing extremities are not parallel to one another. This patterned layer **157B**, e.g., approximating the apodized frame shape in the simplified top plan view in FIG. 1C, is an alternative embodiment corresponding to the patterned layer **157** shown in simplified cross sectional view in FIG. 1A. Top electrical interconnect **171B** extends over (e.g., electrically contacts) top acoustic reflector electrode **115B**. Bottom electrical interconnect **169B** extends over (e.g., electrically contacts) bottom acoustic reflector electrode **113B** through bottom via region **168B**.

In FIGS. 1D and 1E, Nitrogen (N) atoms are depicted with a hatching style, while Aluminum (Al) atoms are depicted without a hatching style. FIG. 1D is a perspective view of an illustrative model of a reverse axis crystal structure **175** of Aluminum Nitride, AlN, in piezoelectric material of layers in FIG. 1A, e.g., having reverse axis orientation of negative polarization. For example, first middle and top piezoelectric layers **107**, **111** discussed previously herein with respect to FIGS. 1A and 1B are reverse axis piezoelectric layers. By convention, when the first layer of normal axis crystal structure **175** is a Nitrogen, N, layer and second layer in an upward direction (in the depicted orientation) is an Aluminum, Al, layer, the piezoelectric material including the reverse axis crystal structure **175** is said to have crystallographic c-axis negative polarization, or reverse axis orientation as indicated by the upward pointing arrow **177**. For example, polycrystalline thin film Aluminum Nitride, AlN, may be grown in the crystallographic c-axis negative polarization, or reverse axis, orientation perpendicular relative to the substrate surface using reactive magnetron sputtering of an aluminum target in a nitrogen atmosphere, and by introducing oxygen into the gas atmosphere of the reaction chamber during fabrication at the position where the flip to the reverse axis is desired. An inert gas, for example, Argon may also be included in a sputtering gas atmosphere, along with the nitrogen and oxygen.

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For example, a predetermined amount of oxygen containing gas may be added to the gas atmosphere over a short predetermined period of time or for the entire time the reverse axis layer is being deposited. The oxygen containing gas may be diatomic oxygen containing gas, such as oxygen (O_2). Proportionate amounts of the Nitrogen gas (N_2) and the inert gas may flow, while the predetermined amount of oxygen containing gas flows into the gas atmosphere over the predetermined period of time. For example, N_2 and Ar gas may flow into the reaction chamber in approximately a 3:1 ratio of N_2 to Ar, as oxygen gas also flows into the reaction chamber. For example, the predetermined amount of oxygen containing gas added to the gas atmosphere may be in a range from about a thousandth of a percent (0.001%) to about ten percent (10%), of the entire gas flow. The entire gas flow may be a sum of the gas flows of argon, nitrogen and oxygen, and the predetermined period of time during which the predetermined amount of oxygen containing gas is added to the gas atmosphere may be in a range from about a quarter (0.25) second to a length of time needed to create an entire layer, for example. For example, based on mass-flows, the oxygen composition of the gas atmosphere may be about 2 percent when the oxygen is briefly injected. This results in an aluminum oxynitride (ALON) portion of the final monolithic piezoelectric layer, integrated in the Aluminum Nitride, AlN, material, having a thickness in a range of about 5 nm to about 20 nm, which is relatively oxygen rich and very thin. Alternatively, the entire reverse axis piezoelectric layer may be aluminum oxynitride.

FIG. 1E is a perspective view of an illustrative model of a normal axis crystal structure **179** of Aluminum Nitride, AlN, in piezoelectric material of layers in FIG. 1A, e.g., having normal axis orientation of positive polarization. For example, bottom and second middle piezoelectric layers **105**, **109** discussed previously herein with respect to FIGS. 1A and 1B are normal axis piezoelectric layers. By convention, when the first layer of the reverse axis crystal structure **179** is an Al layer and second layer in an upward direction (in the depicted orientation) is an N layer, the piezoelectric material including the reverse axis crystal structure **179** is said to have a c-axis positive polarization, or normal axis orientation as indicated by the downward pointing arrow **181**. For example, polycrystalline thin film AlN may be grown in the crystallographic c-axis positive polarization, or normal axis, orientation perpendicular relative to the substrate surface by using reactive magnetron sputtering of an Aluminum target in a nitrogen atmosphere.

FIGS. 2A and 2B show a further simplified view of a bulk acoustic wave resonator similar to the bulk acoustic wave resonator structure shown in FIG. 1A along with its corresponding impedance versus frequency response during its electrical operation, as well as alternative bulk acoustic wave resonator structures with differing numbers of alternating axis piezoelectric layers, and their respective corresponding impedance versus frequency response during electrical operation. FIG. 2C shows additional alternative bulk acoustic wave resonator structures with additional numbers of alternating axis piezoelectric layers. Bulk acoustic wave resonators **2001A** through **2001I** may, but need not be, bulk acoustic millimeter wave resonators **2001A** through **2001I**, operable with a main resonance mode having a main resonant frequency that is a millimeter wave frequency (e.g., twenty-four Gigahertz, 24 GHz) in a millimeter wave frequency band. As defined herein, millimeter wave means a wave having a frequency within a range extending from eight Gigahertz (8 GHz) to three hundred Gigahertz (300 GHz), and millimeter wave band means a frequency band

spanning this millimeter wave frequency range from eight Gigahertz (8 GHz) to three hundred Gigahertz (300 GHz). Bulk acoustic wave resonators **2001A** through **2001I** may, but need not be, bulk acoustic Super High Frequency (SHF) wave resonators **2001A** through **2001I** or bulk acoustic Extremely High Frequency (EHF) wave resonators **2001A** through **2001I**, as the terms Super High Frequency (SHF) and Extremely High Frequency (EHF) are defined by the International Telecommunications Union (ITU). For example, bulk acoustic wave resonators **2001A** through **2001I** may be bulk acoustic Super High Frequency (SHF) wave resonators **2001A** through **2001I** operable with a main resonance mode having a main resonant frequency that is a Super High Frequency (SHF) (e.g., twenty-four Gigahertz, 24 GHz) in a Super High Frequency (SHF) wave frequency band. As discussed previously herein, piezoelectric layer thicknesses may be selected to determine the main resonant frequency of bulk acoustic Super High Frequency (SHF) wave resonators **2001A** through **2001I** in the Super High Frequency (SHF) wave band (e.g., twenty-four Gigahertz, 24 GHz main resonant frequency). Similarly, layer thicknesses of Super High Frequency (SHF) reflector layers (e.g., layer thickness of multilayer metal acoustic SHF wave reflector bottom electrodes **2013A** through **2013I**, e.g., layer thickness of multilayer metal acoustic SHF wave reflector top electrodes **2015A** through **2015I**) may be selected to determine peak acoustic reflectivity of such SHF reflectors at a frequency, e.g., peak reflectivity resonant frequency, within the Super High Frequency (SHF) wave band (e.g., a twenty-four Gigahertz, 24 GHz peak reflectivity resonant frequency). Alternatively, bulk acoustic wave resonators **2001A** through **2001I** may be bulk acoustic Extremely High Frequency (EHF) wave resonators **2001A** through **2001I** operable with a main resonance mode having a main resonant frequency that is an Extremely High Frequency (EHF) wave band (e.g., thirty-nine Gigahertz, 39 GHz main resonant frequency) in an Extremely High Frequency (EHF) wave frequency band. As discussed previously herein, piezoelectric layer thicknesses may be selected to determine the main resonant frequency of bulk acoustic Extremely High Frequency (EHF) wave resonators **2001A** through **2001I** in the Extremely High Frequency (EHF) wave band (e.g., thirty-nine Gigahertz, 39 GHz main resonant frequency). Similarly, layer thicknesses of Extremely High Frequency (EHF) reflector layers (e.g., layer thickness of multilayer metal acoustic EHF wave reflector bottom electrodes **2013A** through **2013I**, e.g., layer thickness of multilayer metal acoustic EHF wave reflector top electrodes **2015A** through **2015I**) may be selected to determine peak acoustic reflectivity of such EHF reflectors at a frequency, e.g., peak reflectivity resonant frequency, within the Extremely High Frequency (EHF) wave band (e.g., a thirty-nine Gigahertz, 39 GHz peak reflectivity resonant frequency). The general structures of the multilayer metal acoustic reflector top electrode and the multilayer metal acoustic reflector bottom electrode have already been discussed previously herein with respect of FIGS. 1A and 1B. As already discussed, these structures are directed to respective pairs of metal electrode layers, in which a first member of the pair has a relatively low acoustic impedance (relative to acoustic impedance of an other member of the pair), in which the other member of the pair has a relatively high acoustic impedance (relative to acoustic impedance of the first member of the pair), and in which the respective pairs of metal electrode layers have layer thicknesses corresponding to one quarter wavelength (e.g., one quarter acoustic wavelength) at a main resonant frequency of the resonator.

Accordingly, it should be understood that the bulk acoustic SHF or EHF wave resonators **2001A**, **2001B**, **2000C** shown in FIG. 2A include respective multilayer metal acoustic SHF or EHF wave reflector top electrodes **2015A**, **2015B**, **2015C** and multilayer metal acoustic SHF or EHF wave reflector bottom electrodes **2013A**, **2013B**, **2013C**, in which the respective pairs of metal electrode layers have layer thicknesses corresponding to a quarter wavelength (e.g., one quarter of an acoustic wavelength) at a SHF or EHF wave main resonant frequency of the respective bulk acoustic SHF or EHF wave resonator **2001A**, **2001B**, **2001C**.

Shown in FIG. 2A is a bulk acoustic SHF or EHF wave resonator **2001A** including a normal axis piezoelectric layer **201A** sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015A** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013A**. Also shown in FIG. 2A is a bulk acoustic SHF or EHF wave resonator **2001B** including a normal axis piezoelectric layer **201B** and a reverse axis piezoelectric layer **202B** arranged in a two piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015B** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013B**. A bulk acoustic SHF or EHF wave resonator **2001C** includes a normal axis piezoelectric layer **201C**, a reverse axis piezoelectric layer **202C**, and another normal axis piezoelectric layer **203C** arranged in a three piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015C** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013C**.

Included in FIG. 2B is bulk acoustic SHF or EHF wave resonator **2001D** in a further simplified view similar to the bulk acoustic wave resonator structure shown in FIGS. 1A and 1B and including a normal axis piezoelectric layer **201D**, a reverse axis piezoelectric layer **202D**, and another normal axis piezoelectric layer **203D**, and another reverse axis piezoelectric layer **204D** arranged in a four piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015D** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013D**. A bulk acoustic SHF or EHF wave resonator **2001E** includes a normal axis piezoelectric layer **201E**, a reverse axis piezoelectric layer **202E**, another normal axis piezoelectric layer **203E**, another reverse axis piezoelectric layer **204E**, and yet another normal axis piezoelectric layer **205E** arranged in a five piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015E** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013E**. A bulk acoustic SHF or EHF wave resonator **2001F** includes a normal axis piezoelectric layer **201F**, a reverse axis piezoelectric layer **202F**, another normal axis piezoelectric layer **203F**, another reverse axis piezoelectric layer **204F**, yet another normal axis piezoelectric layer **205F**, and yet another reverse axis piezoelectric layer **206F** arranged in a six piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015F** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013F**.

In FIG. 2A, shown directly to the right of the bulk acoustic SHF or EHF wave resonator **2001A** including the normal axis piezoelectric layer **201A**, is a corresponding diagram **2019A** depicting its impedance versus frequency

response during its electrical operation, as predicted by simulation. The diagram **2019A** depicts the main resonant peak **2021A** of the main resonant mode of the bulk acoustic SHF or EHF wave resonator **2001A** at its main resonant frequency (e.g., its 24 GHz series resonant frequency). The diagram **2019A** also depicts the satellite resonance peaks **2023A**, **2025A** of the satellite resonant modes of the bulk acoustic SHF or EHF wave resonator **2001A** at satellite frequencies above and below the main resonant frequency **2021A** (e.g., above and below the 24 GHz series resonant frequency). Relatively speaking, the main resonant mode corresponding to the main resonance peak **2021A** is the strongest resonant mode because it is stronger than all other resonant modes of the resonator **2001A**, (e.g., stronger than the satellite modes corresponding to relatively lesser satellite resonance peaks **2023A**, **2025A**).

Similarly, in FIGS. **2A** and **2B**, shown directly to the right of the bulk acoustic SHF or EHF wave resonators **2001B** through **2001F** are respective corresponding diagrams **2019B** through **2019F** depicting corresponding impedance versus frequency response during electrical operation, as predicted by simulation. The diagrams **2019B** through **2019F** depict respective example SHF main resonant peaks **2021B** through **2021F** of respective corresponding main resonant modes of bulk acoustic SHF wave resonators **2001B** through **2001F** at respective corresponding main resonant frequencies (e.g., respective 24 GHz series resonant frequencies). The diagrams **2019B** through **2019F** also depict respective example SHF satellite resonance peaks **2023B** through **2023F**, **2025B** through **2025F** of respective corresponding satellite resonant modes of the bulk acoustic SHF wave resonators **2001B** through **2001F** at respective corresponding SHF satellite frequencies above and below the respective corresponding main SHF resonant frequencies **2021B** through **2021F** (e.g., above and below the corresponding respective 24 GHz series resonant frequencies). Relatively speaking, for the corresponding respective main SHF resonant modes, its corresponding respective SHF main resonance peak **2021B** through **2021F** is the strongest for its bulk acoustic SHF wave resonators **2001B** through **2001F** (e.g., stronger than the corresponding respective SHF satellite modes and corresponding respective lesser SHF satellite resonance peaks **2023B**, **2025B**).

For the bulk acoustic SHF wave resonator **2001F** having the alternating axis stack of six piezoelectric layers, simulation of the 24 GHz design predicts an average passband quality factor of approximately 1,700. Scaling this 24 GHz, six piezoelectric layer design to a 37 GHz, six piezoelectric layer design for a example EHF resonator **2001F**, may have an average passband quality factor of approximately 1,300 as predicted by simulation. Scaling this 24 GHz, six piezoelectric layer design to a 77 GHz, six piezoelectric layer design for another example EHF resonator **2001F**, may have an average passband quality factor of approximately 730 as predicted by simulation.

As mentioned previously, FIG. **2C** shows additional alternative bulk acoustic wave resonator structures with additional numbers of alternating axis piezoelectric layers. A bulk acoustic SHF or EHF wave resonator **2001G** includes four normal axis piezoelectric layers **201G**, **203G**, **205G**, **207G**, and four reverse axis piezoelectric layers **202G**, **204G**, **206G**, **208G** arranged in an eight piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015G** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013G**. A bulk acoustic SHF or EHF wave resonator **2001H** includes five normal

axis piezoelectric layers **201H**, **203H**, **205H**, **207H**, **209H** and five reverse axis piezoelectric layers **202H**, **204H**, **206H**, **208H**, **210H** arranged in a ten piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015H** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013H**. A bulk acoustic SHF or EHF wave resonator **2001I** includes nine normal axis piezoelectric layers **201I**, **203I**, **205I**, **207I**, **209I**, **211I**, **213I**, **215I**, **217I** and nine reverse axis piezoelectric layers **202I**, **204I**, **206I**, **208I**, **210I**, **212I**, **214I**, **216I**, **218I** arranged in an eighteen piezoelectric layer alternating stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **2015I** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013I**.

For the bulk acoustic SHF wave resonator **2001I** having the alternating axis stack of eighteen piezoelectric layers, simulation of the 24 GHz design predicts an average passband quality factor of approximately 2,700. Scaling this 24 GHz, eighteen piezoelectric layer design to a 37 GHz, eighteen piezoelectric layer EHF design, may have an average passband quality factor of approximately 2000 as predicted by simulation. Scaling this 24 GHz, eighteen piezoelectric layer design to a 77 GHz, eighteen piezoelectric layer EHF design, may have an average passband quality factor of approximately 1,130 as predicted by simulation.

In the example resonators, **2001A** through **2001I**, of FIGS. **2A** through **2C**, a notional heavy dashed line is used in depicting respective etched edge region, **253A** through **253I**, associated with the example resonators, **2001A** through **2001I**. Similarly, in the example resonators, **2001A** through **2001I**, of FIGS. **2A** through **2C**, a laterally opposed etched edge region **254A** through **254I** may be arranged laterally opposite from etched edge region, **253A** through **253I**. The respective etched edge region may, but need not, assist with acoustic isolation of the resonators, **2001A** through **2001I**. The respective etched edge region may, but need not, help with avoiding acoustic losses for the resonators, **2001A** through **2001I**. The respective etched edge region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend along the thickness dimension of the respective piezoelectric layer stack. The respective etched edge region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend through (e.g., entirely through or partially through) the respective piezoelectric layer stack. The respective etched edge region, **253A** through **253I** may extend through (e.g., entirely through or partially through) the respective first piezoelectric layer, **201A** through **201I**. The respective etched edge region, **253B** through **253I**, (and the laterally opposed etched edge region **254B** through **254I**) may extend through (e.g., entirely through or partially through) the respective second piezoelectric layer, **202B** through **202I**. The respective etched edge region, **253C** through **253I**, (and the laterally opposed etched edge region **254C** through **254I**) may extend through (e.g., entirely through or partially through) the respective third piezoelectric layer, **203C** through **203I**. The respective etched edge region, **253D** through **253I**, (and the laterally opposed etched edge region **254D** through **254I**) may extend through (e.g., entirely through or partially through) the respective fourth piezoelectric layer, **204D** through **204I**. The respective etched edge region, **253E** through **253I**, (and the laterally opposed etched edge region **254E** through **254I**) may extend through (e.g., entirely through or partially through) the respective additional piezoelectric layers of the resonators, **2001E** through **2001I**. The respective etched edge

region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend along the thickness dimension of the respective multilayer metal acoustic SHF or EHF wave reflector bottom electrode, **2013A** through **2013I**, of the resonators, **2001A** through **2001I**. The respective etched edge region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend through (e.g., entirely through or partially through) the respective multilayer metal acoustic SHF or EHF wave reflector bottom electrode, **2013A** through **2013I**. The respective etched edge region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend along the thickness dimension of the respective multilayer metal acoustic SHF or EHF wave reflector top electrode, **2015A** through **2015I** of the resonators, **2001A** through **2001I**. The etched edge region, **253A** through **253I**, (and the laterally opposed etched edge region **254A** through **254I**) may extend through (e.g., entirely through or partially through) the respective multilayer metal acoustic SHF or EHF wave reflector top electrode, **2015A** through **2015I**.

As shown in FIGS. **2A** through **2C**, first mesa structures corresponding to the respective stacks of piezoelectric material layers may extend laterally between (e.g., may be formed between) etched edge regions **253A** through **253I** and laterally opposing etched edge region **254A** through **254I**. Second mesa structures corresponding to multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013A** through **2013I** may extend laterally between (e.g., may be formed between) etched edge regions **153A** through **153I** and laterally opposing etched edge region **154A** through **154I**. Third mesa structures corresponding to multilayer metal acoustic SHF or EHF wave reflector top electrode **2015A** through **2015I** may extend laterally between (e.g., may be formed between) etched edge regions **153A** through **153I** and laterally opposing etched edge region **154A** through **154I**.

In accordance with the teachings herein, various bulk acoustic SHF or EHF wave resonators may include: a seven piezoelectric layer alternating axis stack arrangement; a nine piezoelectric layer alternating axis stack arrangement; an eleven piezoelectric layer alternating axis stack arrangement; a twelve piezoelectric layer alternating axis stack arrangement; a thirteen piezoelectric layer alternating axis stack arrangement; a fourteen piezoelectric layer alternating axis stack arrangement; a fifteen piezoelectric layer alternating axis stack arrangement; a sixteen piezoelectric layer alternating axis stack arrangement; and a seventeen piezoelectric layer alternating axis stack arrangement; and that these stack arrangements may be sandwiched between respective multilayer metal acoustic SHF or EHF wave reflector top electrodes and respective multilayer metal acoustic SHF or EHF wave reflector bottom electrodes. Mass load layers and lateral features (e.g., step features) as discussed previously herein with respect to FIG. **1A** are not explicitly shown in the simplified diagrams of the various resonators shown in FIGS. **2A**, **2B** and **2C**. However, such mass load layers may be included, and such lateral features may be included, and may be arranged between, for example, top metal electrode layers of the respective top acoustic reflectors of the resonators shown in FIGS. **2A**, **2B** and **2C**. Further, such mass load layers may be included, and such lateral features may be included, and may be arranged between, for example, top metal electrode layers of the respective top acoustic reflectors in the various resonators

having the alternating axis stack arrangements of various numbers of piezoelectric layers, as described in this disclosure.

Further, it should be understood that interposer layers as discussed previously herein with respect to FIG. **1A** are explicitly shown in the simplified diagrams of the various resonators shown in FIGS. **2A**, **2B** and **2C**. Such interposers may be included and interposed between adjacent piezoelectric layers in the various resonators shown in FIGS. **2A**, **2B** and **2C**, and further may be included and interposed between adjacent piezoelectric layers in the various resonators having the alternating axis stack arrangements of various numbers of piezoelectric layers, as described in this disclosure. In some other alternative bulk acoustic wave resonator structures, fewer interposer layers may be employed. For example, FIG. **2D** shows another alternative bulk acoustic wave resonator structure **2001J**, similar to bulk acoustic wave resonator structure **2001I** shown in FIG. **2C**, but with differences. For example, relatively fewer interposer layers may be included in the alternative bulk acoustic wave resonator structure **2001J** shown in FIG. **2D**. For example, FIG. **2D** shows a first interposer layer **261J** interposed between second layer of (reverse axis) piezoelectric material **202J** and third layer of (normal axis) piezoelectric material **203J**, but without an interposer layer interposed between first layer of (normal axis) piezoelectric material **201J** and second layer of (reverse axis) piezoelectric material **202J**. As shown in FIG. **2D** in a first detailed view **220J**, without an interposer layer interposed between first layer of piezoelectric material **201J** and second layer of piezoelectric material **202J**, the first and second piezoelectric layer **201J**, **202J** may be a monolithic layer **222J** of piezoelectric material (e.g., Aluminum Nitride (AlN)) having first and second regions **224J**, **226J**. A central region of monolithic layer **222J** of piezoelectric material (e.g., Aluminum Nitride (AlN)) between first and second regions **224J**, **226J** may be oxygen rich. The first region **224J** of monolithic layer **222J** (e.g., bottom region **224J** of monolithic layer **222J**) has a first piezoelectric axis orientation (e.g., normal axis orientation) as representatively illustrated in detailed view **220J** using a downward pointing arrow at first region **224J**, (e.g., bottom region **224J**). This first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) at first region **224J** of monolithic layer **222J** (e.g., bottom region **224J** of monolithic layer **222J**) corresponds to the first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) of first piezoelectric layer **201J**. The second region **226J** of monolithic layer **222J** (e.g., top region **226J** of monolithic layer **222J**) has a second piezoelectric axis orientation (e.g., reverse axis orientation) as representatively illustrated in detailed view **220J** using an upward pointing arrow at second region **226J**, (e.g., top region **226J**). This second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) at second region **226J** of monolithic layer **222J** (e.g., top region **226J** of monolithic layer **222J**) may be formed to oppose the first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) at first region **224J** of monolithic layer **222J** (e.g., bottom region **224J** of monolithic layer **222J**) by adding gas (e.g., oxygen) to flip the axis while sputtering the second region **226J** of monolithic layer **222J** (e.g., top region **226J** of monolithic layer **222J**) onto the first region **224J** of monolithic layer **222J** (e.g., bottom region **224J** of monolithic layer **222J**). The second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) at second region **226J** of monolithic layer **222J** (e.g., top region **226J** of monolithic layer **222J**)

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monolithic layer 222J) corresponds to the second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) of second piezoelectric layer 202J.

Similarly, as shown in FIG. 2D in a second detailed view 230J, without an interposer layer interposed between third layer of piezoelectric material 203J and fourth layer of piezoelectric material 204J, the third and fourth piezoelectric layer 203J, 204J may be an additional monolithic layer 232J of piezoelectric material (e.g., Aluminum Nitride (AlN)) having first and second regions 234J, 236J. A central region of additional monolithic layer 232J of piezoelectric material (e.g., Aluminum Nitride (AlN)) between first and second regions 234J, 236J may be oxygen rich. The first region 234J of additional monolithic layer 232J (e.g., bottom region 234J of additional monolithic layer 232J) has the first piezoelectric axis orientation (e.g., normal axis orientation) as representatively illustrated in second detailed view 230J using the downward pointing arrow at first region 234J, (e.g., bottom region 224J). This first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) at first region 234J of additional monolithic layer 232J (e.g., bottom region 234J of additional monolithic layer 232J) corresponds to the first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) of third piezoelectric layer 203J. The second region 236J of additional monolithic layer 232J (e.g., top region 236J of additional monolithic layer 232J) has the second piezoelectric axis orientation (e.g., reverse axis orientation) as representatively illustrated in second detailed view 230J using the upward pointing arrow at second region 236J, (e.g., top region 226J). This second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) at second region 236J of additional monolithic layer 232J (e.g., top region 236J of additional monolithic layer 232J) may be formed to oppose the first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) at first region 234J of additional monolithic layer 232J (e.g., bottom region 234J of additional monolithic layer 232J) by adding gas (e.g., oxygen) to flip the axis while sputtering the second region 236J of additional monolithic layer 232J (e.g., top region 236J of additional monolithic layer 232J) onto the first region 234J of additional monolithic layer 232J (e.g., bottom region 234J of additional monolithic layer 232J). The second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) at second region 236J of additional monolithic layer 232J (e.g., top region 236J of additional monolithic layer 232J) corresponds to the second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) of fourth piezoelectric layer 204J.

Similar to what was just discussed, without an interposer layer interposed between fifth layer of piezoelectric material 205J and sixth layer of piezoelectric material 206J, the fifth and sixth piezoelectric layer 205J, 206J may be another additional monolithic layer of piezoelectric material (e.g., Aluminum Nitride (AlN)) having first and second regions. More generally, for example in FIG. 2D, where N is an odd positive integer, without an interposer layer interposed between Nth layer of piezoelectric material and (N+1)th layer of piezoelectric material, the Nth and (N+1)th piezoelectric layer may be an (N+1)/2th monolithic layer of piezoelectric material (e.g., Aluminum Nitride (AlN)) having first and second regions. Accordingly, without an interposer layer interposed between seventeenth layer of piezoelectric material 217J and eighteenth layer of piezoelectric material 218J, the seventeenth and eighteenth piezoelectric

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layer 217J, 218J may be ninth monolithic layer of piezoelectric material (e.g., Aluminum Nitride (AlN)) having first and second regions.

The first interposer layer 261J is shown in FIG. 2D as interposing between a first pair of opposing axis piezoelectric layers 201J, 202J, and a second pair of opposing axis piezoelectric layers 203J, 204J. More generally, for example, where M is a positive integer, an Mth interposer layer is shown in FIG. 2D as interposing between an Mth pair of opposing axis piezoelectric layers and an (M+1)th pair of opposing axis piezoelectric layers. Accordingly, an eighth interposer layer is shown in FIG. 2D as interposing between an eighth pair of opposing axis piezoelectric layers 215J, 216J, and a ninth pair of opposing axis piezoelectric layers 217J, 218J. FIG. 2D shows an eighteen piezoelectric layer alternating axis stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode 2015J and multilayer metal acoustic SHF or EHF wave reflector bottom electrode 2013J. Etched edge region 253J (and laterally opposing etched edge region 254J) may extend through (e.g., entirely through, e.g., partially through) the eighteen piezoelectric layer alternating axis stack arrangement and its interposer layers, and may extend through (e.g., entirely through, e.g., partially through) multilayer metal acoustic SHF or EHF wave reflector top electrode 2015J, and may extend through (e.g., entirely through, e.g., partially through) multilayer metal acoustic SHF or EHF wave reflector bottom electrode 2013J. As shown in FIG. 2D, a first mesa structure corresponding to the stack of eighteen piezoelectric material layers may extend laterally between (e.g., may be formed between) etched edge region 253J and laterally opposing etched edge region 254J. A second mesa structure corresponding to multilayer metal acoustic SHF or EHF wave reflector bottom electrode 2013J may extend laterally between (e.g., may be formed between) etched edge region 153J and laterally opposing etched edge region 154J. Third mesa structure corresponding to multilayer metal acoustic SHF or EHF wave reflector top electrode 2015J may extend laterally between (e.g., may be formed between) etched edge region 153J and laterally opposing etched edge region 154J.

As mentioned previously herein, one or more (e.g., one or a plurality of) interposer layers may be metal interposer layers. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may be dielectric interposer layers. Interposer layers may be metal and/or dielectric interposer layers. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may comprise metal and dielectric for respective interposer layers. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may be formed of different metal layers. For example, high acoustic impedance metal layer such as Tungsten (W) or Molybdenum (Mo) may (but need not) raise effective electromechanical coupling coefficient (Kt2) while subsequently deposited metal layer with hexagonal symmetry such as Titanium (Ti) may (but need not) facilitate higher crystallographic quality of subsequently deposited piezoelectric layer. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may be formed of different dielectric layers. For example, high acoustic impedance dielectric layer such as Hafnium Dioxide (HfO2) may (but need not) raise effective electromechanical coupling coefficient (Kt2). Subsequently deposited amorphous dielectric layer such as Silicon Dioxide (SiO2) may (but need not) facilitate compensating for temperature dependent frequency shifts. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may

comprise metal and dielectric for respective interposer layers. For example, high acoustic impedance metal layer such as Tungsten (W) or Molybdenum (Mo) may (but need not) raise effective electromechanical coupling coefficient (Kt2) while subsequently deposited amorphous dielectric layer such as Silicon Dioxide (SiO2) may (but need not) facilitate compensating for temperature dependent frequency shifts. For example, in FIG. 2D one or more of the interposer layers (e.g., interposer layer **268J**) may comprise metal and dielectric for respective interposer layers. For example, detailed view **240J** of interposer **268J** shows interposer **268J** as comprising metal sublayer **268JB** over dielectric sublayer **268JA**. For interposer **268J**, example thickness of metal sublayer **268JB** may be approximately two hundred Angstroms (200 Å). For interposer **268J**, example thickness of dielectric sublayer **268JA** may be approximately two hundred Angstroms (200 Å). The second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) at region **244J** (e.g., bottom region **244J**) corresponds to the second piezoelectric axis orientation (e.g., reverse axis orientation, e.g., upward pointing arrow) of eighth piezoelectric layer **208J**. The first piezoelectric axis orientation (e.g., normal axis orientation, e.g., downward pointing arrow) at region **246J** (e.g., top region **246J**) corresponds to the first piezoelectric axis orientation (e.g., normal orientation, e.g., downward pointing arrow) of ninth piezoelectric layer **209J**.

As discussed, interposer layers shown in FIG. 1A, and as explicitly shown in the simplified diagrams of the various resonators shown in FIGS. 2A, 2B, 2C and 2D may be included and interposed between adjacent piezoelectric layers in the various resonators. Such interposer layers may laterally extend within the mesa structure of the stack of piezoelectric layers a full lateral extent of the stack, e.g., between the etched edge region of the stack and the opposing etched edge region of the stack. However, in some other alternative bulk acoustic wave resonator structures, interposer layers may be patterned during fabrication of the interposer layers (e.g., patterned using masking and selective etching techniques during fabrication of the interposer layers). Such patterned interposer layers need not extend a full lateral extent of the stack (e.g., need not laterally extend to any etched edge regions of the stack.) For example, FIG. 2E shows another alternative bulk acoustic wave resonator structure **2001K**, similar to bulk acoustic wave resonator structure **2001J** shown in FIG. 2D, but with differences. For example, in the alternative bulk acoustic wave resonator structure **2001K** shown in FIG. 2E, patterned interposer layers (e.g., first patterned interposer layer **261K**) may be interposed between sequential pairs of opposing axis piezoelectric layers (e.g., first patterned interposer layer **295K** may be interposed between a first pair of opposing axis piezoelectric layers **201K**, **202K**, and a second pair of opposing axis piezoelectric layers **203K**, **204K**).

FIG. 2E shows an eighteen piezoelectric layer alternating axis stack arrangement having an active region of the bulk acoustic wave resonator structure **2001K** sandwiched between overlap of multilayer metal acoustic SHF or EHF wave reflector top electrode **2015IK** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013K**. In FIG. 2E, patterned interposer layers (e.g., first patterned interposer layer **261K**) may be patterned to have extent limited to the active region of the bulk acoustic wave resonator structure **2001K** sandwiched between overlap of multilayer metal acoustic SHF or EHF wave reflector top electrode **2015K** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013K**. A planarization

layer **256K** at a limited extent of multilayer metal acoustic SHF or EHF wave reflector bottom electrode **2013K** may facilitate fabrication of the eighteen piezoelectric layer alternating axis stack arrangement (e.g., stack of eighteen piezoelectric layers **201K** through **218K**).

Patterning of interposer layers may be done in various combinations. For example, some interposer layers need not be patterned (e.g., may be unpatterned) within lateral extent of the stack of piezoelectric layers (e.g., some interposer layers may extend to full lateral extent of the stack of piezoelectric layers). For example, first interposer layer **261J** shown in FIG. 2D need not be patterned (e.g., may be unpatterned) within lateral extent of the stack of piezoelectric layers (e.g., first interposer layer **261J** may extend to full lateral extent of the stack of piezoelectric layers). For example, in FIG. 2D interposer layers interposed between adjacent sequential pairs of normal axis and reverse axis piezoelectric layers need not be patterned (e.g., may be unpatterned) within lateral extent of the stack of piezoelectric layers (e.g., interposer layers interposed between sequential pairs of normal axis and reverse axis piezoelectric layers may extend to full lateral extent of the stack of piezoelectric layers). For example in FIG. 2D, first interposer layer **261J** interposed between first sequential pair of normal axis and reverse axis piezoelectric layers **201J**, **202J** and adjacent second sequential pair of normal axis and reverse axis piezoelectric layers **203J**, **204J** need not be patterned within lateral extent of the stack of piezoelectric layers (e.g., first interposer layer **261J** may extend to full lateral extent of the stack of piezoelectric layers). In contrast to these unpatterned interposer layers (e.g., in contrast to unpatterned interposer layer **261J**) as shown in FIG. 2D, in FIG. 2E patterned interposer layers (e.g., first patterned interposer layer **261K**) may be patterned, for example, to have extent limited to the active region of the bulk acoustic wave resonator structure **2001K** shown in FIG. 2E.

FIGS. 2F, 2G, 2H and 2I show simplified diagrams of bulk acoustic wave resonator structures **2000L** through **2000Y** comprising respective pairs of Super or Extremely High Frequency (SHF or EHF) acoustic reflector electrodes **2013L** through **2013Y**, **2015L** through **2015Y**, including one or more respective patterned layers **2057L** through **2057Y**, **2058N**, **2058Q**, **2058R**, **2058S** **2058T**, **2058U**, arranged over a substrate **2001L** through **2001Y** (e.g., silicon substrate **2001L** through **2001Y**). The one or more respective patterned layers **2057L** through **2057Y**, **2058N**, **2058Q**, **2058R**, **2058S** **2058T**, **2058U** may facilitate spurious mode suppression of bulk acoustic wave resonator structures **2000L** through **2000Y**. In an SHF example, in a case where a resonant frequency (e.g., main resonant frequency) of the bulk acoustic wave resonator may be twenty four Gigahertz, 24 GHz, and may include a patterned layer comprising Tungsten (W), a suitable thickness of the patterned layer (e.g., thickness of the step mass feature) may be 200 Angstroms and lateral width of features of the patterned layer **157** (e.g., lateral width of the step mass feature) may be 0.8 microns. This may facilitate suppression of the average strength of the spurious modes in the passband by approximately fifty percent (50%), as estimated by simulation relative to similar designs without the benefit of the patterned layer.

In cases where bulk acoustic wave resonators may have more than one patterned layer (e.g., two patterned layers, e.g., first and second patterned layers, e.g., top and bottom patterned layers), more than one patterned layer may together facilitate spurious mode suppression. Employing more than one patterned layer (e.g., two patterned layers,

e.g., first and second patterned layers, e.g., top and bottom patterned layers) may, but need not, extend a range of step mass features for facilitating spurious mode suppression, relative to employing one patterned layer.

The Super or Extremely High Frequency (SHF or EHF) acoustic reflector electrodes **2013L** through **2013Y**, **2015L** through **2015Y** may have thicknesses of member metal electrode layers selected so as to provide peak acoustic reflectance in the Super High Frequency (SHF) band or the Extreme High Frequency (EHF) band. SHF or EHF bottom acoustic reflector electrodes **2013L** through **2013Y** may include a respective first pair of bottom metal electrode layers **2022L** through **2022Y**, as representatively illustrated by respective horizontal dashed lines drawn between the first pair of bottom metal electrode layers **2022L** through **2022Y** and the inclusive blocks of SHF or EHF bottom acoustic reflector electrodes **2013L** through **2013Y**. SHF or EHF top acoustic reflector electrodes **2015L** through **2015Y** may include a respective first pair of top metal electrode layers **2024L** through **2024Y**, as representatively illustrated by respective horizontal dashed lines drawn between the first pair of top metal electrode layers **2024L** through **2024Y** and the inclusive blocks of SHF or EHF top acoustic reflector electrodes **2015L** through **2015Y**.

As shown in FIGS. 2F, 2G, 2H and 2I, first mesa structures corresponding to the respective stacks of piezoelectric material layers (e.g., stacks of normal axis piezoelectric layer **2005L** through **2005Y**, and reverse axis piezoelectric layer **2007L** through **2007Y**) may extend laterally between (e.g., may be formed between) etched edge regions **2053L** through **2053Y** and laterally opposing etched edge region **2053L** through **2053Y**. Second mesa structures corresponding to SHF or EHF bottom acoustic reflector electrode **2013L** through **2013Y** may extend laterally between (e.g., may be formed between) etched edge regions **2053L** through **2053Y** (vertically highlighted in FIGS. 2F, 2G, 2H and 2I using vertical heavy dashed lines) and laterally opposing etched edge region **2054L** through **2054Y**. Third mesa structures corresponding to SHF or EHF top acoustic reflector electrode **2015L** through **2015Y** may extend laterally between (e.g., may be formed between) etched edge regions **2053L** through **2053Y** and laterally opposing etched edge region **2054L** through **2054Y**.

Bulk acoustic Super High frequency (SHF) wave resonators or bulk acoustic Extremely High Frequency (EHF) resonators **2000L** through **2000Y** may include the normal axis piezoelectric layer **2005L** through **2005Y** (e.g., normal piezoelectric axis Aluminum Nitride piezoelectric layer **2005L** through **2005Y**) and the reverse axis piezoelectric layer **2007L** through **2007Y** (e.g., reverse piezoelectric axis Aluminum Nitride piezoelectric layer **2007L** through **2007Y**) arranged in a two or more piezoelectric layer alternating stack arrangement sandwiched between the multilayer metal acoustic Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y** and the multilayer metal acoustic Super High frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector electrode **2015L** through **2015Y**. The normal axis piezoelectric layer **2005L** through **2005Y** has the normal piezoelectric axis orientation. The reverse axis piezoelectric layer **2007L** through **2007Y** has the reverse piezoelectric axis orientation that opposes the normal piezoelectric axis orientation of the normal axis piezoelectric layer **2005L** through **2005Y**. The normal axis piezoelectric layer **2005L** through **2005Y** and the reverse axis piezoelectric layer **2007L** through **2007Y** may be acoustically coupled with one another to have a piezoelectrically

excitable resonance mode. An interposer layer **2059L** through **2059Y** (e.g., dielectric interposer layer **2059L** through **2059Y**, e.g., metal interposer layer **2059L** through **2059Y**, e.g., metal or dielectric interposer layer **2059L** through **2059Y**, e.g., metal and dielectric interposer layer **2059L** through **2059Y**) may be interposed between normal axis piezoelectric layer **2005L** through **2005Y** and reverse axis piezoelectric layer **2007L** through **2007Y**. As mentioned previously, the Super High frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y** may include the first pair of bottom metal electrode layers **2022L** through **2022Y**. The Super High frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y** may include the first pair of top metal electrode layers **2024L** through **2024Y**. The first pair of bottom metal electrode layers **2022L** through **2022Y** of the Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y** and the first pair of top metal electrode layers **2024L** through **2024Y** of the Super High frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y** may be coupled (e.g., electrically coupled, e.g., acoustically coupled) with the normal axis piezoelectric layer **2005L** through **2005Y** and the reverse axis piezoelectric layer **2007L** through **2007Y** to excite the piezoelectrically excitable resonance mode at a resonant frequency of the bulk acoustic Super High Frequency (SHF) wave resonator **2000L** through **2000Y** in the Super High Frequency (SHF) wave band (e.g., 24 GHz main resonant frequency), or the bulk acoustic Extremely High Frequency (EHF) wave resonator **2000L** through **2000Y** in the Extremely High Frequency (EHF) wave band.

In SHF examples, thicknesses of the normal axis piezoelectric layer **2005L** through **2005Y** and the reverse axis piezoelectric layer **2007L** through **2007Y** may be selected to determine the main resonant frequency of bulk acoustic Super High Frequency (SHF) wave resonator **2000L** through **2000Y** in the Super High Frequency (SHF) wave band (e.g., approximately twenty-four Gigahertz, approximately 24 GHz main resonant frequency). Similarly, in SHF examples, layer thicknesses of Super High Frequency (SHF) acoustic reflector electrode layers (e.g., layer thicknesses of members of the first pair of bottom metal electrode layers **2022L** through **2022Y** of Super High frequency (SHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., layer thickness of members of the first pair of top metal electrode layers **2024L** through **2024Y** of Super High frequency (SHF) top acoustic reflector electrode **2015L** through **2015Y**) may be selected to determine peak acoustic reflectivity of such SHF acoustic reflector electrodes at a frequency, e.g., peak reflectivity resonant frequency, within the Super High Frequency (SHF) wave band (e.g., approximately twenty-four Gigahertz, approximately 24 GHz peak reflectivity resonant frequency). The Super High Frequency (SHF) wave band may include: 1) peak reflectivity resonant frequency (e.g., approximately twenty-four Gigahertz, approximately 24 GHz peak reflectivity resonant frequency) of the Super High Frequency (SHF) acoustic reflector electrode layers (e.g., layer thicknesses of members of the first pair of bottom metal electrode layers **2022L** through **2022Y** of the Super High frequency (SHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., layer thickness of members of the first pair of top metal electrode layers **2024L** through **2024Y** of the Super High frequency (SHF) top acoustic reflector electrode **2015L** through **2015Y**); and 2) the main resonant frequency of bulk acoustic the Super High Frequency (SHF)

wave resonator **2000L** through **2000Y** (e.g., approximately twenty-four Gigahertz, approximately 24 GHz main resonant frequency).

In EHF examples, thicknesses of the normal axis piezoelectric layer **2005L** through **2005Y** and the reverse axis piezoelectric layer **2007L** through **2007Y** may be selected to determine the main resonant frequency of bulk acoustic Extremely High Frequency (EHF) wave resonator **2000L** through **2000Y** in the Extremely High Frequency (EHF) wave band. Similarly, in EHF examples, layer thicknesses of Extremely High Frequency (EHF) acoustic reflector electrode layers (e.g., layer thicknesses of members of the first pair of bottom metal electrode layers **2022L** through **2022Y** of Extremely High frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., layer thickness of members of the first pair of top metal electrode layers **2024L** through **2024Y** of Extremely High frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y**) may be selected to determine peak acoustic reflectivity of such EHF acoustic reflector electrodes at a frequency, e.g., peak reflectivity resonant frequency, within the Extremely High Frequency (EHF) wave band. The Extremely High Frequency (EHF) wave band may include: 1) peak reflectivity resonant frequency of the Extremely High Frequency (EHF) acoustic reflector electrode layers (e.g., layer thicknesses of members of the first pair of bottom metal electrode layers **2022L** through **2022Y** of the Extremely High frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., layer thickness of members of the first pair of top metal electrode layers **2024A** of the Extremely High frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y**); and 2) the main resonant frequency of bulk acoustic the Extremely High Frequency (EHF) wave resonator **2000L** through **2000Y**.

For example, relatively low acoustic impedance titanium (Ti) metal and relatively high acoustic impedance Molybdenum (Mo) metal may be alternated for members of the first pair of bottom metal electrode layers **2022L** through **2022Y** and for members of the first pair of top metal electrode layers **2024L** through **2024Y**. For example, the bottom acoustic reflector electrode **2013L** through **2013Y** may further include a second pair of bottom metal electrode layers, and the top acoustic reflector electrode **2015A** may further include a second pair of top metal electrode layers, in which members of the second pair of top and bottom metal electrode layers likewise alternate relatively low acoustic impedance titanium (Ti) metal and relatively high acoustic impedance Molybdenum (Mo) metal. Accordingly, the first pair of top metal electrode layers of the top and bottom acoustic reflector electrode **2022L** through **2022Y**, **2024L** through **2024Y** may be different metals from one another having respective acoustic impedances that are different from one another so as to provide a reflective acoustic impedance mismatch at the resonant frequency of the resonator. For example, a first member of the first pair of top metal electrode layers may have an acoustic impedance, and a second member of the first pair of top metal electrodes may have a relatively higher acoustic impedance that is at least about twice (e.g., twice) as high as the acoustic impedance of the first member. For example, a first member of the first pair of top metal electrode layers may have an acoustic impedance, and a second member of the first pair of top metal electrodes may have a relatively higher acoustic impedance that is at least about twice (e.g., twice) as high as the acoustic impedance of the first member.

Thicknesses of member layers of the acoustic reflector electrodes may be related to resonator resonant frequency.

Member layers of the acoustic reflector electrodes may be made thinner as resonators are made to extend to higher resonant frequencies, and as acoustic reflector electrodes are made to extend to higher peak reflectivity resonant frequencies. In accordance with teachings of this disclosure, to compensate for this member layer thinning, number of member layers of the acoustic reflector electrodes may be increased in designs extending to higher resonant frequencies, to facilitate thermal conductivity through acoustic reflector electrodes, and to facilitate electrical conductivity through acoustic reflectivity at higher resonant frequencies. Operation of the example bulk acoustic wave resonators **2000L** through **2000Y** at a resonant Super High Frequency (SHF) or resonant Extremely High Frequency (EHF) may generate heat to be removed from bulk acoustic wave resonators **2000L** through **2000Y** through the acoustic reflector electrodes. The acoustic reflector electrodes (e.g., Super High Frequency (SHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., Super High Frequency (SHF) top acoustic reflector electrode **2015L** through **2015Y**, e.g., Extremely High Frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., Extremely High Frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y**) may have thermal resistance of three thousand degrees Kelvin per Watt or less at the given frequency (e.g., at the resonant frequency of the BAW resonator in the Super High Frequency (SHF) band or the Extremely High Frequency (EHF) band, e.g., at the peak reflectivity resonant frequency of the acoustic reflector electrode in the super high frequency band or the extremely high frequency band). For example, a sufficient number of member layers may be employed to provide for this thermal resistance at the given frequency (e.g., at the resonant frequency of the BAW resonator in the super high frequency band or the extremely high frequency band, e.g., at the peak reflectivity resonant frequency of the acoustic reflector electrode in the Super High Frequency band or the Extremely High Frequency band).

Further, quality factor (Q factor) is a figure of merit for bulk acoustic wave resonators that may be related, in part, to acoustic reflector electrode conductivity. In accordance with the teachings of this disclosure, without an offsetting compensation that increases number of member layers, member layer thinning with increasing frequency may otherwise diminish acoustic reflector electrode conductivity, and may otherwise diminish quality factor (Q factor) of bulk acoustic wave resonators. In accordance with the teachings of this disclosure, number of member layers of the acoustic reflector electrodes may be increased in designs extending to higher resonant frequencies, to facilitate electrical conductivity through acoustic reflector electrodes. The acoustic reflector electrodes (e.g., Super High Frequency (SHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., Super High Frequency (SHF) top acoustic reflector electrode **2015L** through **2015Y**, e.g., Extremely High Frequency (EHF) bottom acoustic reflector electrode **2013L** through **2013Y**, e.g., Extremely High Frequency (EHF) top acoustic reflector electrode **2015L** through **2015Y**) may have sheet resistance of less than one Ohm per square at the given frequency (e.g., at the resonant frequency of the BAW resonator in the Super High Frequency band or the Extremely High Frequency band, e.g., at the peak reflectivity resonant frequency of the acoustic reflector electrode in the Super High Frequency band or the Extremely High Frequency band). For example, a sufficient number of member layers may be employed to provide for this sheet resistance at the given frequency (e.g., at the resonant

frequency of the BAW resonator in the Super High Frequency band or the Extremely High Frequency band, e.g., at the peak reflectivity resonant frequency of the acoustic reflector electrode in the Super High Frequency band or the Extremely High Frequency band). This may, but need not, facilitate enhancing quality factor (Q factor) to a quality factor (Q factor) that may be above a desired one thousand (1000).

Bulk acoustic wave resonator **2000L** through **2000Y** comprising respective pairs of Super or Extremely High Frequency (SHF or EHF) acoustic reflector electrodes **2013L** through **2013Y**, **2015L** through **2015Y**, may include one or more respective patterned layers **2057L** through **2057Y**, **2058N**, **2058Q**, **2058R**, **2058S** **2058T**, **2058U**. For example, in FIG. 2F the patterned layer may be interposed between a first pair of metal electrode layers and one of the layers of piezoelectric material. As shown in example bulk acoustic wave resonator **2000L**, patterned layer **2057L** may be interposed between the first pair of top metal electrode layers **2024L** and one of the layers of piezoelectric material (e.g., top piezoelectric layer **2007L**, e.g., reverse axis piezoelectric layer **2007L**.) As shown in example bulk acoustic wave resonator **2000M**, patterned layer **2057M** may be interposed between the first pair of bottom metal electrode layers **2022M** and one of the layers of piezoelectric material (e.g., bottom piezoelectric layer **2007M**, e.g., normal axis piezoelectric layer **2007M**.) As shown in example bulk acoustic wave resonator **2000N**, a first patterned layer **2057N** (e.g., top patterned layer **2057N**) may be interposed between the first pair of top metal electrode layers **2024N** and a first one of the layers of piezoelectric material (e.g., top piezoelectric layer **2007N**, e.g., reverse axis piezoelectric layer **2007N**). Bulk acoustic wave resonator **2000N** also includes second patterned layer **2058N** (e.g., bottom patterned layer **2058N**), interposed between the first pair of bottom metal electrode layers **2022N** and another one of the layers of piezoelectric material (e.g., bottom piezoelectric layer **2005N**, e.g., normal axis piezoelectric layer **2005N**).

For example, in FIG. 2G a first pair of metal electrode layers may be interposed between a patterned layer and one of the piezoelectric layers. As shown in bulk acoustic resonator **2000O**, a first pair of top metal electrode layers **2034O** may be interposed between patterned layer **2057O** (e.g., top patterned layer **2057O**) and one of the piezoelectric layers (e.g., top piezoelectric layer **2007O**, e.g., reverse axis piezoelectric layer **2007O**). In bulk acoustic resonator **2000O**, patterned layer **2057O** (e.g., top patterned layer **2057O**) may be interposed between the first pair of top metal electrode layers **2024O** and a remainder of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015O** (e.g., top patterned layer **2057O** may be interposed between the first pair of top metal electrode layers **2024O** and a second pair of top electrodes, not shown, of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015O**).

As shown in bulk acoustic resonator **2000P**, a first pair of bottom metal electrode layers **2022P** may be interposed between patterned layer **2057P** (e.g., bottom patterned layer **2057P**) and one of the piezoelectric layers (e.g., bottom piezoelectric layer **2005P**, e.g., normal axis piezoelectric layer **2005P**). In bulk acoustic resonator **2000P**, patterned layer **2057P** (e.g., bottom patterned layer **2057P**) may be interposed between first pair of bottom metal electrode layers **2022P** and a remainder of the Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector **2013P** (e.g., bottom patterned layer **2057P** may be

interposed between first pair of bottom metal electrode layers **2022P** and a second pair of bottom electrodes, not shown, of the Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector **2013P**).

As shown in bulk acoustic resonator **2000Q**, a first pair of top metal electrode layers **2034Q** may be interposed between first patterned layer **2057Q** (e.g., top patterned layer **2057Q**) and one of the piezoelectric layers (e.g., top piezoelectric layer **2007Q**, e.g., reverse axis piezoelectric layer **2007Q**). In bulk acoustic resonator **2000Q**, first patterned layer **2057Q** (e.g., top patterned layer **2057Q**) may be interposed between the first pair of top metal electrode layers **2024Q** and a remainder of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015Q** (e.g., top patterned layer **2057Q** may be interposed between the first pair of top metal electrode layers **2024Q** and a second pair of top electrodes, not shown, of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015Q**). As shown in bulk acoustic resonator **2000Q**, a first pair of bottom metal electrode layers **2022Q** may be interposed between a second patterned layer **2058Q** (e.g., bottom patterned layer **2058Q**) and one of the piezoelectric layers (e.g., bottom piezoelectric layer **2005Q**, e.g., normal axis piezoelectric layer **2005Q**). In bulk acoustic resonator **2000Q**, second patterned layer **2058Q** (e.g., bottom patterned layer **2058Q**) may be interposed between first pair of bottom metal electrode layers **2022Q** and a remainder of the Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector **2013Q** (e.g., bottom patterned layer **2058Q** may be interposed between first pair of bottom metal electrode layers **2022Q** and a second pair of bottom electrodes, not shown, of the Super High Frequency (SHF) or Extremely High Frequency (EHF) bottom acoustic reflector **2013Q**).

FIG. 2H shows example bulk acoustic wave resonators **2000R**, **2000S**, **2000T**, **2000U** in which first and second patterned layers **2057R** through **2057U**, **2058R** through **2058U** (e.g., top patterned layer **2057R** through **2057U** and bottom patterned layer **2058R** through **2058U**) contribute substantially differently to facilitating spurious mode suppression in the bulk acoustic wave resonators **2000R**, **2000S**, **2000T**, **2000U**. In accordance with the teachings of this disclosure, one of the patterned layers may be arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material than another one of the patterned layers, to contribute more to facilitating spurious mode suppression than what the another one of the patterned layers contributes. In the example bulk acoustic wave resonators **2000R**, **2000S**, **2000T**, **2000U** shown in FIG. 2H, one of the top and bottom patterned layers is arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material than another one of the top and bottom patterned layers, to contribute more to facilitating spurious mode suppression than what the another one of the top and bottom patterned layers contributes. Having relatively more contribution to facilitating spurious mode suppression, the one the patterned layers that is relatively nearer to piezoelectric layer stack including the first and second layers of piezoelectric material may provide a coarse facilitating of spurious mode suppression. Having relatively less contribution to facilitating spurious mode suppression, the one the patterned layers that is relatively further from the piezoelectric layer stack including the first and second layers of piezoelectric material may provide a fine facilitating of spurious mode suppression.

For example, in the bulk acoustic wave resonator **2000R**, a first pair of top metal electrodes **2024R** may be interposed between a first patterned layer **2057R** (e.g., patterned layer **2057R**) and a first piezoelectric layer **2007R** (e.g., top piezoelectric layer **2007R**, e.g., reverse axis piezoelectric layer **2007R**). A second patterned layer **2058R** (e.g., bottom patterned layer **2058R**) may be interposed between the first pair of bottom metal electrode layers **2022R** and a second piezoelectric layer **2005R** (e.g., bottom piezoelectric layer **2005R**, e.g., normal axis piezoelectric layer **2005R**). In the example bulk acoustic wave resonator **2000R**, bottom patterned layer **2058R** is arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material **2005R**, **2007R** than the top patterned layer **2057R**, to contribute more to facilitating spurious mode suppression than what the top patterned layer contributes. First and second patterned layers **2057R**, **2058R** (e.g., top patterned layer **2057R** and bottom patterned layer **2058R**) contribute substantially differently to facilitating spurious mode suppression in the bulk acoustic wave resonator **2000R**.

In the bulk acoustic wave resonator **2000S**, the first pair of top metal electrode layers **2024S** comprises first and second members **2025S**, **2027S** of the first pair of top metal electrode layers **2024S**. First patterned layer **2057S** (e.g., top patterned layer **2057S**) may be interposed between first and second members **2025S**, **2027S** of the first pair of top metal electrode layers **2024S**. A second patterned layer **2058S** (e.g., bottom patterned layer **2058S**) may be interposed between the first pair of bottom metal electrode layers **2022S** and a second piezoelectric layer **2005S** (e.g., bottom piezoelectric layer **2005S**, e.g., normal axis piezoelectric layer **2005S**). In the example bulk acoustic wave resonator **2000S**, bottom patterned layer **2058R** is arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material **2005S**, **2007S** than the top patterned layer **2057S**, to contribute more to facilitating spurious mode suppression than what the top patterned layer contributes. First and second patterned layers **2057S**, **2058S** (e.g., top patterned layer **2057S** and bottom patterned layer **2058S**) contribute substantially differently to facilitating spurious mode suppression in the bulk acoustic wave resonator **2000S**.

In the bulk acoustic wave resonator **2000U**, the first pair of top metal electrode layers **2024U** comprises first and second members **2025U**, **2027U** of the first pair of top metal electrode layers **2024U**. First patterned layer **2057U** (e.g., top patterned layer **2057U**) may be interposed between first and second members **2025U**, **2027U** of the first pair of top metal electrode layers **2024U**. A first pair of bottom metal electrodes **2022U** may be interposed between a second patterned layer **2058U** (e.g., bottom patterned layer **2058U**) and a piezoelectric layer **2005U** (e.g., bottom piezoelectric layer **2005U**, e.g., normal axis piezoelectric layer **2005U**). In the example bulk acoustic wave resonator **2000U**, top patterned layer **2057U** is arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material **2005U**, **2007U** than the bottom patterned layer **2058U**, to contribute more to facilitating spurious mode suppression than what the bottom patterned layer contributes. First and second patterned layers **2057U**, **2058U** (e.g., top patterned layer **2057U** and bottom patterned layer **2058U**) contribute substantially differently to facilitating spurious mode suppression in the bulk acoustic wave resonator **2000U**.

For example, in FIG. 2I a first pair of metal electrode layers may be interposed between a patterned layer and one

of the piezoelectric layers. As shown in bulk acoustic resonators **2000V** through **2000Y**, a first pair of top metal electrode layers **2034V** through **2034Y** may be interposed between patterned layer **2057V** through **2057Y** (e.g., top patterned layer **2057V** through **2057Y**) and one of the piezoelectric layers (e.g., top piezoelectric layer **2007V** through **2007Y**, e.g., reverse axis piezoelectric layer **2007V** through **2007Y**). In bulk acoustic resonators **2000V** through **2000Y**, patterned layer **2057V** through **2057Y** (e.g., top patterned layer **2057V** through **2057Y**) may be interposed between the first pair of top metal electrode layers **2024V** through **2024Y** and a remainder of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015V** through **2015Y** (e.g., top patterned layer **2057V** through **2057Y** may be interposed between the first pair of top metal electrode layers **2024V** through **2024Y** and a second pair of top electrodes, not shown, of the Super High Frequency (SHF) or Extremely High Frequency (EHF) top acoustic reflector **2015V** through **2015Y**).

In addition to facilitating spurious mode suppression, differing patterns of patterned layers **2057V** through **2057Y** may have differing influence on quality factor (e.g., to facilitate enhanced average quality factor below a series resonant frequency of the bulk acoustic wave resonator, e.g., to facilitate enhanced average quality factor above the series resonant frequency of the bulk acoustic wave resonator, e.g., to facilitate balancing enhanced average quality factor below a series resonant frequency of the bulk acoustic wave resonator and facilitate enhanced average quality factor above the series resonant frequency of the bulk acoustic wave resonator).

For example, patterned layer **2057V** may comprise gaps or thinned regions adjacent to a lateral perimeter of top acoustic reflector electrode **2015V** of bulk acoustic wave resonator **2000V**. Design of width of gaps or thinned regions may be determined by simulation. These perimeter gaps in patterned layer **2057** may facilitate enhanced average quality factor below the series resonant frequency of the bulk acoustic wave resonator.

For example patterned layer **2057W** may comprise gaps or thinned regions spaced away from a lateral perimeter of top acoustic reflector electrode **2015W** of bulk acoustic wave resonator **2000W**. Design of width of gaps or thinned regions may be determined by simulation. Further, a central region of patterned layer **2057W** may have reduced thickness (e.g., less than 500 Å, e.g., less than thickness of extremities of patterned layer **2057W**). Design of reduced thickness of central region of patterned layer **2057W** may be determined by simulation. Thickness of extremities of patterned layer **2057W** may be e.g., less than 500 Å, e.g., greater than the reduced thickness of the central region of patterned layer **2057W**). Design of thickness of extremities of patterned layer **2057W** may be determined by simulation. Pattern features (e.g., gaps as just described, e.g., reduced thickness of central region as just described, e.g., extremities as just described) of patterned layer **2057W** may facilitate balancing enhanced average quality factor below a series resonant frequency of the bulk acoustic wave resonator and facilitate enhanced average quality factor above the series resonant frequency of the bulk acoustic wave resonator.

For example patterned layer **2057X** may comprise gaps or thinned regions adjacent to a lateral perimeter of top acoustic reflector electrode **2015X** of bulk acoustic wave resonator **2000X**. Design of width of gaps or thinned regions may be determined by simulation. Further, a central region of patterned layer **2057X** may have reduced thickness (e.g., less than 300 Å, e.g., less than thickness of extremities of

patterned layer **2057X** that are spaced away from the lateral perimeter of top acoustic reflector electrode **2015X** of bulk acoustic wave resonator **2000X**. Design of reduced thickness of central region of patterned layer **2057X** may be determined by simulation. Thickness of extremities of patterned layer **2057X** may be e.g., less than 500 Å, e.g., greater than the reduced thickness of the central region of patterned layer **2057X**). Design of thickness of extremities of patterned layer **2057X** may be determined by simulation. Pattern features (e.g., gaps as just described, e.g., reduced thickness of central region as just described, e.g., extremities as just described) of patterned layer **2057X** may facilitate balancing enhanced average quality factor below a series resonant frequency of the bulk acoustic wave resonator and facilitate enhanced average quality factor above the series resonant frequency of the bulk acoustic wave resonator.

For example periodic (or semi-periodic, or aperiodic) patterned layer **2057Y** may comprise periodic (or semi-periodic, or aperiodic) gaps or periodic (or semi-periodic, or aperiodic) thinned regions extending between extremities of a lateral perimeter of top acoustic reflector electrode **2015Y** of bulk acoustic wave resonator **2000Y**. Design of width of periodic (or semi-periodic) gaps or periodic (or semi-periodic, or aperiodic) thinned regions may be determined by simulation. Further, a central region of patterned layer **2057Y** may have a gap or thinned region of periodic (or semi-periodic, or aperiodic) patterned layer **2057Y** that is spaced away from extremities of the lateral perimeter of top acoustic reflector electrode **2015Y** of bulk acoustic wave resonator **2000Y**. Design of the gap or thinned region of central region of periodic (or semi-periodic, or aperiodic) patterned layer **2057Y** may be determined by simulation. Thickness of extremities of patterned layer **2057Y** may be e.g., less than 500 Å. Design of thickness of extremities of patterned layer **2057Y** may be determined by simulation. Periodic (or semi-periodic, or aperiodic) pattern features (e.g., gaps as just described, e.g., reduced thickness as just described, e.g., extremities as just described) of periodic (or semi-periodic, or aperiodic) patterned layer **2057Y** may facilitate balancing enhanced average quality factor below a series resonant frequency of the bulk acoustic wave resonator and facilitate enhanced average quality factor above the series resonant frequency of the bulk acoustic wave resonator. In addition, patterned layers **2057V** through **2057Y** (e.g., top patterned layer **2057V** through **2057Y**) may be placed at different locations and in different combinations as discussed in relation to FIGS. 2F, 2G and 2H and as will be appreciated by one skilled in the art.

FIG. 2J shows two diagrams for different patterned layer mass load materials (e.g., Tungsten (W), e.g., Titanium (Ti)) and different patterned layer electrode placement (e.g., patterned layer mass load of initial electrode, e.g., patterned layer mass load of first pair of electrodes, e.g., patterned layer mass loaded of second pair of electrodes) shown with bulk acoustic wave resonator patterned layer mass load sensitivity in MegaHertz per Angstrom (MHz/Å) units versus number of alternating axis half wavelength thickness piezoelectric layers (e.g., 1 through 6 piezoelectric layers) as predicted by simulation. In accordance with the teachings of this disclosure, it is theorized that that patterned mass load layer sensitivity (e.g., patterned mass load layer potency in facilitating spurious mode suppression) as expressed in MegaHertz per Angstrom (MHz/Å) units may increase with a square of resonant frequency of the bulk acoustic wave resonator. For example, the twenty-four Gigahertz (24 GHz) single piezoelectric layer resonator **2001A** may have a patterned layer mass load sensitivity that may be one hun-

dred times higher (100× higher) than a more modest 2.4 GHz scaled version of the single piezoelectric layer resonator **2001A**.

It is also the teaching of this disclosure that patterned layer mass load sensitivity may decrease with the square of the distance of location of the mass load of the patterned layer from the stack of piezoelectric layers. Further, it is the teaching of this disclosure that the very high patterned layer mass load sensitivity (e.g., relative 100× higher sensitivity) may be mitigated by placement of the patterned layer further from the stack of piezoelectric layers. In other words, for an example resonator near the high end of the Super High Frequency (SHF) band (e.g., the example twenty four Gigahertz (24 GHz) resonator), very high patterned layer mass load sensitivity (e.g., relative 100× higher sensitivity than a 2 GHz resonator) may be mitigated by placement of the patterned layer to acoustically modify the adjacent first pair of electrodes of the acoustic reflector (e.g., acoustically modify the adjacent member of the first pair of electrodes that is further from the piezoelectric layer stack than the other member of the first pair). For example, with reference to FIG. 1A, patterned layer **157** is placed above second member **139** that is further from the piezoelectric stack **104** than the first member **137** of the first pair of electrodes **137**, **139**. Correspondingly, a first diagram **2019V** shows a trace **2021V** for an example Tungsten (W) patterned mass load layer, as predicted by simulation for such patterned layer mass loaded second member of the first pair of electrodes, providing what may be a desired patterned layer mass load sensitivity extending from about 1 MegaHertz per Angstrom (MHz/Å) to about two tenths (0.02) MegaHertz per Angstrom (MHz/Å) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in first diagram **2019V**).

In contrast, first diagram **2019V** shows a trace **2023V** for an example Tungsten (W) patterned mass load layer, as predicted by simulation for an alternate design, in which the patterned layer is placed substantially nearer to the piezoelectric stack to acoustically modify the adjacent initial electrode layer (e.g., the initial electrode layer **135** shown in FIG. 1A). Trace **2023V** shows what may be excess patterned layer mass load sensitivity extending from about fifty (50) MegaHertz per Angstrom (MHz/Å) to about eight (8) MegaHertz per Angstrom (MHz/Å) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in first diagram **2019V**).

However, moving the patterned layer too far from the piezoelectric stack may result in insufficient patterned layer mass load sensitivity. First diagram **2019V** shows a trace **2025V** for an example Tungsten (W) patterned layer, as predicted by simulation for another alternate design, in which the patterned layer is placed substantially further from the piezoelectric stack to acoustically modify the adjacent second pair of electrode layers (e.g., the second member **143** electrode layer of the second pair or electrodes **141**, **143** shown in FIG. 1A). Trace **2025V** shows what may be insufficient patterned layer mass load sensitivity extending from about two tenths (0.2) MegaHertz per Angstrom (MHz/Å) to about 0.006 MegaHertz per Angstrom (MHz/Å) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in first diagram **2019V**). For example, with six piezoelectric layers, placement of the mass loaded layer to acoustically modify the adjacent second member of the second pair of electrode may provide insufficient patterned layer mass load sensitivity.

It is the teaching of this disclosure that patterned layer mass load sensitivity may decrease with increasing number of alternating axis half wavelength thickness piezoelectric layers. It is theorized that this may be because of increasing fraction of acoustic energy confined in the stack of piezoelectric layers and corresponding decreasing fraction of acoustic energy in the electrode layers of the acoustic reflector. As shown in the first diagram **2019V**, in all three traces **2021V**, **2023V**, **2025V** simulation predicts patterned layer mass load sensitivity decreasing with increasing number of alternating axis half wavelength thickness piezoelectric layers. Accordingly, it is the teaching of this disclosure that number of piezoelectric layers may be a consideration in selecting patterned layer location for desired patterned layer mass load sensitivity. For the example resonator near the high end of the Super High Frequency (SHF) band (e.g., the example twenty four Gigahertz (24 GHz) resonator): designs having one or two piezoelectric layers may achieve desired patterned layer mass load sensitivity by placement of the patterned layer to acoustically modify the adjacent second member of the second pair of electrode layers; and designs having three to six piezoelectric layers may achieve desired patterned layer mass load sensitivity by placement of the mass load to acoustically modify the adjacent second member of the first pair of electrode layers.

Titanium (Ti) has lower mass than Tungsten (W). Using Titanium (Ti) instead of Tungsten (W) in the patterned layer may help to mitigate increasing patterned layer mass load sensitivity with increasing frequency (e.g., for the example twenty-four Gigahertz (24 GHz) resonator). A second diagram **2019W** shows a trace **2021W** for an example Titanium (Ti) patterned layer, as predicted by simulation for the patterned mass loaded second member of the first pair of electrodes, providing what may be a desired patterned mass load sensitivity extending from about three tenths (0.03) MegaHertz per Angstrom (MHz/A) to about seven hundredths (0.07) MegaHertz per Angstrom (MHz/A) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in second diagram **2019W**). For Titanium patterned mass load sensitivity as predicted by simulation for the patterned mass loaded initial electrode, trace **2023W** shows what may be excessive patterned mass load sensitivity, extending from about nine (9) MegaHertz per Angstrom (MHz/A) to about two (2) MegaHertz per Angstrom (MHz/A) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in first diagram **2019W**). For Titanium patterned mass load sensitivity as predicted by simulation for the patterned mass loaded second member of the second pair of electrodes, trace **2025W** shows what may be insufficient patterned mass load sensitivity, extending from about three hundredths (0.03) MegaHertz per Angstrom (MHz/A) to about five thousandths (0.005) MegaHertz per Angstrom (MHz/A) depending on number of alternating axis half wavelength thickness piezoelectric layers (e.g., from one to six layers in first diagram **2019W**).

FIGS. 3A through 3E illustrate example integrated circuit structures used to form the example bulk acoustic wave resonator structure of FIG. 1A. As shown in FIG. 3A, magnetron sputtering may sequentially deposit layers on silicon substrate **101**. Initially, a seed layer **103** of suitable material (e.g., aluminum nitride (AlN), e.g., silicon dioxide (SiO₂), e.g., aluminum oxide (Al₂O₃), e.g., silicon nitride (Si₃N₄), e.g., amorphous silicon (a-Si), e.g., silicon carbide (SiC)) may be deposited, for example, by sputtering from a respective target (e.g., from an aluminum, silicon, or silicon

carbide target). The seed layer may have a layer thickness in a range from approximately one hundred Angstroms (100 Å) to approximately one micron (1 μm). Next, successive pairs of alternating layers of high acoustic impedance metal and low acoustic impedance metal may be deposited by alternating sputtering from targets of high acoustic impedance metal and low acoustic impedance metal. For example, sputtering targets of high acoustic impedance metal such as Molybdenum or Tungsten may be used for sputtering the high acoustic impedance metal layers, and sputtering targets of low acoustic impedance metal such as Aluminum or Titanium may be used for sputtering the low acoustic impedance metal layers. For example, the fourth pair of bottom metal electrode layers, **133**, **131**, may be deposited by sputtering the high acoustic impedance metal for a first bottom metal electrode layer **133** of the pair on the seed layer **103**, and then sputtering the low acoustic impedance metal for a second bottom metal electrode layer **131** of the pair on the first layer **133** of the pair. Similarly, the third pair of bottom metal electrode layers, **129**, **127**, may then be deposited by sequentially sputtering from the high acoustic impedance metal target and the low acoustic impedance metal target. Similarly, the second pair of bottom metal electrodes **125**, **123**, may then be deposited by sequentially sputtering from the high acoustic impedance metal target and the low acoustic impedance metal target. Similarly, the first pair of bottom metal electrodes **121**, **119**, may then be deposited by sequentially sputtering from the high acoustic impedance metal target and the low acoustic impedance metal target. Respective layer thicknesses of bottom metal electrode layers of the first, second, third and fourth pairs **119**, **121**, **123**, **125**, **127**, **129**, **131**, **133** may correspond to approximately a quarter wavelength (e.g., a quarter of an acoustic wavelength) of the resonant frequency at the resonator (e.g., respective layer thickness of about six hundred Angstroms (660 Å) for the example 24 GHz resonator.) Initial bottom electrode layer **119** may then be deposited by sputtering from the high acoustic impedance metal target. Thickness of the initial bottom electrode layer may be, for example, about an eighth wavelength (e.g., an eighth of an acoustic wavelength) of the resonant frequency of the resonator (e.g., layer thickness of about three hundred Angstroms (300 Å) for the example 24 GHz resonator.)

A stack of four layers of piezoelectric material, for example, four layers of Aluminum Nitride (AlN) having the wurzite structure may be deposited by sputtering. For example, bottom piezoelectric layer **105**, first middle piezoelectric layer **107**, second middle piezoelectric layer **109**, and top piezoelectric layer **111** may be deposited by sputtering. The four layers of piezoelectric material in the stack **104**, may have the alternating axis arrangement in the respective stack **104**. For example the bottom piezoelectric layer **105** may be sputter deposited to have the normal axis orientation, which is depicted in FIG. 3A using the downward directed arrow. The first middle piezoelectric layer **107** may be sputter deposited to have the reverse axis orientation, which is depicted in the FIG. 3A using the upward directed arrow. The second middle piezoelectric layer **109** may have the normal axis orientation, which is depicted in the FIG. 3A using the downward directed arrow. The top piezoelectric layer may have the reverse axis orientation, which is depicted in the FIG. 3A using the upward directed arrow. As mentioned previously herein, polycrystalline thin film AlN may be grown in the crystallographic c-axis negative polarization, or normal axis orientation perpendicular relative to the substrate surface using reactive magnetron sputtering of the Aluminum target in the nitrogen atmosphere. As was

discussed in greater detail previously herein, changing sputtering conditions, for example by adding oxygen, may reverse the axis to a crystallographic c-axis positive polarization, or reverse axis, orientation perpendicular relative to the substrate surface.

Interposer layers may be sputtered between sputtering of piezoelectric layers, so as to be sandwiched between piezoelectric layers of the stack. For example, first interposer layer **159**, may sputtered between sputtering of bottom piezoelectric layer **105**, and the first middle piezoelectric layer **107**, so as to be sandwiched between the bottom piezoelectric layer **105**, and the first middle piezoelectric layer **107**. For example, second interposer layer **161** may be sputtered between sputtering first middle piezoelectric layer **107** and the second middle piezoelectric layer **109** so as to be sandwiched between the first middle piezoelectric layer **107**, and the second middle piezoelectric layer **109**. For example, third interposer layer **163**, may be sputtered between sputtering of second middle piezoelectric layer **109** and the top piezoelectric layer **111** so as to be sandwiched between the second middle piezoelectric layer **109** and the top piezoelectric layer **111**.

As discussed previously, one or more of the interposer layers may be metal interposer layers, e.g., high acoustic impedance metal interposer layers, e.g., Molybdenum metal interposer layers. These may be deposited by sputtering from a metal target. As discussed previously, one or more of the interposer layers may be dielectric interposer layers, e.g., silicon dioxide interposer layers. These may be deposited by reactive sputtering from a Silicon target in an oxygen atmosphere. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may comprise metal and dielectric for respective interposer layers. Sputtering thickness of interposer layers may be as discussed previously herein.

Initial top electrode layer **135** may be deposited on the top piezoelectric layer **111** by sputtering from the high acoustic impedance metal target. Thickness of the initial top electrode layer may be, for example, about an eighth wavelength (e.g., an eighth of an acoustic wavelength) of the resonant frequency of the resonator (e.g., layer thickness of about three hundred Angstroms (300 Å) for the example 24 GHz resonator.) The first pair of top metal electrode layers, **137**, **139**, may then be deposited by sputtering the low acoustic impedance metal for a first top metal electrode layer **137** of the pair, and then sputtering the high acoustic impedance metal for a second top metal electrode layer **139** of the pair on the first layer **137** of the pair. Layer thicknesses of top metal electrode layers of the first pair **137**, **139** may correspond to approximately a quarter wavelength (e.g., a quarter acoustic wavelength) of the resonant frequency of the resonator (e.g., respective layer thickness of about six hundred Angstroms (600 Å) for the example 24 GHz resonator.) The mass load layer **155** may be sputtered from a high acoustic impedance metal target onto the second top metal electrode layer **139** of the pair. Thickness of the mass load layer may be as discussed previously herein. As mentioned previously, the mass load layer **155** may be an additional mass layer to increase electrode layer mass, so as to facilitate the preselected frequency compensation down in frequency (e.g., compensate to decrease resonant frequency). Alternatively, the mass load layer **155** may be a mass load reduction layer, e.g., ion milled mass load reduction layer **155**, to decrease electrode layer mass, so as to facilitate the preselected frequency compensation up in frequency (e.g., compensate to increase resonant frequency). Accordingly, in such case, in FIG. 3A mass load reduction layer **155** may representa-

tively illustrate, for example, an ion milled region of the second member **139** of the first pair of electrodes **137**, **139** (e.g., ion milled region of high acoustic impedance metal electrode **139**).

The plurality of lateral features **157** (e.g., patterned layer **157**) may be formed by sputtering a layer of additional mass loading having a layer thickness as discussed previously herein. The plurality of lateral features **157** (e.g., patterned layer **157**) may be made by patterning the layer of additional mass loading after it is deposited by sputtering. The patterning may done by photolithographic masking, layer etching, and mask removal. Initial sputtering may be sputtering of a metal layer of additional mass loading from a metal target (e.g., a target of Tungsten (W), Molybdenum (Mo), Titanium (Ti) or Aluminum (Al)). In alternative examples, the plurality of lateral features **157** may be made of a patterned dielectric layer (e.g., a patterned layer of Silicon Nitride (SiN), Silicon Dioxide (SiO₂) or Silicon Carbide (SiC)). For example Silicon Nitride, and Silicon Dioxide may be deposited by reactive magnetron sputtering from a silicon target in an appropriate atmosphere, for example Nitrogen, Oxygen or Carbon Dioxide. Silicon Carbide may be sputtered from a Silicon Carbide target.

Once the plurality of lateral features **157** have been patterned (e.g., patterned layer **157**) as shown in FIG. 3A, sputter deposition of successive additional pairs of alternating layers of high acoustic impedance metal and low acoustic impedance metal may continue as shown in FIG. 3B by alternating sputtering from targets of high acoustic impedance metal and low acoustic impedance metal. For example, sputtering targets of high acoustic impedance metal such as Molybdenum or Tungsten may be used for sputtering the high acoustic impedance metal layers, and sputtering targets of low acoustic impedance metal such as Aluminum or Titanium may be used for sputtering the low acoustic impedance metal layers. For example, the second pair of top metal electrode layers, **141**, **143**, may be deposited by sputtering the low acoustic impedance metal for a first bottom metal electrode layer **141** of the pair on the plurality of lateral features **157**, and then sputtering the high acoustic impedance metal for a second top metal electrode layer **143** of the pair on the first layer **141** of the pair. Similarly, the third pair of top metal electrode layers, **145**, **147**, may then be deposited by sequentially sputtering from the low acoustic impedance metal target and the high acoustic impedance metal target. Similarly, the fourth pair of top metal electrodes **149**, **151**, may then be deposited by sequentially sputtering from the low acoustic impedance metal target and the high acoustic impedance metal target. Respective layer thicknesses of top metal electrode layers of the first, second, third and fourth pairs **137**, **139**, **141**, **143**, **145**, **147**, **149**, **151** may correspond to approximately a quarter wavelength (e.g., a quarter acoustic wavelength) at the resonant frequency of the resonator (e.g., respective layer thickness of about six hundred Angstroms (600 Å) for the example 24 GHz resonator.)

As mentioned previously, and as shown in FIG. 3B, after the lateral features **157** are formed (e.g., patterned layer **157**), they may function as a step feature template, so that subsequent top metal electrode layers formed on top of the lateral features **157** may retain step patterns imposed by step features of the lateral features **157**. For example, the second pair of top metal electrode layers **141**, **143**, the third pair of top metal electrode layers **145**, **147**, and the fourth pair of top metal electrodes **149**, **151**, may retain step patterns imposed by step features of the lateral features **157**.

After depositing layers of the fourth pair of top metal electrodes **149**, **151** as shown in FIG. 3B, suitable photolithographic masking and etching may be used to form a first portion of etched edge region **153C** for the top acoustic reflector **115** as shown in FIG. 3C. A notional heavy dashed line is used in FIG. 3C depicting the first portion of etched edge region **153C** associated with the top acoustic reflector **115**. The first portion of etched edge region **153C** may extend along the thickness dimension **T25** of the top acoustic reflector **115**. The first portion etched edge region **153C** may extend through (e.g., entirely through or partially through) the top acoustic reflector **115**. The first portion of the etched edge region **153C** may extend through (e.g., entirely through or partially through) the initial top metal electrode layer **135**. The first portion of the etched edge region **153C** may extend through (e.g., entirely through or partially through) the first pair of top metal electrode layers **137**, **139**. The first portion of the etched edge region **153C** may extend through (e.g., entirely through or partially through) the mass load layer **155**. The first portion of the etched edge region **153C** may extend through (e.g., entirely through or partially through) at least one of the lateral features **157** (e.g., through patterned layer **157**). The first portion of etched edge region **153C** may extend through (e.g., entirely through or partially through) the second pair of top metal electrode layers, **141**, **143**. The first portion etched edge region **153C** may extend through (e.g., entirely through or partially through) the third pair of top metal electrode layers, **145**, **147**. The first portion of etched edge region **153C** may extend through (e.g., entirely through or partially through) the fourth pair of top metal electrode layers, **149**, **151**. Just as suitable photolithographic masking and etching may be used to form the first portion of etched edge region **153C** at a lateral extremity the top acoustic reflector **115** as shown in FIG. 3C, such suitable photolithographic masking and etching may likewise be used to form another first portion of a laterally opposing etched edge region **154C** at an opposing lateral extremity the top acoustic reflector **115**, e.g., arranged laterally opposing or opposite from the first portion of etched edge region **153C**, as shown in FIG. 3C. The another first portion of the laterally opposing etched edge region **154C** may extend through (e.g., entirely through or partially through) the opposing lateral extremity of the top acoustic reflector **115**, e.g., arranged laterally opposing or opposite from the first portion of etched edge region **153C**, as shown in FIG. 3C. The mesa structure (e.g., third mesa structure) corresponding to the top acoustic reflector **115** may extend laterally between (e.g., may be formed between) etched edge region **153C** and laterally opposing etched edge region **154C**. Dry etching may be used, e.g., reactive ion etching may be used to etch the materials of the top acoustic reflector. Chlorine based reactive ion etch may be used to etch Aluminum, in cases where Aluminum is used in the top acoustic reflector. Fluorine based reactive ion etch may be used to etch Tungsten (W), Molybdenum (Mo), Titanium (Ti), Silicon Nitride (SiN), Silicon Dioxide (SiO₂) and/or Silicon Carbide (SiC) in cases where these materials are used in the top acoustic reflector.

After etching to form the first portion of etched edge region **153C** for top acoustic reflector **115** as shown in FIG. 3C, additional suitable photolithographic masking and etching may be used to form elongated portion of etched edge region **153D** for top acoustic reflector **115** and for the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111** as shown in FIG. 3D. A notional heavy dashed line is used in FIG. 3D depicting the elongated portion of etched edge region **153D** associated with the stack **104** of four piezoelectric layers

105, **107**, **109**, **111** and with the top acoustic reflector **115**. Accordingly, the elongated portion of etched edge region **153D** shown in FIG. 3D may extend through (e.g., entirely through or partially through) the fourth pair of top metal electrode layers, **149**, **151**, the third pair of top metal electrode layers, **145**, **147**, the second pair of top metal electrode layers, **141**, **143**, at least one of the lateral features **157** (e.g., patterned layer **157**), the mass load layer **155**, the first pair of top metal electrode layers **137**, **139** and the initial top metal electrode layer **135** of the top acoustic reflector **115**. The elongated portion of etched edge region **153D** may extend through (e.g., entirely through or partially through) the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111**. The elongated portion of etched edge region **153D** may extend through (e.g., entirely through or partially through) the first piezoelectric layer, **105**, e.g., having the normal axis orientation, first interposer layer **159**, first middle piezoelectric layer, **107**, e.g., having the reverse axis orientation, second interposer layer **161**, second middle interposer layer, **109**, e.g., having the normal axis orientation, third interposer layer **163**, and top piezoelectric layer **111**, e.g., having the reverse axis orientation. The elongated portion of etched edge region **153D** may extend along the thickness dimension **T25** of the top acoustic reflector **115**. The elongated portion of etched edge region **153D** may extend along the thickness dimension **T27** of the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111**. Just as suitable photolithographic masking and etching may be used to form the elongated portion of etched edge region **153D** at the lateral extremity the top acoustic reflector **115** and at a lateral extremity of the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111** as shown in FIG. 3D, such suitable photolithographic masking and etching may likewise be used to form another elongated portion of the laterally opposing etched edge region **154D** at the opposing lateral extremity the top acoustic reflector **115** and the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111**, e.g., arranged laterally opposing or opposite from the elongated portion of etched edge region **153D**, as shown in FIG. 3D. The another elongated portion of the laterally opposing etched edge region **154D** may extend through (e.g., entirely through or partially through) the opposing lateral extremity of the top acoustic reflector **115** and the stack of four piezoelectric layers **105**, **107**, **109**, **111**, e.g., arranged laterally opposing or opposite from the elongated portion of etched edge region **153D**, as shown in FIG. 3D. The mesa structure (e.g., third mesa structure) corresponding to the top acoustic reflector **115** may extend laterally between (e.g., may be formed between) etched edge region **153D** and laterally opposing etched edge region **154D**. The mesa structure (e.g., first mesa structure) corresponding to stack **104** of the example four piezoelectric layers may extend laterally between (e.g., may be formed between) etched edge region **153D** and laterally opposing etched edge region **154D**. Dry etching may be used, e.g., reactive ion etching may be used to etch the materials of the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111** and any interposer layers. For example, Chlorine based reactive ion etch may be used to etch Aluminum Nitride piezoelectric layers. For example, Fluorine based reactive ion etch may be used to etch Tungsten (W), Molybdenum (Mo), Titanium (Ti), Silicon Nitride (SiN), Silicon Dioxide (SiO₂) and/or Silicon Carbide (SiC) in cases where these materials are used in interposer layers.

After etching to form the elongated portion of etched edge region **153D** for top acoustic reflector **115** and the stack **104** of four piezoelectric layers **105**, **107**, **109**, **111** as shown in FIG. 3D, further additional suitable photolithographic mask-

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ing and etching may be used to form etched edge region 153D for top acoustic reflector 115 and for the stack 104 of four piezoelectric layers 105, 107, 109, 111 and for bottom acoustic reflector 113 as shown in FIG. 3E. The notional heavy dashed line is used in FIG. 3E depicting the etched edge region 153 associated with the stack 104 of four piezoelectric layers 105, 107, 109, 111 and with the top acoustic reflector 115 and with the bottom acoustic reflector 113. The etched edge region 153 may extend along the thickness dimension T25 of the top acoustic reflector 115. The etched edge region 153 may extend along the thickness dimension T27 of the stack 104 of four piezoelectric layers 105, 107, 109, 111. The etched edge region 153 may extend along the thickness dimension T23 of the bottom acoustic reflector 113. Just as suitable photolithographic masking and etching may be used to form the etched edge region 153 at the lateral extremity the top acoustic reflector 115 and at the lateral extremity of the stack 104 of four piezoelectric layers 105, 107, 109, 111 and at a lateral extremity of the bottom acoustic reflector 113 as shown in FIG. 3E, such suitable photolithographic masking and etching may likewise be used to form another laterally opposing etched edge region 154 at the opposing lateral extremity of the top acoustic reflector 115 and the stack 104 of four piezoelectric layers 105, 107, 109, 111, and the bottom acoustic reflector 113, e.g., arranged laterally opposing or opposite from the etched edge region 153, as shown in FIG. 3E. The laterally opposing etched edge region 154 may extend through (e.g., entirely through or partially through) the opposing lateral extremity of the top acoustic reflector 115 and the stack of four piezoelectric layers 105, 107, 109, 111, and the bottom acoustic reflector 113 e.g., arranged laterally opposing or opposite from the etched edge region 153, as shown in FIG. 3E.

After the foregoing etching to form the etched edge region 153 and the laterally opposing etched edge region 154 of the resonator 100 shown in FIG. 3E, a planarization layer 165 may be deposited. A suitable planarization material (e.g., Silicon Dioxide (SiO₂), Hafnium Dioxide (HfO₂), Polyimide, or BenzoCyclobutene (BCB)). These materials may be deposited by suitable methods, for example, chemical vapor deposition, standard or reactive magnetron sputtering (e.g., in cases of SiO₂ or HfO₂) or spin coating (e.g., in cases of Polyimide or BenzoCyclobutene (BCB)). An isolation layer 167 may also be deposited over the planarization layer 165. A suitable low dielectric constant (low-k), low acoustic impedance (low-Z_a) material may be used for the isolation layer 167, for example polyimide, or BenzoCyclobutene (BCB). These materials may be deposited by suitable methods, for example, chemical vapor deposition, standard or reactive magnetron sputtering or spin coating. After planarization layer 165 and the isolation layer 167 have been deposited, additional procedures of photolithographic masking, layer etching, and mask removal may be done to form a pair of etched acceptance locations 183A, 183B for electrical interconnections. Reactive ion etching or inductively coupled plasma etching with a gas mixture of argon, oxygen and a fluorine containing gas such as tetrafluoromethane (CF₄) or Sulfur hexafluoride (SF₆) may be used to etch through the isolation layer 167 and the planarization layer 165 to form the pair of etched acceptance locations 183A, 183B for electrical interconnections. Photolithographic masking, sputter deposition, and mask removal may then be used form electrical interconnects in the pair of etched acceptance locations 183A, 183B shown in FIG. 3E, so as to provide for the bottom electrical interconnect 169 and top electrical interconnect 171 that are shown explicitly

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in FIG. 1A. A suitable material, for example Gold (Au) may be used for the bottom electrical interconnect 169 and top electrical interconnect 171.

FIGS. 4A through 4G show alternative example bulk acoustic wave resonators 400A through 400G to the example bulk acoustic wave resonator 100A shown in FIG. 1A. For example, the bulk acoustic wave resonator 400A, 400E shown in FIG. 4A, 4E may have a cavity 483A, 483E, e.g., an air cavity 483A, 483E, e.g., extending into substrate 401A, 401E, e.g., extending into silicon substrate 401A, 401E, e.g., arranged below bottom acoustic reflector 413A, 413E. The cavity 483A, 483E may be formed using techniques known to those with ordinary skill in the art. For example, the cavity 483A, 483E may be formed by initial photolithographic masking and etching of the substrate 401A, 401E (e.g., silicon substrate 401A, 401E), and deposition of a sacrificial material (e.g., phosphosilicate glass (PSG)). The phosphosilicate glass (PSG) may comprise 8% phosphorous and 92% silicon dioxide. The resonator 400A, 400E may be formed over the sacrificial material (e.g., phosphosilicate glass (PSG)). The sacrificial material may then be selectively etched away beneath the resonator 400A, 400E, leaving cavity 483A, 483E beneath the resonator 400A, 400E. For example phosphosilicate glass (PSG) sacrificial material may be selectively etched away by hydrofluoric acid beneath the resonator 400A, 400E, leaving cavity 483A, 483E beneath the resonator 400A, 400E. The cavity 483A, 483E may, but need not, be arranged to provide acoustic isolation of the structures, e.g., bottom acoustic reflector 413A, 413E, e.g., stack 404A, 404E of piezoelectric layers, e.g., resonator 400A, 400E from the substrate 401A, 401E.

Similarly, in FIGS. 4B, 4C, 4F and 4G, a via 485B, 485C, 485F, 485G (e.g., through silicon via 485B, 485F, e.g., through silicon carbide via 485C, 485G) may, but need not, be arranged to provide acoustic isolation of the structures, e.g., bottom acoustic reflector 413B, 413C, 413F, 413G, e.g., stack 404B, 404C, 404F, 404G, of piezoelectric layers, e.g., resonator 400B, 400C, 400F, 400G from the substrate 401B, 401C, 401F, 401G. The via 485B, 485C, 485F, 485G (e.g., through silicon via 485B, 485F, e.g., through silicon carbide via 485C, 485G) may be formed using techniques (e.g., using photolithographic masking and etching techniques) known to those with ordinary skill in the art. For example, in FIGS. 4B and 4F, backside photolithographic masking and etching techniques may be used to form the through silicon via 485B, 485F, and an additional passivation layer 487B, 487F may be deposited, after the resonator 400B, 400F is formed. For example, in FIGS. 4C and 4G, backside photolithographic masking and etching techniques may be used to form the through silicon carbide via 485C, 485G, after the top acoustic reflector 415C, 415G and stack 404C, 404G of piezoelectric layers are formed. In FIGS. 4C and 4G, after the through silicon carbide via 485C, 485G, is formed, backside photolithographic masking and deposition techniques may be used to form bottom acoustic reflector 413C, 413G, and additional passivation layer 487C, 487G.

In FIGS. 4A, 4B, 4C, 4E, 4F, 4G, bottom acoustic reflector 413A, 413B, 413C, 413E, 413F, 413G, may include the acoustically reflective bottom electrode stack of the plurality of bottom metal electrode layers, in which thicknesses of the bottom metal electrode layers may be related to wavelength (e.g., acoustic wavelength) at the main resonant frequency of the example resonator 400A, 400B, 400C, 400E, 400F, 400G. As mentioned previously herein, the layer thickness of the initial bottom metal electrode layer 417A, 417B, 417C, 417E, 417F, 417G, may be about one

eighth of a wavelength (e.g., one eighth acoustic wavelength) at the main resonant frequency of the example resonator 400A. Respective layer thicknesses, (e.g., T01 through T04, explicitly shown in FIGS. 4A, 4B, 4C) for members of the pairs of bottom metal electrode layers may be about one quarter of the wavelength (e.g., one quarter acoustic wavelength) at the main resonant frequency of the example resonators 400A, 400B, 400C, 400E, 400F, 400G. Relatively speaking, in various alternative designs of the example resonators 400A, 400B, 400C, 400E, 400F, 400G, for relatively lower main resonant frequencies (e.g., five Gigahertz (5 GHz)) and having corresponding relatively longer wavelengths (e.g., longer acoustic wavelengths), may have relatively thicker bottom metal electrode layers in comparison to other alternative designs of the example resonators 400A, 400B, 400C, 400E, 400F, 400G, for relatively higher main resonant frequencies (e.g., twenty-four Gigahertz (24 GHz)). There may be corresponding longer etching times to form, e.g., etch through, the relatively thicker bottom metal electrode layers in designs of the example resonator 400A, 400B, 400C, 400E, 400F, 400G, for relatively lower main resonant frequencies (e.g., five Gigahertz (5 GHz)). Accordingly, in designs of the example resonators 400A, 400B, 400C, 400E, 400F, 400G, for relatively lower main resonant frequencies (e.g., five Gigahertz (5 GHz)) having the relatively thicker bottom metal electrode layers, there may (but need not) be an advantage in etching time in having a relatively fewer number (e.g., five (5)) of bottom metal electrode layers, shown in 4A, 4B, 4C, 4E, 4F, 4G, in comparison to a relatively larger number (e.g., nine (9)) of bottom metal electrode layers, shown in FIGS. 1A and 1n FIG. 4D. The relatively larger number (e.g., nine (9)) of bottom metal electrode layers, shown in FIGS. 1A and 1n FIG. 4D may (but need not) provide for relatively greater acoustic isolation than the relatively fewer number (e.g., five (5)) of bottom metal electrode layers. However, in FIGS. 4A and 4E the cavity 483A, 483E, (e.g., air cavity 483A, 483E) may (but need not) be arranged to provide acoustic isolation enhancement relative to some designs without the cavity 483A, 483E. Similarly, in FIGS. 4B, 4C, 4F, 4G, the via 483B, 483C, 483F, 483G, (e.g., through silicon via 485B, 485F, e.g., through silicon carbide via 485C, 485G) may (but need not) be arranged to provide acoustic isolation enhancement relative to some designs without the via 483B, 483C, 483F, 483G.

In FIGS. 4A and 4E, the cavity 483A, 483E may (but need not) be arranged to compensate for relatively lesser acoustic isolation of the relatively fewer number (e.g., five (5)) of bottom metal electrode layers. In FIGS. 4A and 4E, the cavity 483A, 483E may (but need not) be arranged to provide acoustic isolation benefits, while retaining possible electrical conductivity improvements and etching time benefits of the relatively fewer number (e.g., five (5)) of bottom metal electrode layers, e.g., particularly in designs of the example resonator 400A, 400E, for relatively lower main resonant frequencies (e.g., five Gigahertz (5 GHz)). Similarly, in FIGS. 4B, 4C, 4F, 4G, the via 483B, 483C, 483F, 483G, may (but need not) be arranged to compensate for relatively lesser acoustic isolation of the relatively fewer number (e.g., five (5)) of bottom metal electrode layers. In FIGS. 4B, 4C, 4F, 4G, the via 483B, 483C, 483F, 483G, may (but need not) be arranged to provide acoustic isolation benefits, while retaining possible electrical conductivity improvement benefits and etching time benefits of the relatively fewer number (e.g., five (5)) of bottom metal electrode layers, e.g., particularly in designs of the example resonator 400B, 400C, 400F, 400G, for relatively lower main resonant

frequencies (e.g., five Gigahertz (5 GHz), e.g., below six Gigahertz (6 GHz), e.g., below five Gigahertz (5 GHz)).

FIGS. 4D through 4G show alternative example bulk acoustic wave resonators 400D through 400G to the example bulk acoustic wave resonator 100A shown in FIG. 1A, in which the top acoustic reflector, 415D through 415G, may comprise a lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of the top acoustic reflector, 415D through 415G. A gap, 491D through 491G, may be formed beneath the lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of the top acoustic reflector 415D through 415G. The gap, 491D through 491G, may be arranged adjacent to the etched edge region, 453D through 453G, of the example resonators 400D through 400G.

For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the stack 404D through 404G, of piezoelectric layers, for example along the thickness dimension T27 of the stack 404D through 404G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the bottom piezoelectric layer 405D through 405G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the bottom piezoelectric layer 405D through 405G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the first middle piezoelectric layer 407D through 407G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the second middle piezoelectric layer 409D through 409G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the top piezoelectric layer 411D through 411G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) one or more interposer layers (e.g., first interposer layer, 495D through 495G, second interposer layer, 461D through 461G, third interposer layer 411D through 411G).

For example, as shown in FIGS. 4D through 4G, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends partially through) the top acoustic reflector 415D through 415G, for example partially along the thickness dimension T25 of the top acoustic reflector 415D through 415G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the initial top electrode layer 435D through 435G. For example, the gap, 491D through 491G, may be arranged adjacent to where the etched edge region, 453D through 453G, extends through (e.g., extends entirely through or extends partially through) the first member, 437D through 437G, of the first pair of top electrode layers, 437D through 437G, 439D through 439G.

For example, as shown in FIGS. 4D through 4F, the gap, 491D through 491F, may be arranged adjacent to where the etched edge region, 453D through 453F, extends through (e.g., extends entirely through or extends partially through) the bottom acoustic reflector 413D through 413F, for example along the thickness dimension T23 of the bottom acoustic reflector 413D through 413F. For example, the gap, 491D through 491F, may be arranged adjacent to where the etched edge region, 453D through 453F, extends through (e.g., extends entirely through or extends partially through) the first pair of bottom electrode layers, 419D through 419F, 421D through 421F. For example, the gap, 491D through 491F, may be arranged adjacent to where the etched edge region, 453D through 453F, extends through (e.g., extends entirely through or extends partially through) the second pair of bottom electrode layers, 423D through 423F, 425D through 425F. For example, as shown in FIGS. 4D through 4F, the etched edge region, 453D through 453F, may extend through (e.g., entirely through or partially through) the bottom acoustic reflector, 413D through 413F, and through (e.g., entirely through or partially through) one or more of the piezoelectric layers, 405D through 405F, 407D through 407F, 409D through 409F, 411D through 411F, to the lateral connection portion, 489D through 489G, (e.g., to the bridge portion, 489D through 489G), of the top acoustic reflector, 415D through 415F.

As shown in FIGS. 4D-4G, lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, may be a multilayer lateral connection portion, 415D through 415G, (e.g., a multilayer metal bridge portion, 415D through 415G, comprising differing metals, e.g., metals having differing acoustic impedances.) For example, lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, may comprise the second member, 439D through 439G, (e.g., comprising the relatively high acoustic impedance metal) of the first pair of top electrode layers, 437D through 437G, 439D through 439G. For example, lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, may comprise the second pair of top electrode layers, 441D through 441G, 443D through 443G.

Gap 491D-491G may be an air gap 491D-491G, or may be filled with a relatively low acoustic impedance material (e.g., BenzoCyclobutene (BCB)), which may be deposited using various techniques known to those with skill in the art. Gap 491D-491G may be formed by depositing a sacrificial material (e.g., phosphosilicate glass (PSG)) after the etched edge region, 453D through 453G, is formed. The lateral connection portion, 489D through 489G, (e.g., bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, may then be deposited (e.g., sputtered) over the sacrificial material. The sacrificial material may then be selectively etched away beneath the lateral connection portion, 489D through 489G, (e.g., beneath the bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, leaving gap 491D-491G beneath the lateral connection portion, 489D through 489G, (e.g., beneath the bridge portion, 489D through 489G). For example the phosphosilicate glass (PSG) sacrificial material may be selectively etched away by hydrofluoric acid beneath the lateral con-

nection portion, 489D through 489G, (e.g., beneath the bridge portion, 489D through 489G), of top acoustic reflector, 415D through 415G, leaving gap 491D-491G beneath the lateral connection portion, 489D through 489G, (e.g., beneath the bridge portion, 489D through 489G).

Although in various example resonators, 100A, 400A, 400B, 400D, 400E, 400F, polycrystalline piezoelectric layers (e.g., polycrystalline Aluminum Nitride (AlN)) may be deposited (e.g., by sputtering), in other example resonators 400C, 400G, alternative single crystal or near single crystal piezoelectric layers (e.g., single/near single crystal Aluminum Nitride (AlN)) may be deposited (e.g., by metal organic chemical vapor deposition (MOCVD)). Normal axis piezoelectric layers (e.g., normal axis Aluminum Nitride (AlN) piezoelectric layers) may be deposited by MOCVD using techniques known to those with skill in the art. As discussed previously herein, the interposer layers may be deposited by sputtering, but alternatively may be deposited by MOCVD. Reverse axis piezoelectric layers (e.g., reverse axis Aluminum Nitride (AlN) piezoelectric layers) may likewise be deposited via MOCVD. For the respective example resonators 400C, 400G shown in FIGS. 4C and 4G, the alternating axis piezoelectric stack 404C, 404G comprised of piezoelectric layers 405C, 407C, 409C, 411C, 405G, 407G, 409G, 411G as well as interposer layers 459C, 461C, 463C, 459G, 461G, 453G extending along stack thickness dimension T27 fabricated using MOCVD on a silicon carbide substrate 401C, 401G. For example, aluminum nitride of piezoelectric layers 405C, 407C, 409C, 411C, 405G, 407G, 409G, 411G may grow nearly epitaxially on silicon carbide (e.g., 4H SiC) by virtue of the small lattice mismatch between the polar axis aluminum nitride wurtzite structure and specific crystal orientations of silicon carbide. Alternative small lattice mismatch substrates may be used (e.g., sapphire, e.g., aluminum oxide). By varying the ratio of the aluminum and nitrogen in the deposition precursors, an aluminum nitride film may be produced with the desired polarity (e.g., normal axis, e.g., reverse axis). For example, normal axis aluminum nitride may be synthesized using MOCVD when a nitrogen to aluminum ratio in precursor gases approximately 1000. For example, reverse axis aluminum nitride may synthesized when the nitrogen to aluminum ratio is approximately 27000. In accordance with the foregoing, FIGS. 4C and 4G show MOCVD synthesized normal axis piezoelectric layer 405C, 405G, MOCVD synthesized reverse axis piezoelectric layer 407C, 407G, MOCVD synthesized normal axis piezoelectric layer 409C, 409G, and MOCVD synthesized reverse axis piezoelectric layer 411C, 411G. For example, normal axis piezoelectric layer 405C, 405G may be synthesized by MOCVD in a deposition environment where the nitrogen to aluminum gas ratio is relatively low, e.g., 1000 or less. Next an oxyaluminum nitride layer, 459C at lower temperature, may be deposited by MOCVD that may reverse axis (e.g., reverse axis polarity) of the growing aluminum nitride under MOCVD growth conditions, and has also been shown to be able to be deposited by itself under MOCVD growth conditions. Increasing the nitrogen to aluminum ratio into the several thousands during the MOCVD synthesis may enable the reverse axis piezoelectric layer 407C, 407G to be synthesized. Interposer layer 461C, 461G may be an oxide layer such as, but not limited to, aluminum oxide or silicon dioxide. This oxide layer may be deposited in a low temperature physical vapor deposition process such as sputtering or in a higher temperature chemical vapor deposition process. Normal axis piezoelectric layer 409C, 409G may be grown by MOCVD on top of interposer layer 461C, 461G using growth conditions similar to the normal axis

layer **405C**, **405G**, as discussed previously, namely MOCVD in a deposition environment where the nitrogen to aluminum gas ratio is relatively low, e.g., 1000 or less. Next an aluminum oxynitride, interposer layer **463C**, **463G** may be deposited in a low temperature MOCVD process followed by a reverse axis piezoelectric layer **411C**, **411G**, synthesized in a high temperature MOCVD process and an atmosphere of nitrogen to aluminum ratio in the several thousand range. Upon conclusion of these depositions, the piezoelectric stack **404C**, **404G** shown in FIGS. **4C** and **4G** may be realized.

FIG. **5** shows a schematic of an example ladder filter **500A** (e.g., SHF or EHF wave ladder filter **500A**) using three series resonators of the bulk acoustic wave resonator structure of FIG. **1A** (e.g., three bulk acoustic SHF or EHF wave resonators), and two mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. **1A** (e.g., two mass loaded bulk acoustic SHF or EHF wave resonators), along with a simplified view of the three series resonators. Accordingly, the example ladder filter **500A** (e.g., SHF or EHF wave ladder filter **500A**) is an electrical filter, comprising a plurality of bulk acoustic wave (BAW) resonators, e.g., on a substrate, in which the plurality of BAW resonators may comprise a respective first layer (e.g., bottom layer) of piezoelectric material having a respective piezoelectrically excitable resonance mode. The plurality of BAW resonators of the filter **500A** may comprise a respective top acoustic reflector (e.g., top acoustic reflector electrode) including a respective initial top metal electrode layer and a respective first pair of top metal electrode layers electrically and acoustically coupled with the respective first layer (e.g., bottom layer) of piezoelectric material to excite the respective piezoelectrically excitable resonance mode at a respective resonant frequency. For example, the respective top acoustic reflector (e.g., top acoustic reflector electrode) may include the respective initial top metal electrode layer and the respective first pair of top metal electrode layers, and the foregoing may have a respective peak acoustic reflectivity in the Super High Frequency (SHF) band or the Extremely High Frequency (EHF) band that includes the respective resonant frequency of the respective BAW resonator. The plurality of BAW resonators of the filter **500A** may comprise a respective bottom acoustic reflector (e.g., bottom acoustic reflector electrode) including a respective initial bottom metal electrode layer and a respective first pair of bottom metal electrode layers electrically and acoustically coupled with the respective first layer (e.g., bottom layer) of piezoelectric material to excite the respective piezoelectrically excitable resonance mode at the respective resonant frequency. For example, the respective bottom acoustic reflector (e.g., bottom acoustic reflector electrode) may include the respective initial bottom metal electrode layer and the respective first pair of bottom metal electrode layers, and the foregoing may have a respective peak acoustic reflectivity in the super high frequency band or the extremely high frequency band that includes the respective resonant frequency of the respective BAW resonator. The respective first layer (e.g., bottom layer) of piezoelectric material may be sandwiched between the respective top acoustic reflector and the respective bottom acoustic reflector. Further, the plurality of BAW resonators may comprise at least one respective additional layer of piezoelectric material, e.g., first middle piezoelectric layer. The at least one additional layer of piezoelectric material may have the piezoelectrically excitable main resonance mode with the respective first layer (e.g., bottom layer) of piezoelectric material. The respective first layer (e.g., bottom layer) of piezoelectric material may have a

respective first piezoelectric axis orientation (e.g., normal axis orientation) and the at least one respective additional layer of piezoelectric material may have a respective piezoelectric axis orientation (e.g., reverse axis orientation) that opposes the first piezoelectric axis orientation of the respective first layer of piezoelectric material. Further discussion of features that may be included in the plurality of BAW resonators of the filter **500A** is present previously herein with respect to previous discussion of FIG. **1A**.

As shown in the schematic appearing at an upper section of FIG. **5**, the example ladder filter **500A** may include an input port comprising a first node **521A** (InA), and may include a first series resonator **501A** (Series1A) (e.g., first bulk acoustic SHF or EHF wave resonator **501A**) coupled between the first node **521A** (InA) associated with the input port and a second node **522A**. The example ladder filter **500A** may also include a second series resonator **502A** (Series2A) (e.g., second bulk acoustic SHF or EHF wave resonator **502A**) coupled between the second node **522A** and a third node **523A**. The example ladder filter **500A** may also include a third series resonator **503A** (Series3A) (e.g., third bulk acoustic SHF or EHF wave resonator **503A**) coupled between the third node **523A** and a fourth node **524A** (OutA), which may be associated with an output port of the ladder filter **500A**. The example ladder filter **500A** may also include a first mass loaded shunt resonator **511A** (Shunt1A) (e.g., first mass loaded bulk acoustic SHF or EHF wave resonator **511A**) coupled between the second node **522A** and ground. The example ladder filter **500A** may also include a second mass loaded shunt resonator **512A** (Shunt2A) (e.g., second mass loaded bulk acoustic SHF or EHF wave resonator **512A**) coupled between the third node **523** and ground.

Appearing at a lower section of FIG. **5** is the simplified view of the three series resonators **501B** (Series1B), **502B** (Series2B), **503B** (Series3B) in a serial electrically interconnected arrangement **500B**, for example, corresponding to series resonators **501A**, **502A**, **503A**, of the example ladder filter **500A**. The three series resonators **501B** (Series1B), **502B** (Series2B), **503B** (Series3B), may be constructed as shown in the arrangement **500B** and electrically interconnected in a way compatible with integrated circuit fabrication of the ladder filter. Although the first mass loaded shunt resonator **511A** (Shunt1A) and the second mass loaded shunt resonator **512A** are not explicitly shown in the arrangement **500B** appearing at a lower section of FIG. **5**, it should be understood that the first mass loaded shunt resonator **511A** (Shunt1A) and the second mass loaded shunt resonator **512A** are constructed similarly to what is shown for the series resonators in the lower section of FIG. **5**, but that the first and second mass loaded shunt resonators **511A**, **512A** may include mass layers, in addition to layers corresponding to those shown for the series resonators in the lower section of FIG. **5** (e.g., the first and second mass loaded shunt resonators **511A**, **512A** may include respective mass layers, in addition to respective top acoustic reflectors of respective top metal electrode layers, may include respective alternating axis stacks of piezoelectric material layers, and may include respective bottom acoustic reflectors of bottom metal electrode layers.) For example, all of the resonators of the ladder filter may be co-fabricated using integrated circuit processes (e.g., Complementary Metal Oxide Semiconductor (CMOS) compatible fabrication processes) on the same substrate (e.g., same silicon substrate). The example ladder filter **500A** and serial electrically interconnected arrangement **500B** of series resonators **501A**, **502A**, **503A**, may

respectively be relatively small in size, and may respectively have a lateral dimension (x5) of less than approximately one millimeter.

For example, the serial electrically interconnected arrangement 500B of three series resonators 501B (Series1B), 502B (Series2B), 503B (Series3B), may include an input port comprising a first node 521B (InB) and may include a first series resonator 501B (Series1B) (e.g., first bulk acoustic SHF or EHF wave resonator 501B) coupled between the first node 521B (InB) associated with the input port and a second node 522B. The first node 521B (InB) may include bottom electrical interconnect 569B electrically contacting a first bottom acoustic reflector of first series resonator 501B (Series1B) (e.g., first bottom acoustic reflector electrode of first series resonator 501B (Series1B)). Accordingly, in addition to including bottom electrical interconnect 569, the first node 521B (InB) may also include the first bottom acoustic reflector of first series resonator 501B (Series1B) (e.g., first bottom acoustic reflector electrode of first series resonator 501B (Series1B)). The first bottom acoustic reflector of first series resonator 501B (Series1B) (e.g., first bottom acoustic reflector electrode of first series resonator 501B (Series1B)) may include a stack of the plurality of bottom metal electrode layers 517 through 525. The serial electrically interconnected arrangement 500B of three series resonators 501B (Series1B), 502B (Series2B), 503B (Series3B), may include the second series resonator 502B (Series2B) (e.g., second bulk acoustic SHF or EHF wave resonator 502B) coupled between the second node 522B and a third node 523B. The third node 523B may include a second bottom acoustic reflector of second series resonator 502B (Series2B) (e.g., second bottom acoustic reflector electrode of second series resonator 502B (Series2B)). The second bottom acoustic reflector of second series resonator 502B (Series2B) (e.g., second bottom acoustic reflector electrode of second series resonator 502B (Series2B)) may include an additional stack of an additional plurality of bottom metal electrode layers. The serial electrically interconnected arrangement 500B of three series resonators 501B (Series1B), 502B (Series2B), 503B (Series3B), may also include the third series resonator 503B (Series3B) (e.g., third bulk acoustic SHF or EHF wave resonator 503B) coupled between the third node 523B and a fourth node 524B (OutB). The third node 523B, e.g., including the additional plurality of bottom metal electrode layers, may electrically interconnect the second series resonator 502B (Series2B) and the third series resonator 503B (Series3B). The second bottom acoustic reflector (e.g., second bottom acoustic reflector electrode) of second series resonator 502B (Series2B) of the third node 523B, e.g., including the additional plurality of bottom metal electrode layers, may be a mutual bottom acoustic reflector (e.g., mutual bottom acoustic reflector electrode), and may likewise serve as bottom acoustic reflector (e.g., bottom acoustic reflector) of third series resonator 503B (Series3B). The fourth node 524B (OutB) may be associated with an output port of the serial electrically interconnected arrangement 500B of three series resonators 501B (Series1B), 502B (Series2B), 503B (Series3B). The fourth node 524B (OutB) may include electrical interconnect 571C.

The stack of the plurality of bottom metal electrode layers 517 through 525 are associated with the first bottom acoustic reflector (e.g., first bottom acoustic reflector electrode) of first series resonator 501B (Series1B). The additional stack of the additional plurality of bottom metal electrode layers (e.g., of the third node 523B) may be associated with the mutual bottom acoustic reflector (e.g., mutual bottom acous-

tic reflector electrode) of both the second series resonant 502B (Series2B) and the third series resonator 503B (Series3B). Although stacks of respective five bottom metal electrode layers are shown in simplified view in FIG. 5, in should be understood that the stacks may include respective larger numbers of bottom metal electrode layers, e.g., respective nine top metal electrode layers. Further, the first series resonator (Series1B), and the second series resonant 502B (Series2B) and the third series resonator 503B (Series3B) may all have the same, or approximately the same, or different (e.g., achieved by means of additional mass loading layers) resonant frequency (e.g., the same, or approximately the same, or different main resonant frequency). For example, small additional massloads (e.g., a tenth of the main shunt mass-load) of series and shunt resonators may help to reduce pass-band ripples in insertion loss, as may be appreciated by one with skill in the art. The bottom metal electrode layers 517 through 525 and the additional plurality of bottom metal electrode layers (e.g., of the mutual bottom acoustic reflector, e.g., of the third node 523B) may have respective thicknesses that are related to wavelength (e.g., acoustic wavelength) for the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator 501B (Series1B), e.g., second series resonator 502B, e.g., third series resonator (503B)).

One or more patterned layers 557C, 557D, 557E may be placed at various different locations in various different bulk acoustic wave resonators 501B Series 1B, 502B Series 2B, 503B Series 3B of the arrangement 500B for filter 500A. Patterned layers 557C, 557D, 557E in various different resonators 501B Series 1B, 502B Series 2B, 503B Series 3B of the arrangement 500A for filter 500B may contribute substantially differently to respective facilitation of spurious mode suppression in various different bulk acoustic wave resonators 501B Series 1B, 502B Series 2B, 503B Series 3B. In the various different resonators 501B Series 1B, 502B Series 2B, 503B Series 3B, some patterned layers 557E, 557D may be arranged relatively nearer to a respective piezoelectric layer stack including the respective first and second layers of piezoelectric material than other patterned layers, e.g., patterned layer 555C, arranged relatively further from the respective piezoelectric layer stack, to contribute more to the respective facilitation of spurious mode suppression than what the other patterned layers 557D, 557E contribute in the arrangement 500B for filter 500A. For example, first patterned layer 557C (e.g., top patterned layer 557C) of bulk acoustic wave resonator 501B Series 1B may be interposed between members 537C, 539C of the first pair of top metal electrode layers of bulk acoustic wave resonator 501B Series 1B of the arrangement 500B for filter 500A. First patterned layer 557D (e.g., top patterned layer 557D) of bulk acoustic wave resonator 502B Series 2B may be interposed between initial top electrode layer and the first pair of top metal electrode layers 537D, 539D of bulk acoustic wave resonator 501B Series 1B of the arrangement 500B for filter 500A. In other words, first patterned layer 557D (e.g., top patterned layer 557D) of bulk acoustic wave resonator 502B Series 2B may be interposed between the first pair of top metal electrode layers 537D, 539D and a piezoelectric layer stack including the first and second layers of piezoelectric material of bulk acoustic wave resonator 502B Series 2B of the arrangement 500B for filter 500A. First patterned layer 557E (e.g., top patterned layer 557E) of bulk acoustic wave resonator 503B Series 3B may be interposed between the plurality of top metal electrode layers of top acoustic reflector and a piezoelectric layer stack

including the first and second layers of piezoelectric material of bulk acoustic wave resonator **503B** Series **3B** of the arrangement **500B** for filter **500A**. Notably, various materials may be used to form patterned layers **557C**, **557D**, **557E**. In particular, both for series resonators **501B** Series **1B**, **502B** Series **2B**, **503B** Series **3B** and for shunt resonators (not shown) a combination of relatively heavy material, like, for example, Tungsten in combination with relatively light material like, for example, Titanium (Ti), may be used to achieve desired facilitation of spurious mode suppression, as discussed in relation with FIG. 2J and as will be appreciated by one skilled in the art.

Various embodiments for series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively higher resonant frequency (e.g., higher main resonant frequency) may have relatively thinner bottom metal electrode thicknesses, e.g., scaled thinner with relatively higher resonant frequency (e.g., higher main resonant frequency). Similarly, various embodiments of the series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively lower resonant frequency (e.g., lower main resonant frequency) may have relatively thicker bottom metal electrode layer thicknesses, e.g., scaled thicker with relatively lower resonant frequency (e.g., lower main resonant frequency). The bottom metal electrode layers **517** through **525** and the additional plurality of bottom metal electrode layers (e.g., of the mutual bottom acoustic reflector, e.g., of the third node **523B**) may include members of pairs of bottom metal electrodes having respective thicknesses of one quarter wavelength (e.g., one quarter acoustic wavelength) at the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)). The stack of bottom metal electrode layers **517** through **525** and the stack of additional plurality of bottom metal electrode layers (e.g., of the mutual bottom acoustic reflector, e.g., of the third node **523B**) may include respective alternating stacks of different metals, e.g., different metals having different acoustic impedances (e.g., alternating relatively high acoustic impedance metals with relatively low acoustic impedance metals). The foregoing may provide acoustic impedance mismatches for facilitating acoustic reflectivity (e.g., SHF or EHF acoustic wave reflectivity) of the first bottom acoustic reflector (e.g., first bottom acoustic reflector electrode) of the first series resonator **501B** (Series**1B**) and the mutual bottom acoustic reflector (e.g., of the third node **523B**) of the second series resonator **502B** (Series**2B**) and the third series resonator **503B** (Series**3B**).

A first top acoustic reflector (e.g., first top acoustic reflector electrode) comprises a first stack of a first plurality of top metal electrode layers **535C** through **543C** of the first series resonator **501B** (Series**1B**). A second top acoustic reflector (e.g., second top acoustic reflector electrode) comprises a second stack of a second plurality of top metal electrode layers **535D** through **543D** of the second series resonator **502B** (Series**2B**). A third top acoustic reflector (e.g., third top acoustic reflector electrode) comprises a third stack of a third plurality of top metal electrode layers **535E** through **543E** of the third series resonator **503B** (Series**3B**). Although stacks of respective five top metal electrode layers are shown in simplified view in FIG. 5, it should be understood that the stacks may include respective larger numbers of top metal electrode layers, e.g., respective nine bottom metal electrode layers. Further, the first plurality of

top metal electrode layers **535C** through **543C**, the second plurality of top metal electrode layers **535D** through **543D**, and the third plurality of top metal electrode layers **535E** through **543E** may have respective thicknesses that are related to wavelength (e.g., acoustic wavelength) for the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)). Various embodiments for series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively higher resonant frequency (e.g., higher main resonant frequency) may have relatively thinner top metal electrode thicknesses, e.g., scaled thinner with relatively higher resonant frequency (e.g., higher main resonant frequency). Similarly, various embodiments of the series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively lower resonant frequency (e.g., lower main resonant frequency) may have relatively thicker top metal electrode layer thicknesses, e.g., scaled thicker with relatively lower resonant frequency (e.g., lower main resonant frequency). The first plurality of top metal electrode layers **535C** through **543C**, the second plurality of top metal electrode layers **535D** through **543D**, and the third plurality of top metal electrode layers **535E** through **543E** may include members of pairs of bottom metal electrodes having respective thicknesses of one quarter wavelength (e.g., one quarter acoustic wavelength) of the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator **501B** (Series**1B**), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)). The first stack of the first plurality of top metal electrode layers **535C** through **543C**, the second stack of the second plurality of top metal electrode layers **535D** through **543D**, and the third stack of the third plurality of top metal electrode layers **535E** through **543E** may include respective alternating stacks of different metals, e.g., different metals having different acoustic impedances (e.g., alternating relatively high acoustic impedance metals with relatively low acoustic impedance metals). The foregoing may provide acoustic impedance mismatches for facilitating acoustic reflectivity (e.g., SHF or EHF acoustic wave reflectivity) of the top acoustic reflectors (e.g., the first top acoustic reflector of the first series resonator **501B** (Series**1B**), e.g., the second top acoustic reflector of the second series resonator **502B** (Series**2B**), e.g., the third top acoustic reflector of the third series resonator **503B** (Series**3B**)). Explicitly shown in the FIG. 5 simplified views of metal electrode layers of the series resonators, respective pluralities of lateral features **557C**, **557D**, **557E** (e.g., respective patterned layers **557C**, **557D**, **557E**, e.g., respective pluralities of step mass features **557C**, **557D**, **557E**) may be sandwiched between metal electrode layers (e.g., between respective pairs of top metal electrode layers, e.g., between respective first pairs of top metal electrode layers **537C**, **539C**, **537D**, **539D**, **537E**, **539E**, and respective second pairs of top metal electrode layers **541C**, **543C**, **541D**, **543D**, **541E**, **543E**). The respective pluralities of lateral features **557C**, **557D**, **557E** (e.g., patterned layers **557C**, **557D**, **557E**) may, but need not, limit parasitic lateral acoustic modes (e.g., facilitate suppression of spurious modes) of the bulk acoustic wave resonators of FIG. 5 (e.g., of the series resonators, the mass loaded series resonators, and the mass loaded shunt resonators).

The first series resonator **501B** (Series**1B**) may comprise a first alternating axis stack, e.g., an example first stack of

four layers of alternating axis piezoelectric material, **505C** through **511C**. The second series resonator **502B** (Series2B) may comprise a second alternating axis stack, e.g., an example second stack of four layers of alternating axis piezoelectric material, **505D** through **511D**. The third series resonator **503B** (Series3B) may comprise a third alternating axis stack, e.g., an example third stack of four layers of alternating axis piezoelectric material, **505E** through **511E**. The first, second and third alternating axis piezoelectric stacks may comprise layers of Aluminum Nitride (AlN) having alternating C-axis wurtzite structures. For example, piezoelectric layers **505C**, **505D**, **505E**, **509C**, **509D**, **509E** have normal axis orientation. For example, piezoelectric layers **507C**, **507D**, **507E**, **511C**, **511D**, **511E** have reverse axis orientation. Members of the first stack of four layers of alternating axis piezoelectric material, **505C** through **511C**, and members of the second stack of four layers of alternating axis piezoelectric material, **505D** through **511D**, and members of the third stack of four layers of alternating axis piezoelectric material, **505E** through **511E**, may have respective thicknesses that are related to wavelength (e.g., acoustic wavelength) for the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator **501B** (Series1B), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)). Various embodiments for series resonators (e.g., first series resonator **501B** (Series1B), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively higher resonant frequency (e.g., higher main resonant frequency) may have relatively thinner piezoelectric layer thicknesses, e.g., scaled thinner with relatively higher resonant frequency (e.g., higher main resonant frequency). Similarly, various embodiments of the series resonators (e.g., first series resonator **501B** (Series1B), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)) having various relatively lower resonant frequency (e.g., lower main resonant frequency) may have relatively thicker piezoelectric layer thicknesses, e.g., scaled thicker with relatively lower resonant frequency (e.g., lower main resonant frequency). The example first stack of four layers of alternating axis piezoelectric material, **505C** through **511C**, the example second stack of four layers of alternating axis piezoelectric material, **505D** through **511D** and the example third stack of four layers of alternating axis piezoelectric material, **505D** through **511D** may include stack members of piezoelectric layers having respective thicknesses of approximately one half wavelength (e.g., one half acoustic wavelength) at the resonant frequency (e.g., main resonant frequency) of the series resonators (e.g., first series resonator **501B** (Series1B), e.g., second series resonator **502B**, e.g., third series resonator (**503B**)).

The example first stack of four layers of alternating axis piezoelectric material, **505C** through **511C**, may include a first three members of interposer layers **559C**, **561C**, **563C** respectively sandwiched between the corresponding four layers of alternating axis piezoelectric material, **505C** through **511C**. The example second stack of four layers of alternating axis piezoelectric material, **505D** through **511D**, may include a second three members of interposer layers **559D**, **561D**, **563D** respectively sandwiched between the corresponding four layers of alternating axis piezoelectric material, **505D** through **511D**. The example third stack of four layers of alternating axis piezoelectric material, **505E** through **511E**, may include a third three members of interposer layers **559E**, **561E**, **563E** respectively sandwiched between the corresponding four layers of alternating axis piezoelectric material, **505E** through **511E**. One or more

(e.g., one or a plurality of) interposer layers may be metal interposer layers. The metal interposer layers may be relatively high acoustic impedance metal interposer layers (e.g., using relatively high acoustic impedance metals such as Tungsten (W) or Molybdenum (Mo)). Such metal interposer layers may (but need not) flatten stress distribution across adjacent piezoelectric layers, and may (but need not) raise effective electromechanical coupling coefficient (K_{t2}) of adjacent piezoelectric layers. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may be dielectric interposer layers. The dielectric of the dielectric interposer layers may be a dielectric that has a positive acoustic velocity temperature coefficient, so acoustic velocity increases with increasing temperature of the dielectric. The dielectric of the dielectric interposer layers may be, for example, silicon dioxide. Dielectric interposer layers may, but need not, facilitate compensating for frequency response shifts with increasing temperature. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may comprise metal and dielectric for respective interposer layers. The first series resonator **501B** (Series1B), the second series resonator **502B** (Series2B) and the third series resonator **503B** (Series3B) may have respective etched edge regions **553C**, **553D**, **553E**, and respective laterally opposing etched edge regions **554C**, **554D**, **554E**. Reference is made to resonator mesa structures as have already been discussed in detail previously herein. Accordingly, they are not discussed again in detail at this point. Briefly, respective first, second and third mesa structures of the respective first series resonator **501B** (Series1B), the respective second series resonator **502B** (Series2B) and the respective third series resonator **503B** (Series3B) may extend between respective etched edge regions **553C**, **553D**, **553E**, and respective laterally opposing etched edge regions **554C**, **554D**, **554E** of the respective first series resonator **501B** (Series1B), the respective second series resonator **502B** (Series2B) and the respective third series resonator **503B** (Series3B). The second bottom acoustic reflector of second series resonator **502B** (Series2B) of the third node **523B**, e.g., including the additional plurality of bottom metal electrode layers may be a second mesa structure. For example, this may be a mutual second mesa structure bottom acoustic reflector **523B**, and may likewise serve as bottom acoustic reflector of third series resonator **503B** (Series3B). Accordingly, this mutual second mesa structure bottom acoustic reflector **523B** may extend between etched edge region **553E** of the third series resonator **503B** (Series3B) and the laterally opposing etched edge region **554D** of the third series resonator **503B** (Series3B).

FIG. 6 shows a schematic of an example ladder filter **600A** (e.g., SHF or EHF wave ladder filter **600A**) using five series resonators of the bulk acoustic wave resonator structure of FIG. 1A (e.g., five bulk acoustic SHF or EHF wave resonators), and four mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. 1A (e.g., four mass loaded bulk acoustic SHF or EHF wave resonators), along with a simplified top view of the nine resonators interconnected in the example ladder filter **600B**, and lateral dimensions of the example ladder filter **600B**. As shown in the schematic appearing at an upper section of FIG. 6, the example ladder filter **600A** may include an input port comprising a first node **621A** (InputA E1TopA), and may include a first series resonator **601A** (Ser1A) (e.g., first bulk acoustic SHF or EHF wave resonator **601A**) coupled between the first node **621A** (InputA E1TopA) associated with the input port and a second node **622A** (E1BottomA). The example ladder filter **600A** may also include a second

series resonator **602A** (Ser2A) (e.g., second bulk acoustic SHF or EHF wave resonator **602A**) coupled between the second node **622A** (E1BottomA) and a third node **623A** (E3TopA). The example ladder filter **600A** may also include a third series resonator **603A** (Ser3A) (e.g., third bulk acoustic SHF or EHF wave resonator **603A**) coupled between the third node **623A** (E3TopA) and a fourth node **624A** (E2BottomA). The example ladder filter **600A** may also include a fourth series resonator **604A** (Ser4A) (e.g., fourth bulk acoustic SHF or EHF wave resonator **604A**) coupled between the fourth node **624A** (E2BottomA) and a fifth node **625A** (E4TopA). The example ladder filter **600A** may also include a fifth series resonator **605A** (Ser5A) (e.g., fifth bulk acoustic SHF or EHF wave resonator **605A**) coupled between the fifth node **625A** (E4TopA) and a sixth node **626A** (OutputA E4BottomA), which may be associated with an output port of the ladder filter **600A**. The example ladder filter **600A** may also include a first mass loaded shunt resonator **611A** (Sh1A) (e.g., first mass loaded bulk acoustic SHF or EHF wave resonator **611A**) coupled between the second node **622A** (E1BottomA) and a first grounding node **631A** (E2TopA). The example ladder filter **600A** may also include a second mass loaded shunt resonator **612A** (Sh2A) (e.g., second mass loaded bulk acoustic SHF or EHF wave resonator **612A**) coupled between the third node **623A** (E3TopA) and a second grounding node **632A** (E3BottomA). The example ladder filter **600A** may also include a third mass loaded shunt resonator **613A** (Sh3A) (e.g., third mass loaded bulk acoustic SHF or EHF wave resonator **613A**) coupled between the fourth node **624A** (E2BottomA) and the first grounding node **631A** (E2TopA). The example ladder filter **600A** may also include a fourth mass loaded shunt resonator **614A** (Sh4A) (e.g., fourth mass loaded bulk acoustic SHF or EHF wave resonator **614A**) coupled between the fifth node **625A** (E4TopA) and the second grounding node **632A** (E3BottomA). The first grounding node **631A** (E2TopA) and the second grounding node **632A** (E3BottomA) may be interconnected to each other, and may be connected to ground, through an additional grounding connection (AdditionalConnection).

Appearing at a lower section of FIG. 6 is the simplified top view of the nine resonators interconnected in the example ladder filter **600B**, and lateral dimensions of the example ladder filter **600B**. The example ladder filter **600B** may include an input port comprising a first node **621B** (InputA E1TopB), and may include a first series resonator **601B** (Ser1B) (e.g., first bulk acoustic SHF or EHF wave resonator **601B**) coupled between (e.g., sandwiched between) the first node **621B** (InputA E1TopB) associated with the input port and a second node **622B** (E1BottomB). The example ladder filter **600B** may also include a second series resonator **602B** (Ser2B) (e.g., second bulk acoustic SHF or EHF wave resonator **602B**) coupled between (e.g., sandwiched between) the second node **622B** (E1BottomB) and a third node **623B** (E3TopB).

The example ladder filter **600B** may also include a third series resonator **603B** (Ser3B) (e.g., third bulk acoustic SHF or EHF wave resonator **603B**) coupled between (e.g., sandwiched between) the third node **623B** (E3TopB) and a fourth node **624B** (E2BottomB). The example ladder filter **600B** may also include a fourth series resonator **604B** (Ser4B) (e.g., fourth bulk acoustic SHF or EHF wave resonator **604B**) coupled between (e.g., sandwiched between) the fourth node **624B** (E2BottomB) and a fifth node **625B** (E4TopB). The example ladder filter **600B** may also include a fifth series resonator **605B** (Ser5B) (e.g., fifth bulk acoustic SHF or EHF wave resonator **605B**) coupled between (e.g.,

sandwiched between) the fifth node **625B** (E4TopB) and a sixth node **626B** (OutputB E4BottomB), which may be associated with an output port of the ladder filter **600B**. The example ladder filter **600B** may also include a first mass loaded shunt resonator **611B** (Sh1B) (e.g., first mass loaded bulk acoustic SHF or EHF wave resonator **611B**) coupled between (e.g., sandwiched between) the second node **622B** (E1BottomB) and a first grounding node **631B** (E2TopB). The example ladder filter **600B** may also include a second mass loaded shunt resonator **612B** (Sh2B) (e.g., second mass loaded bulk acoustic SHF or EHF wave resonator **612B**) coupled between (e.g., sandwiched between) the third node **623B** (E3TopB) and a second grounding node **632B** (E3BottomB). The example ladder filter **600B** may also include a third mass loaded shunt resonator **613B** (Sh3B) (e.g., third mass loaded bulk acoustic SHF or EHF wave resonator **613B**) coupled between (e.g., sandwiched between) the fourth node **624B** (E2BottomB) and the first grounding node **631B** (E2TopB). The example ladder filter **600B** may also include a fourth mass loaded shunt resonator **614B** (Sh4B) (e.g., fourth mass loaded bulk acoustic SHF or EHF wave resonator **614B**) coupled between (e.g., sandwiched between) the fifth node **625B** (E4TopB) and the second grounding node **632B** (E3BottomB). The first grounding node **631B** (E2TopB) and the second grounding node **632B** (E3BottomB) may be interconnected to each other, and may be connected to ground, through an additional grounding connection, not shown in the lower section of FIG. 6. The example ladder filter **600B** may respectively be relatively small in size, and may respectively have lateral dimensions (X6 by Y6) of less than approximately one millimeter by one millimeter.

FIG. 7 shows a schematic of example inductors modifying an example lattice filter **700** using a first pair of series resonators **701A** (Se1T), **702A** (Se2T), (e.g., two bulk acoustic SHF or EHF wave resonators) of the bulk acoustic wave resonator structure of FIG. 1A, a second pair of series resonators **701B** (Se2B), **702B** (Se2B), (e.g., two additional bulk acoustic SHF or EHF wave resonators) of the bulk acoustic wave resonator structure of FIG. 1A and two pairs of cross coupled mass loaded shunt resonators **701C** (Sh1C), **702D** (Sh2C), **703C** (Sh3C), **704C** (Sh4C), (e.g., four mass loaded bulk acoustic SHF or EHF wave resonators) of the bulk acoustic wave resonator structure of FIG. 1A. As shown in the schematic of FIG. 7, the example inductor modified lattice filter **700** may include a first top series resonator **701A** (Se1T) (e.g., first top bulk acoustic SHF or EHF wave resonator **701A**) coupled between a first top node **721A** and a second top node **722A**. The example inductor modified lattice filter **700** may also include a second top series resonator **702A** (Se2T) (e.g., second top bulk acoustic SHF or EHF wave resonator **702A**) coupled between the second top node **722A** and a third top node **723A**.

The example inductor modified lattice filter **700** may include a first bottom series resonator **701B** (Se1B) (e.g., first bottom bulk acoustic SHF or EHF wave resonator **701B**) coupled between a first bottom node **721B** and a second bottom node **722B**. The example inductor modified lattice filter **700** may also include a second bottom series resonator **702B** (Se2B) (e.g., second bottom bulk acoustic SHF or EHF wave resonator **702B**) coupled between the second bottom node **722B** and a third bottom node **723B**. The example inductor modified lattice filter **700** may include a first cross-coupled mass loaded shunt resonator **701C** (Sh1C) (e.g., first mass loaded bulk acoustic SHF or EHF wave resonator **701C**) coupled between the first top node **721A** and the second bottom node **722B**. The example

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inductor modified lattice filter **700** may also include a second cross-coupled mass loaded shunt resonator **702C** (Sh2C) (e.g., second mass loaded bulk acoustic SHF or EHF wave resonator **702C**) coupled between the second top node **722A** and the first bottom node **721B**. The example inductor modified lattice filter **700** may include a third cross-coupled mass loaded shunt resonator **703C** (Sh3C) (e.g., third mass loaded bulk acoustic SHF or EHF wave resonator **703C**) coupled between the second top node **722A** and the third bottom node **723B**. The example inductor modified lattice filter **700** may also include a fourth cross-coupled mass loaded shunt resonator **704C** (Sh4C) (e.g., fourth mass loaded bulk acoustic SHF or EHF wave resonator **704C**) coupled between the third top node **723A** and the second bottom node **722B**. The example inductor modified lattice filter **700** may include a first inductor **711** (L1) coupled between the first top node **721A** and the first bottom node **721B**. The example inductor modified lattice filter **700** may include a second inductor **712** (L2) coupled between the second top node **722A** and the second bottom node **722B**. The example inductor modified lattice filter **700** may include a third inductor **713** (L3) coupled between the third top node **723A** and the third bottom node **723B**.

FIGS. **8A** and **8B** show an example oscillator **800A**, **800B** (e.g., millimeter wave oscillator **800A**, **800B**, e.g., Super High Frequency (SHF) wave oscillator **800A**, **800B**, e.g., Extremely High Frequency (EHF) wave oscillator **800A**, **800B**) using the bulk acoustic wave resonator structure of FIG. **1A**. For example, FIGS. **8A** and **8B** shows simplified views of bulk acoustic wave resonator **801A**, **801B** electrically coupled with electrical oscillator circuitry (e.g., active oscillator circuitry **802A**, **802B**) through phase compensation circuitry **803A**, **803B** (Φ comp). The example oscillator **800A**, **800B** may be a negative resistance oscillator, e.g., in accordance with a one-port model as shown in FIGS. **8A** and **8B**. The electrical oscillator circuitry, e.g., active oscillator circuitry may include one or more suitable active devices (e.g., one or more suitably configured amplifying transistors) to generate a negative resistance commensurate with resistance of the bulk acoustic wave resonator **801A**, **801B**. In other words, energy lost in bulk acoustic wave resonator **801A**, **801B** may be replenished by the active oscillator circuitry, thus allowing steady oscillation, e.g., steady SHF or EHF wave oscillation. To ensure oscillation start-up, active gain (e.g., negative resistance) of active oscillator circuitry **802A**, **802B** may be greater than one. As illustrated on opposing sides of a notional dashed line in FIGS. **8A** and **8B**, the active oscillator circuitry **802A**, **802B** may have a complex reflection coefficient of the active oscillator circuitry (Γ_{amp}), and the bulk acoustic wave resonator **801A**, **801B** together with the phase compensation circuitry **803A**, **803B** (Φ comp) may have a complex reflection coefficient (Γ_{res}). To provide for the steady oscillation, e.g., steady SHF or EHF wave oscillation, a magnitude may be greater than one for $|\Gamma_{amp} \Gamma_{res}|$, e.g., magnitude of a product of the complex reflection coefficient of the active oscillator circuitry (Γ_{amp}) and the complex reflection coefficient (Γ_{res}) of the resonator to bulk acoustic wave resonator **801A**, **801B** together with the phase compensation circuitry **803A**, **803B** (Φ comp) may be greater than one. Further, to provide for the steady oscillation, e.g., steady SHF or EHF wave oscillation, phase angle may be an integer multiple of three-hundred-sixty degrees for $\angle \Gamma_{amp} \Gamma_{res}$, e.g., a phase angle of the product of the complex reflection coefficient of the active oscillator circuitry (Γ_{amp}) and the complex reflection coefficient (Γ_{res}) of the resonator to bulk acoustic wave resonator **801A**, **801B** together with the phase compensation

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circuitry **803A**, **803B** (Φ comp) may be an integer multiple of three-hundred-sixty degrees. The foregoing may be facilitated by phase selection, e.g., electrical length selection, of the phase compensation circuitry **803A**, **803B** (Φ comp).

In the simplified view of FIG. **8A**, the bulk acoustic wave resonator **801A** (e.g., bulk acoustic SHF or EHF wave resonator) includes first normal axis piezoelectric layer **805A**, first reverse axis piezoelectric layer **807A**, and another normal axis piezoelectric layer **809A**, and another reverse axis piezoelectric layer **811A** arranged in a four piezoelectric layer alternating axis stack arrangement sandwiched between multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **813A**. General structures and applicable teaching of this disclosure for the multilayer metal acoustic SHF or EHF reflector top electrode **815A** and the multilayer metal acoustic SHF or EHF reflector bottom electrode **813A** have already been discussed in detail previously herein with respect of FIGS. **1A** and **4A** through **4G**, which for brevity are incorporated by reference rather than repeated fully here. As already discussed, these structures are directed to respective pairs of metal electrode layers, in which a first member of the pair has a relatively low acoustic impedance (relative to acoustic impedance of an other member of the pair), in which the other member of the pair has a relatively high acoustic impedance (relative to acoustic impedance of the first member of the pair), and in which the respective pairs of metal electrode layers have layer thicknesses corresponding to approximately one quarter wavelength (e.g., approximately one quarter acoustic wavelength) at a main resonant frequency of the resonator. Accordingly, it should be understood that the bulk acoustic SHF or EHF wave resonator **801A** shown in FIG. **8A** may include multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** and multilayer metal acoustic SHF or EHF wave reflector bottom electrode **815B** in which the respective pairs of metal electrode layers may include layer thicknesses corresponding to approximately a quarter wavelength (e.g., approximately one quarter of an acoustic wavelength) at a SHF or EHF wave main resonant frequency of the bulk acoustic SHF or EHF wave resonator **801A**. Initial top metal electrode layer **835A** and initial bottom metal electrode layer **817A** may have respective layer thickness of about one eighth of a wavelength (e.g., one eighth of an acoustic wavelength) at the main resonant frequency of the bulk acoustic SHF or EHF wave resonator **801A**. The multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** may include the initial top metal electrode layer **835A** and the first pair of top metal electrode layers **824A** electrically and acoustically coupled with the four piezoelectric layer alternating axis stack arrangement (e.g., with the first normal axis piezoelectric layer **805A**, e.g., with first reverse axis piezoelectric layer **807A**, e.g., with another normal axis piezoelectric layer **809A**, e.g., with another reverse axis piezoelectric layer **811A**) to excite the piezoelectrically excitable resonance mode at the resonant frequency. For example, the multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** may include the initial top metal electrode layer **835A** and the first pair of top metal electrode layers **824A**, and the foregoing may have a respective peak acoustic reflectivity in the Super High Frequency (SHF) band or the Extremely High Frequency (EHF) band that includes the respective resonant frequency of the respective BAW resonator. The multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** may include a first patterned layer **857A**. The first pair of top metal electrodes

824A may be interposed between the first patterned layer **857A** and a stack of layers of piezoelectric material including the first layer of piezoelectric material **805A** (e.g., normal axis layer of piezoelectric material **805A**) and the second layer of piezoelectric material **807A** (e.g., reverse axis layer of piezoelectric material **807A**). First and second patterned layers **857A**, **858A** (e.g., top patterned layer **857A** and bottom patterned layer **858A**) may contribute substantially differently to facilitating spurious mode suppression in the bulk acoustic wave resonator **801A** for oscillator **800A**. In accordance with the teachings of this disclosure, one of the patterned layers (e.g., second patterned layer **858A**, e.g., bottom patterned layer **858A**) may be arranged substantially nearer to a piezoelectric layer stack including the first and second layers of piezoelectric material than another one of the mass load layers (e.g., first patterned layer **857A**, e.g., top patterned layer **857A**), to contribute more to facilitating spurious mode suppression for bulk acoustic wave resonator **801A** for oscillator **800A** than what the another one of the patterned layers contributes.

Similarly, the multilayer metal acoustic SHF or EHF wave reflector bottom electrode **813A** may include the initial bottom metal electrode layer **817A** and the first pair of bottom metal electrode layers **822A** electrically and acoustically coupled with the four piezoelectric layer alternating axis stack arrangement (e.g., with the first normal axis piezoelectric layer **805A**, e.g., with first reverse axis piezoelectric layer **807A**, e.g., with another normal axis piezoelectric layer **809A**, e.g., with another reverse axis piezoelectric layer **811A**) to excite the piezoelectrically excitable resonance mode at the resonant frequency. For example, the multilayer metal acoustic SHF or EHF wave reflector bottom electrode **817A** may include the initial bottom metal electrode layer **817A** and the first pair of bottom metal electrode layers **822A**, and the foregoing may have a respective peak acoustic reflectivity in the Super High Frequency (SHF) band or the Extremely High Frequency (EHF) band that includes the resonant frequency of the BAW resonator **801A**. The second patterned layer **858A** may be interposed between the first pair of bottom metal electrodes **822A** and the initial bottom metal electrode layer **817A**.

An output **816A** of the oscillator **800A** may be coupled to the bulk acoustic wave resonator **801A** (e.g., coupled to multilayer metal acoustic SHF or EHF wave reflector top electrode **815A**). Interposer layers as discussed previously herein with respect to FIG. 1A are explicitly shown in the simplified view the example resonator **801A** shown in FIG. 8A. Such interposer layers may be included and interposed between adjacent piezoelectric layers. For example, a first interposer layer may be arranged between first normal axis piezoelectric layer **805A** and first reverse axis piezoelectric layer **807A**. For example, a second interposer layer may be arranged between first reverse axis piezoelectric layer **807A** and another normal axis piezoelectric layer **809A**. For example, a third interposer may be arranged between the another normal axis piezoelectric layer **809A** and another reverse axis piezoelectric layer **807A**. As discussed previously herein, such interposer may be metal and/or dielectric, and may, but need not provide various benefits, as discussed previously herein. Alternatively or additionally, one or more (e.g., one or a plurality of) interposer layers may comprise metal and dielectric for respective interposer layers.

A notional heavy dashed line is used in depicting an etched edge region **853A** associated with example resonator **801A**. The example resonator **801A** may also include a laterally opposing etched edge region **854A** arranged opposite from the etched edge region **853A**. The etched edge

region **853A** (and the laterally opposing etch edge region **854A**) may similarly extend through various members of the example resonator **801A** of FIG. 8A, in a similar fashion as discussed previously herein with respect to the etched edge region **253D** (and the laterally opposing etch edge region **254D**) of example resonator **2001D** shown in FIG. 2B. As shown in FIG. 8A, a first mesa structure corresponding to the stack of four piezoelectric material layers **805A**, **807A**, **809A**, **811A** may extend laterally between (e.g., may be formed between) etched edge region **853A** and laterally opposing etched edge region **854A**. A second mesa structure corresponding to multilayer metal acoustic SHF or EHF wave reflector bottom electrode **813A** may extend laterally between (e.g., may be formed between) etched edge region **853A** and laterally opposing etched edge region **854A**. Third mesa structure corresponding to multilayer metal acoustic SHF or EHF wave reflector top electrode **815A** may extend laterally between (e.g., may be formed between) etched edge region **853A** and laterally opposing etched edge region **854A**. Although not explicitly shown in the FIG. 8A simplified view of metal electrode layers, e.g., multilayer metal acoustic SHF or EHF wave reflector top electrode **815A**, a plurality of lateral features (e.g., plurality of step features) may be sandwiched between metal electrode layers (e.g., between pairs of top metal electrode layers. The plurality of lateral features may, but need not, limit parasitic lateral acoustic modes of the example bulk acoustic wave resonator of FIG. 8A.

FIG. 8B shows a schematic of and example circuit implementation of the oscillator shown in FIG. 8A. Active oscillator circuitry **802B** may include active elements, symbolically illustrated in FIG. 8B by alternating voltage source **804B** (Vs) coupled through negative resistance **806B** (Rneg), e.g., active gain element **806B**, to example bulk acoustic wave resonator **801B** (e.g., bulk acoustic SHF or EHF wave resonator) via phase compensation circuitry **803B** (Φcomp). The representation of example bulk acoustic wave resonator **801B** (e.g., bulk acoustic SHF or EHF wave resonator) may include passive elements, symbolically illustrated in FIG. 8B by electrode ohmic loss parasitic series resistance **808B** (Rs), motional capacitance **810B** (Cm), acoustic loss motional resistance **812B** (Rm), motional inductance **814B** (Lm), static or plate capacitance **816B** (Co), and acoustic loss parasitic **818B** (Ro). An output **816B** of the oscillator **800B** may be coupled to the bulk acoustic wave resonator **801B** (e.g., coupled to a multilayer metal acoustic SHF or EHF wave reflector top electrode of bulk acoustic wave resonator **801B**).

FIGS. 9A and 9B are simplified diagrams of a frequency spectrum illustrating application frequencies and application frequency bands of the example bulk acoustic wave resonators shown in FIG. 1A and FIGS. 4A through 4G, and the example filters shown in FIGS. 5 through 7, and the example oscillators shown in FIGS. 8A and 8B. A widely used standard to designate frequency bands in the microwave range by letters is established by the United States Institute of Electrical and Electronic Engineers (IEEE). In accordance with standards published by the IEEE, as defined herein, and as shown in FIGS. 9A and 9B are application bands as follows: S Band (2 GHz-4 GHz), C Band (4 GHz-8 GHz), X Band (8 GHz-12 GHz), Ku Band (12 GHz-18 GHz), K Band (18 GHz-27 GHz), Ka Band (27 GHz-40 GHz), V Band (40 GHz-75 GHz), and W Band (75 GHz-110 GHz). FIG. 9A shows a first frequency spectrum portion **9000A** in a range from three Gigahertz (3 GHz) to eight Gigahertz (8 GHz), including application bands of S Band (2 GHz-4 GHz) and C Band (4 GHz-8 GHz). As described

subsequently herein, the 3rd Generation Partnership Project standards organization (e.g., 3GPP) has standardized various 5G frequency bands. For example, included is a first application band **9010** (e.g., 3GPP 5G n77 band) (3.3 GHz-4.2 GHz) configured for fifth generation broadband cellular network (5G) applications. As described subsequently herein, the first application band **9010** (e.g., 5G n77 band) includes a 5G sub-band **9011** (3.3 GHz-3.8 GHz). The 3GPP 5G sub-band **9011** includes Long Term Evolution broadband cellular network (LTE) application sub-bands **9012** (3.4 GHz-3.6 GHz), **9013** (3.6 GHz-3.8 GHz), and **9014** (3.55 GHz-3.7 GHz). A second application band **9020** (4.4 GHz-5.0 GHz) includes a sub-band **9021** for China specific applications. Discussed next are Unlicensed National Information Infrastructure (UNII) bands. A third application band **9030** includes a UNII-1 band **9031** (5.15 GHz-5.25 GHz) and a UNII-2A band **9032** (5.25 GHz-5.33 GHz). An LTE band **9033** (LTE Band **252**) overlaps the same frequency range as the UNII-1 band **6031**. A fourth application band **9040** includes a UNII-2C band **9041** (5.490 GHz-5.735 GHz), a UNII-3 band **9042** (5.735 GHz-5.85 GHz), a UNII-4 band **9043** (5.85 GHz-5.925 GHz), a UNII-5 band **9044** (5.925 GHz-6.425 GHz), a UNII-6 band **9045** (6.425 GHz-6.525 GHz), a UNII-7 band **9046** (6.525 GHz-6.875 GHz), and a UNII-8 band **9047** (6.875 GHz-7.125 GHz). An LTE band **9048** overlaps the same frequency range (5.490 GHz-5.735 GHz) as the UNII-3 band **9042**. A sub-band **9049A** shares the same frequency range as the UNII-4 band **9043**. An LTE band **9049B** shares a subsection of the same frequency range (5.855 GHz-5.925 GHz).

FIG. 9B shows a second frequency spectrum portion **9000B** in a range from eight Gigahertz (8 GHz) to one-hundred and ten Gigahertz (110 GHz), including application bands of X Band (8 GHz-12 GHz), Ku Band (12 GHz-18 GHz), K Band (18 GHz-27 GHz), Ka Band (27 GHz-40 GHz), V Band (40 GHz-75 GHz), and W Band (75 GHz-110 GHz). A fifth application band **9050** includes 3GPP 5G bands configured for fifth generation broadband cellular network (5G) applications, e.g., 3GPP 5G n258 band **9051** (24.25 GHz-27.5 GHz), e.g., 3GPP 5G n261 band **9052** (27.5 GHz-28.35 GHz), e.g., 3GPP 5G n257 band **9053** (26.5 GHz-29.5). A sixth application band **9060** includes the 3GPP 5G n260 band **9060** (37 GHz-40 GHz). A seventh application band **9070** includes United States WiGig Band for IEEE 802.11ad and IEEE 802.11ay **9071** (57 GHz-71 GHz), European Union and Japan WiGig Band for IEEE 802.11ad and IEEE 802.11ay **9072** (57 GHz-66 GHz), South Korea WiGig Band for IEEE 802.11ad and IEEE 802.11ay **9073** (57 GHz-64 GHz), and China WiGig Band for IEEE 802.11ad and IEEE 802.11ay **9074** (59 GHz-64 GHz). An eighth application band **9080** includes an automobile radar band **9080** (76 GHz-81 GHz).

Accordingly, it should be understood from the foregoing that the acoustic wave devices (e.g., resonators, e.g., filters, e.g., oscillators) of this disclosure may be implemented in the respective application frequency bands just discussed. For example, the layer thicknesses of the acoustic reflector electrodes and piezoelectric layers in alternating axis arrangement for the example acoustic wave devices (e.g., the example 24 GHz bulk acoustic wave resonators) of this disclosure may be scaled up and down as needed to be implemented in the respective application frequency bands just discussed. This is likewise applicable to the example filters (e.g., bulk acoustic wave based filters) and example oscillators (e.g., bulk acoustic wave resonator based oscillators) of this disclosure to be implemented in the respective application frequency bands just discussed. For example, the

layer thicknesses of the acoustic reflector electrodes, for example, thickness of piezoelectric layers in alternating axis arrangement of acoustic wave devices of this disclosure (e.g., resonators, e.g., filters, e.g., oscillators) may be sufficiently thin to be implemented in the respective application frequency bands just discussed. The following examples pertain to further embodiments for acoustic wave devices, including but not limited to, e.g., bulk acoustic wave resonators, e.g., filters incorporating such bulk acoustic wave resonators, e.g., oscillators incorporating such bulk acoustic wave resonators, and from which numerous permutations and configurations will be apparent. A first example is an acoustic wave device comprising first and second layers of piezoelectric material acoustically coupled with one another to have a piezoelectrically excitable resonance mode, in which the first layer of piezoelectric material has a first piezoelectric axis orientation, and the second layer of piezoelectric material has a second piezoelectric axis orientation that substantially opposes the first piezoelectric axis orientation of the first layer of piezoelectric material, and in which the first and second layers of piezoelectric material have respective thicknesses so that the acoustic wave device has a resonant frequency, and an acoustic reflector electrode including a first pair of metal electrode layers electrically and acoustically coupled with the first and second layer of piezoelectric material to excite the piezoelectrically excitable resonance mode at the resonant frequency, the acoustic reflector electrode including at least one patterned layer. A second example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in a 3rd Generation Partnership Project (3GPP) band. A third example is an acoustic wave device as described in the first example in which a frequency of a peak acoustic reflectivity of the first pair of top metal electrode layers is in a 3rd Generation Partnership Project (3GPP) band. A fourth example is an acoustic wave device as the first example, in which a peak the resonant frequency of the acoustic wave device is in a 3GPP n77 band **9010** as shown in FIG. 9A. A fifth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in a 3GPP n79 band **9020** as shown in FIG. 9A. A sixth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in a 3GPP n258 band **9051** as shown in FIG. 9B. A seventh example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in a 3GPP n261 band **9052** as shown in FIG. 9B. An eighth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in a 3GPP n260 band **9060** as shown in FIG. 9B. A ninth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) C band as shown in FIG. 9A. A tenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) X band as shown in FIG. 9B. An eleventh example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) Ku band as shown in FIG. 9B. An twelfth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE)

K band as shown in FIG. 9B. A thirteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) Ka band as shown in FIG. 9B. A fourteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) V band as shown in FIG. 9B. A fifteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in an Institute of Electrical and Electronic Engineers (IEEE) W band as shown in FIG. 9B. A sixteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-1 band **9031**, as shown in FIG. 9A. A seventeenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-2A band **9032**, as shown in FIG. 9A. A eighteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-2C band **9041**, as shown in FIG. 9A. A nineteenth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-3 band **9042**, as shown in FIG. 9A. A twentieth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-4 band **9043**, as shown in FIG. 9A. A twenty first example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-5 band **9044**, as shown in FIG. 9A. A twenty second example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-6 band **9045**, as shown in FIG. 9A. A twenty third example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-7 band **9046**, as shown in FIG. 9A. A twenty fourth example is an acoustic wave device as described in the first example, in which the resonant frequency of the acoustic wave device is in UNII-8 band **9047**, as shown in FIG. 9A.

FIGS. 9C and 9D are first and second diagrams **9100**, **9200** illustrating respective simulated bandpass characteristics **9101**, **9201** of insertion loss versus frequency for example SHF wave filters. For example, FIG. 9C is a first diagram **9100** illustrating a first simulated bandpass characteristic **9101** of insertion loss versus frequency for a first example SHF wave filter configured as in FIG. 7 (e.g., inductors modifying an example lattice filter using a first pair of series resonators of the bulk acoustic wave resonator structure of FIG. 1A, a second pair of series resonators of the bulk acoustic wave resonator structure of FIG. 1A and two pairs of cross coupled mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. 1A). For example, the first example SHF wave filter having the simulated bandpass characteristic **9101** may be a 3GPP 5G n258 band filter (e.g., filter corresponding to the FIG. 9B 3GPP 5G n258 band **9051** (24.25 GHz-27.5 GHz)). For example, the first example SHF wave filter having the simulated bandpass characteristic **9101** may have a fractional bandwidth of about twelve percent (12%), and may include resonators having electromechanical coupling coefficient (K_{t2}) of about six and a half percent (6.5%). For example, the simulated bandpass characteristic **9101** of FIG. 9C shows a first 3GPP 5G n258 band edge feature **9103**

having an insertion loss of -1.6328 decibels (dB) at an initial 24.25 GHz extremity of the 3GPP 5G n258 band. For example, the simulated bandpass characteristic **9101** of FIG. 9C shows an opposing 3GPP 5G n258 band edge feature **9105** having an insertion loss of -1.648 decibels (dB) at an opposing 27.5 GHz extremity of the 3GPP 5G n258 band. The first example SHF wave filter having the simulated bandpass characteristic **9101** may have a pass band that is configured for 3GPP 5G n258 applications. For example, the simulated bandpass characteristic **9101** of FIG. 9C shows a first 3GPP 5G n258 band roll off feature **9107** having an insertion loss of -21.664 decibels (dB) at an initial 23.56 GHz roll off extremity of the 3GPP 5G n258 band. At the initial 23.56 GHz roll off extremity of the 3GPP 5G n258 band, the first 3GPP 5G n258 band roll off feature **9107** may provide about twenty dB of roll off at about 690 MHz from the first 3GPP 5G n258 band edge feature **9103** at the initial 24.25 GHz extremity of the 3GPP 5G n258 band. For example, the simulated bandpass characteristic **9101** of FIG. 9C shows an opposing 3GPP 5G n258 band roll off feature **9109** having an insertion loss of -21.764 decibels (dB) at an opposing 28.02 GHz roll off extremity of the 3GPP 5G n258 band. At the opposing 28.02 GHz roll off extremity of the 3GPP 5G n258 band, the opposing 3GPP 5G n258 band roll off feature **9109** may provide about twenty dB of roll off at about 580 MHz from the opposing 3GPP 5G n258 band edge feature **9105** at the opposing 27.5 GHz extremity of the 3GPP 5G n258 band.

For example, FIG. 9D is a second diagram **9200** illustrating a second simulated bandpass characteristic **9201** of insertion loss versus frequency for a second example SHF wave filter configured as two external shunt inductors modifying the example ladder filter of FIG. 6 (e.g., an input port shunt inductor and an output port shunt inductor modifying the ladder configuration using five series resonators of the bulk acoustic wave resonator structure of FIG. 1A, and four mass loaded shunt resonators of the bulk acoustic wave resonator structure of FIG. 1A). The shunt inductors may be, for example, 0.8 nanohenry inductors having a quality factor of twenty (Q of 20). For example, the second example SHF wave filter having the simulated bandpass characteristic **9201** may be a 3GPP 5G n258 band channel filter (e.g., filter corresponding to a channel in the FIG. 9B 3GPP 5G n258 band **9051** (24.25 GHz-27.5 GHz)). For example, the second example SHF wave filter having the simulated bandpass characteristic **9201** may be a two hundred Megahertz (200 MHz) channel filter of the 3GPP 5G n258, e.g., the filter may have a fractional bandwidth of about nine tenths of a percent (0.9%), and may include resonators having electromechanical coupling coefficient (K_{t2}) of about one and seven tenths percent (1.7%). For example, the simulated bandpass characteristic **9201** of FIG. 9D shows a first 3GPP 5G n258 band channel edge feature **9203** having an insertion loss of -2.9454 decibels (dB) at an initial 24.25 GHz channel extremity of the 3GPP 5G n258 band. For example, the simulated bandpass characteristic **9201** of FIG. 9D shows an opposing 3GPP 5G n258 band channel edge feature **9205** having an insertion loss of -2.9794 decibels (dB) at an opposing 24.45 GHz extremity of the 3GPP 5G n258 band channel. The second example SHF wave filter having the simulated bandpass characteristic **9201** may have a channel pass band that is configured for 3GPP 5G n258 applications. For example, the simulated bandpass characteristic **9201** of FIG. 9D shows a first 3GPP 5G n258 band channel roll off feature **9207** having an insertion loss of -22.406 decibels (dB) at an initial 24.203 GHz roll off extremity of the 3GPP 5G n258 band channel. At the initial 24.203 GHz roll off

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extremity of the 3GPP 5G n258 band channel, the first 3GPP 5G n258 band channel roll off feature **9207** may provide about twenty dB of roll off at about 50 Mhz from the first 3GPP 5G n258 band channel edge feature **9203** at the initial 24.25 GHz extremity of the 3GPP 5G n258 band channel. For example, the simulated bandpass characteristic **9201** FIG. 9D shows an opposing 3GPP 5G n258 band channel roll off feature **9209** having an insertion loss of -22.291 decibels (dB) at an opposing 24.497 GHz channel roll off extremity of the 3GPP 5G n258 band channel. At the opposing 24.497 GHz channel roll off extremity of the 3GPP 5G n258 band channel, the opposing 3GPP 5G n258 band roll off channel feature **9209** may provide about twenty dB of roll off at about 50 Mhz from the opposing 3GPP 5G n258 band channel edge feature **9205** at the opposing 24.45 GHz extremity of the 3GPP 5G n258 band channel.

FIG. 10 illustrates a computing system implemented with integrated circuit structures or devices formed using the techniques disclosed herein, in accordance with an embodiment of the present disclosure. As can be seen, the computing system **1000** houses a motherboard **1002**. The motherboard **1002** may include a number of components, including, but not limited to, a processor **1004** and at least one communication chip **1006A**, **1006B** each of which may be physically and electrically coupled to the motherboard **1002**, or otherwise integrated therein. As will be appreciated, the motherboard **1002** may be, for example, any printed circuit board, whether a main board, a daughterboard mounted on a main board, or the only board of system **1000**, etc.

Depending on its applications, computing system **1000** may include one or more other components that may or may not be physically and electrically coupled to the motherboard **1002**. These other components may include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth). Any of the components included in computing system **1000** may include one or more integrated circuit structures or devices formed using the disclosed techniques in accordance with an example embodiment. In some embodiments, multiple functions may be integrated into one or more chips (e.g., for instance, note that the communication chips **1006A**, **1006B** may be part of or otherwise integrated into the processor **1004**).

The communication chips **1006A**, **1006B** enables wireless communications for the transfer of data to and from the computing system **1000**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication chips **1006A**, **1006B** may implement any of a number of wireless standards or protocols, including, but not limited to, Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing system **1000** may

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include a plurality of communication chips **1006A**, **1006B**. For instance, a first communication chip **1006A** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **1006A** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, 5G and others. In some embodiments, communication chips **1006A**, **1006B** may include one or more acoustic wave devices **1008A**, **1008B** (e.g., resonators, e.g., filters, e.g., oscillators) as variously described herein (e.g., acoustic wave devices including a stack of alternating axis piezoelectric material). Acoustic wave devices may be included in various ways, e.g., one or more resonators, e.g., one or more filters, and e.g., one or more oscillators. Further, such acoustic wave devices **1008A**, **1008B**, e.g., resonators, e.g., filters, e.g., oscillators may be configured to be Super High Frequency (SHF) acoustic wave devices **1008A**, **1008B** or Extremely High Frequency (EHF) acoustic wave devices **1008A**, **1008B**, e.g., resonators, e.g., filters, e.g., oscillators (e.g., operating at greater than 3, 4, 5, 6, 7, or 8 GHz, e.g., operating at greater than 23, 24, 25, 26, 27, 28, 29, or 30 GHz, e.g., operating at greater than 36, 37, 38, 39, or 40 GHz). Further still, such Super High Frequency (SHF) acoustic wave devices or Extremely High Frequency (EHF) resonators, filters, and/or oscillators may be included in the RF front end of computing system **1000** and they may be used for 5G wireless standards or protocols, for example.

The processor **1004** of the computing system **1000** includes an integrated circuit die packaged within the processor **1004**. In some embodiments, the integrated circuit die of the processor includes onboard circuitry that is implemented with one or more integrated circuit structures or devices formed using the disclosed techniques, as variously described herein. The term “processor” may refer to any device or portion of a device that processes, for instance, electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

The communication chips **1006A**, **1006B** also may include an integrated circuit die packaged within the communication chips **1006A**, **1006B**. In accordance with some such example embodiments, the integrated circuit die of the communication chip includes one or more integrated circuit structures or devices formed using the disclosed techniques as variously described herein. As will be appreciated in light of this disclosure, note that multi-standard wireless capability may be integrated directly into the processor **1004** (e.g., where functionality of any chips **1006** is integrated into processor **1004**, rather than having separate communication chips). Further note that processor **1004** may be a chip set having such wireless capability. In short, any number of processor **1004** and/or communication chips **1006** may be used. Likewise, any one chip or chip set may have multiple functions integrated therein.

In various implementations, the computing device **1000** may be a laptop, a netbook, a notebook, a smartphone, a tablet, a personal digital assistant (PDA), an ultra-mobile PC, a mobile phone, a desktop computer, a server, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a digital camera, a portable music player, a digital video recorder, or any other electronic device that processes data or employs one or more integrated circuit structures or devices formed using the disclosed techniques, as variously described herein.

Further Example Embodiments

The following examples pertain to further embodiments, from which numerous permutations and configurations will

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be apparent. The foregoing description of example embodiments has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed. Many modifications and variations are possible in light of this disclosure. It is intended that the scope of the present disclosure be limited not by this detailed description, but rather by the claims appended hereto. Future filed applications claiming priority to this application may claim the disclosed subject matter in a different manner, and may generally include any set of one or more limitations as variously disclosed or otherwise demonstrated herein.

What is claimed is:

1. A bulk acoustic millimeter wave resonator comprising:
 - a substrate;
 - a piezoelectric stack including at least a first piezoelectric layer, a second piezoelectric layer, and a third piezoelectric layer acoustically coupled with one another, in which the first piezoelectric layer and the second piezoelectric layer and the third piezoelectric layer have respective thicknesses to facilitate a main resonant frequency of the bulk acoustic millimeter wave resonator;
 - an electrode electrically and acoustically coupled with the third piezoelectric layer; and
 - a patterned layer associated with the electrode, in which the patterned layer is acoustically coupled with the third piezoelectric to facilitate a suppression of a spurious mode of the bulk acoustic millimeter wave resonator.
2. The bulk acoustic millimeter wave resonator as in claim 1 in which the electrode is an acoustically reflective electrode stack including at least a first pair of metal electrode layers, in which the first pair of metal electrode layers is electrically and acoustically coupled with the third piezoelectric layer.
3. The bulk acoustic millimeter wave resonator as in claim 1 in which the main resonant frequency of the bulk acoustic millimeter wave resonator is in an Institute of Electrical and Electronic Engineers (IEEE) band in one of a Ku band, a K band, a Ka band, a V band, and a W band.
4. The bulk acoustic millimeter wave resonator as in claim 1 in which the patterned layer is arranged to mitigate a patterned layer mass load sensitivity of the bulk acoustic millimeter wave resonator at the main resonant frequency of the bulk acoustic millimeter wave resonator.
5. The bulk acoustic millimeter wave resonator as in claim 1 in which the patterned layer includes at least titanium.
6. The bulk acoustic millimeter wave resonator as in claim 1 in which the patterned layer includes at least a first dielectric.
7. The bulk acoustic millimeter wave resonator as in claim 1 in which:
 - the patterned layer is a first patterned layer;
 - the bulk acoustic millimeter wave resonator includes at least a second patterned layer; and
 - the first patterned layer and the second patterned layer contribute differently to mitigating a patterned layer mass load sensitivity of the bulk acoustic millimeter wave resonator at the main resonant frequency of the bulk acoustic millimeter wave resonator.
8. The bulk acoustic millimeter wave resonator as in claim 1 in which:
 - the patterned layer is a first patterned layer including at least a first material;

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the bulk acoustic millimeter wave resonator includes at least a second patterned layer, the second patterned layer including at least a second material; and
 the first material of the first patterned layer and the second material of the second patterned layer are different from one another.

9. The bulk acoustic millimeter wave resonator as in claim 1 in which:

- the patterned layer includes at least a first step mass feature having a thickness;

- the bulk acoustic millimeter wave resonator includes at least an intermediate step mass feature having an intermediate thickness, in which the intermediate thickness of the intermediate step mass feature is thinner than the thickness of the first step mass feature; and
- the first step mass feature and the intermediate step mass feature are arranged to facilitate the suppression of the spurious mode of the bulk acoustic millimeter wave resonator.

10. The bulk acoustic millimeter wave resonator as in claim 1 in which:

- a first step mass feature is associated with the patterned layer and arranged proximate to a lateral perimeter of the electrode; and

- the bulk acoustic millimeter wave resonator includes at least an intermediate step mass feature arranged relatively further from the lateral perimeter of the electrode than the first step mass feature.

11. The bulk acoustic millimeter wave resonator as in claim 1 in which the first piezoelectric layer has a first piezoelectric axis orientation, the second piezoelectric layer has a second piezoelectric axis orientation, and the third piezoelectric layer has a third piezoelectric axis orientation that substantially opposes the second piezoelectric axis orientation of the second piezoelectric layer.

12. The bulk acoustic millimeter wave resonator as in claim 1 in which:

- the electrode is a top electrode;

- the bulk acoustic millimeter wave resonator includes at least a bottom electrode;

- the piezoelectric stack includes at least an additional pair of piezoelectric layers;

- the piezoelectric stack includes at least the first piezoelectric layer, the second piezoelectric layer, the third piezoelectric layer, and the additional pair of piezoelectric layers coupled between the top electrode and the bottom electrode; and

- the piezoelectric stack is free of any interposing electrode.

13. The bulk acoustic millimeter wave resonator as in claim 1 in which:

- the electrode is an acoustically reflective electrode stack including at least a first pair of metal electrode layers and a second pair of metal electrode layers, in which the first pair of metal electrode layers and the second pair of metal electrode layers are acoustically coupled with the third piezoelectric layer; and

- members of the first pair of metal electrode layers and the second pair of metal electrode layers have respective acoustic impedances in an alternating arrangement to facilitate a plurality of reflective acoustic impedance mismatches.

14. The bulk acoustic millimeter wave resonator as in claim 13 in which the patterned layer is coupled between the first pair of metal electrode layers and the second pair of metal electrode layers of the electrode.

15. The bulk acoustic millimeter wave resonator as in claim 1 in which:

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the electrode is an acoustically reflective top electrode stack;

the acoustically reflective top electrode stack includes at least a first pair of top metal electrode layers; the patterned layer is a top patterned layer; and a first step mass feature is associated with the top patterned layer.

16. The bulk acoustic millimeter wave resonator as in claim 15 comprising an acoustically reflective bottom electrode stack including at least a first pair of bottom metal electrode layers electrically and acoustically coupled with the first piezoelectric layer and the second piezoelectric layer.

17. The bulk acoustic millimeter wave resonator as in claim 16 in which:

a first mesa structure includes at least the third piezoelectric layer;

a second mesa structure includes at least the acoustically reflective bottom electrode stack; and

a third mesa structure includes at least the acoustically reflective top electrode stack.

18. The bulk acoustic millimeter wave resonator as in claim 1 in which:

the electrode is an acoustically reflective electrode stack including at least a first pair of electrode layers; and the first pair of electrode layers is coupled between the patterned layer and the piezoelectric stack.

19. The bulk acoustic millimeter wave resonator as in claim 1 in which:

the electrode includes at least a first pair of electrode layers; and

the patterned layer is coupled between members of the first pair of electrode layers.

20. The bulk acoustic millimeter wave resonator as in claim 1 in which the bulk acoustic millimeter wave resonator has a quality factor of approximately 730 or greater at the main resonant frequency of the bulk acoustic millimeter wave resonator.

21. The bulk acoustic millimeter wave resonator as in claim 1 in which the electrode has sheet resistance of less than one Ohm per square at the main resonant frequency of the bulk acoustic millimeter wave resonator.

22. The bulk acoustic millimeter wave resonator as in claim 1 in which the main resonant frequency of the bulk acoustic millimeter wave resonator is in a millimeter wave band.

23. The bulk acoustic millimeter wave resonator as in claim 1 in which:

an acoustic wavelength is associated with the main resonant frequency of the bulk acoustic millimeter wave resonator;

the electrode includes at least a metal electrode layer having a metal electrode layer thickness; and

the metal electrode layer thickness is in a range from an eighth of the acoustic wavelength to three quarters of the acoustic wavelength at the main resonant frequency of the bulk acoustic millimeter wave resonator.

24. The bulk acoustic millimeter wave resonator as in claim 1 in which:

an acoustic wavelength is associated with the main resonant frequency of the bulk acoustic millimeter wave resonator;

the electrode includes at least a metal electrode layer having a metal electrode layer thickness; and

the metal electrode layer thickness is approximately half of the acoustic wavelength at the main resonant frequency of the bulk acoustic millimeter wave resonator.

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25. A millimeter wave filter comprising a plurality of bulk acoustic millimeter wave resonators over a substrate, in which at least one bulk acoustic millimeter wave resonator of the plurality of bulk acoustic millimeter wave resonators includes at least:

a first piezoelectric layer, a second piezoelectric layer, and a third piezoelectric layer having respective thicknesses to facilitate a main resonant frequency in a millimeter wave frequency band;

an electrode electrically and acoustically coupled with the third piezoelectric layer; and

a patterned layer associated with the electrode, in which the patterned layer is acoustically coupled with the third piezoelectric layer to facilitate a suppression of a spurious mode of the at least one bulk acoustic millimeter wave resonator.

26. The millimeter wave filter as in claim 25 in which:

the electrode is an acoustically reflective electrode stack including at least a first pair of metal electrode layers electrically and acoustically coupled with the third piezoelectric layer; and

a step mass feature is associated with the patterned layer.

27. The millimeter wave filter as in claim 25 in which:

the at least one bulk acoustic millimeter wave resonator includes at least an additional piezoelectric layer; the third piezoelectric layer has a third piezoelectric axis orientation; and

the additional piezoelectric layer has a piezoelectric axis orientation that substantially opposes the third piezoelectric axis orientation of the third piezoelectric layer.

28. A millimeter wave oscillator comprising:

electrical oscillator circuitry; and

a bulk acoustic millimeter wave resonator coupled with the electrical oscillator circuitry to excite electrical oscillation in the bulk acoustic millimeter wave resonator, in which the bulk acoustic millimeter wave resonator includes at least:

a first piezoelectric layer, a second piezoelectric layer, and a third piezoelectric layer;

an electrode electrically and acoustically coupled with the third piezoelectric layer to facilitate a main resonant frequency of the bulk acoustic millimeter wave resonator; and

a patterned layer associated with the electrode, in which the patterned layer is acoustically coupled with the third piezoelectric layer to facilitate a suppression of a spurious mode of the bulk acoustic millimeter wave resonator.

29. The millimeter wave oscillator as in claim 28 in which:

the electrode is an acoustically reflective electrode stack including at least a first pair of metal electrode layers electrically and acoustically coupled with the third piezoelectric layer; and

a step mass feature is associated with the patterned layer.

30. The millimeter wave oscillator as in claim 28 in which the bulk acoustic millimeter wave resonator includes at least an additional piezoelectric layer.

31. An acoustic millimeter wave device comprising:

a substrate;

a first piezoelectric layer having a first piezoelectric axis orientation;

a second piezoelectric layer having a second piezoelectric axis orientation;

a third piezoelectric layer acoustically coupled with the second piezoelectric layer, the third piezoelectric layer

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having a third piezoelectric axis orientation that is antiparallel to the second piezoelectric axis orientation; a first electrode electrically interfacing with the third piezoelectric layer; and

a patterned layer associated with the first electrode, in which the patterned layer is acoustically coupled with the third piezoelectric layer to facilitate a suppression of a spurious mode of the acoustic millimeter wave device.

32. The acoustic millimeter wave device as in claim **31** in which:

the first electrode includes at least a first pair of metal layers interfacing with the third piezoelectric layer; and a step mass feature is associated with the patterned layer.

33. The acoustic millimeter wave device of claim **32**, comprising a second electrode electrically interfacing with the first piezoelectric layer, the second electrode including at least a second pair of metal layers.

34. The acoustic millimeter wave device of claim **31**, comprising a fourth piezoelectric layer disposed between the first piezoelectric layer and the second piezoelectric layer, in which the fourth piezoelectric layer is acoustically coupled with the first piezoelectric layer and the second piezoelectric layer.

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